# The Unification of Matter, Forces, and Spacetime: Supplementary Proofs

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# Guide to This Supplement

This supplement accompanies *The Unification of Matter, Forces, and Spacetime from the Bost–Connes System*; together they constitute the rewritten manuscript. The main paper presents the physical relations, core formulas, and conditions under which they hold. This supplement retains the complete dependency index and provides the corresponding detailed dynamical proofs.

Section, equation, and theorem numbers prefixed by S refer to this supplement; all other chapter, section, and equation numbers refer to the accompanying main paper. Reference numbers follow the main paper. Q numbers identify dependency items. Conditional constructions, established consequences, and physical identifications are stated within the scope of each item.

This supplement follows the physical sequence of the main paper, placing constructions shared by later proofs first. S1–S3 develop the thermal cycle, formation phase, and convergence. S4–S7 establish records and nodes, spectral references, common-source participation, and the common matter–background carrier. S8 proves the normalization preservation used in resonance corrections; S9 and S10 treat metric processes with persistent replenishment and with complete spatial round trips, respectively. S11–S14 then give the specific proofs for the nucleon scale, up-type return, neutrino mass matrix, and nucleon phase. S15 gives the joint representation of quantum attributes and conditional selection by weak contact. S16 connects formation records, node visits, and actual returns into common persistent down-type and up-type processes, and proves their motion and common-source background response. Each proof retains its actual preparations, operations, and scope of readings. Specific examples cited ahead of their main treatment remain grouped with the corresponding physical objects.

# S0 Construction and dependency index

| Proof location | Question addressed | Main-paper sections |
|---|---|---|
| S1 Cycle with fixed thermal preparation | Complete outputs, thermal exchange, uninterrupted survival, and conserved energy | 2.3.1, 2.4.6 |
| S2 Return of the formation phase | Net phase and elapsed time of the actual round trip | 2.3.3, 8.1 |
| S3 Convergence of the formation process | Feedback convergence, persistent orbit, and spectral separation | 2.3.7, 2.4.6, 11.5 |
| S4 Records, nodes, and coherent return | Path coherence, node and whole-group shares, and reversible return | 2.3.4, 2.4.2–2.4.4, 3.3–3.5 |
| S5 Spectral reference and normalization | Analytic response, positive spectral weights, thermal preparation, and truncation | 2.4.1, 2.5, 7.1 |
| S6 Common-source participation and common frequency | Occupation, return columns, configuration formation, and common frequency | 2.5.3, 6.1–6.2 |
| S7 Common carrier of matter and background | Common transport of resource occupation, mass phase, and background gradient | 2.3.5, 2.5.3, 4.1 |
| S8 Matching normalization of the output state | Terminal ratios under retries and their electromagnetic output-state realization | 3.4.4, 7.3–7.4 |
| S9 Matter sources, background propagation, and metric strength | Positive-source delivery, original thermal position kernel, resource replenishment, and strength | 4.1–4.5.6, 7.6 |
| S10 Spatial outward and return paths and common response | Complete arrival histories, return budget, common timing, and strength | 4.5.7, 7.6 |
| S11 Approximation regime of nucleon spectral preparation | Nucleon-scale prescription, finite cutoff, and error comparison | 6.3 |
| S12 Finite loading and persistent-return enhancement | Finite loading, persistent frequency, and complete elapsed time for up-type quarks | 8.4 |
| S13 Complete neutrino round trip and mass matrix | Two spectral transport stages, double filtering, common cycle period, and survival records | 8.5–8.6 |
| S14 Nucleon phase and electromagnetic share | Nucleon-phase prescription and electromagnetic allocation | 9.4 |

| Proof location | Question addressed | Main-paper sections |
|---|---|---|
| S15 Joint representation of quantum attributes | Charge, color, propagation spinors, and conditional selection by weak contact | 5.2.3, 9.2 |
| S16 Formation structure and common family returns | Complete formation records, common down-type and up-type persistent actions, motion, and common-source background | 2.3–2.5, 8.3–8.4 |

Q01–Q22 summarize model inputs, constructed results, physical identifications, and their scope by function, for cross-chapter reference. A single item may contain content of these different kinds; the numbers identify dependency items, not the number of independent parameters or unproved propositions. Specific constructions and proofs are listed in the last column.

Masses expressed on the $v$ scale share Q06 and Q07; weak occupation and the remaining background are realized by the outputs from a common source in 6.1.2. Results that additionally invoke corrected couplings inherit the corresponding conditions listed in Q10. Subsequent uses of these results inherit their specified preparation, action, and calibration; paths and readouts added for each family are listed under the corresponding items.

## S0.1 Objects and spectral inputs

Table S0-1 Objects, spectral references, and matching.

| Item | Assumptions, results, and identifications | Applicable results and main-text locations |
|---|---|---|
| Q01 Selection of the p-channel | **Assumed:** All nontrivial labels of a complete single CRT factor satisfy $\chi^2$=1 or $\chi^2$=$\bar{\chi}$, and both classes are present. **Obtained:** These selection requirements give p=7. **Identification:** The three groups serve as three generations, each connected to an actual maintenance branch. | Numerical values at p=7; 5.2.1. |
| Q02 Representation of matter | **Obtained:** The natural SU(3) decomposition of the species space is 3⊕3. **Identification:** Color singlets, color carriers, and charge use a complete representation compatible with the actual exchange. S15 proves compatibility with the constructed propagation spinor; a complete embedding of the formation state and natural weak handedness selection remain to be established. | Physical names of matter and interactions; 5.2.2–5.2.3, 9.2, S15. |
| Q03 Analytic spectral reference | **Assumed:** Generator 3, the character convention, $s_0$=1/2, and a common analytic spectral reference, with $R_\ell = L(s_0, \chi_\ell)/\zeta(s_0)$. **Scope:** This analytic spectral reference, positive spectral preparation, and shell thermal preparation are defined separately. | Spectral weights and response ratios; 7.1, S5. |

| Item | Assumptions, results, and identifications | Applicable results and main-text locations |
|---|---|---|
| Q04 Matching and events | **Assumed:** Equal-weight preparation of independent group actions, point pairs, and orientation labels, together with rank-one matching. **Obtained:** N=3, 6, 18 and $\alpha^{(0)}$=w/N. **Identification:** Event shares are matched through the complete effect; terminal thermal exit additionally contains $t_u$. | Born couplings and event shares; 3.3, 7.2, S5.1. |

## S0.2 Formation and common readings

Table S0-2 Formation, persistent generators, and common calibration.

| Item | Assumptions, results, and identifications | Applicable results and main-text locations |
|---|---|---|
| Q12 Formation phase and frequency shift | **Assumed:** Label alignment, full feedback on the positive branch, the background–direction sign action, and independent two-spectrum auxiliaries and relays. **Obtained:** A coherent source, unit-spectral instrument capacity, the single-address quantities $w_{\mathrm{sp}} = \alpha_0$ and $a = w_{\mathrm{sp}} - w_{\mathrm{sp}}^2/\sqrt{3}$, and 1−a from an independent maintenance insertion. **Identification:** The return is identified with the unit-spectral kernel, whose comparison determines s=1. | Formation angle and frequency shift; 2.3.2–2.3.4, 2.4.5, 7.1.3, 8.1. |
| Q05 Phase mass | **Assumed:** Complete preparation, simultaneous equal-strength arms, reflection protection, and a shared motion port; feedback using a product-form isometric embedding, auxiliary replacement, and scalar thermal return over the entire transient sector. **Obtained:** A persistent generator, a common mass parameter for the rest-frame half-gap and inertia at the origin, and orbital attraction. **Reading:** An internal reference, unwrapped phase, and total elapsed duration. | Mass, survival, and energy compatibility; 2.3.5–2.4.7, S1.5, S3. |

| Item | Assumptions, results, and identifications | Applicable results and main-text locations |
|---|---|---|
| Q06 Common energy scale | **Assumed:** A common log W response, background squared matching, and a common drive history; elementary states use the common matrix element of 2.5.3, while collective ports actually deploy complete output columns and simultaneous reflection exchange. **Obtained:** The common capacity cancels in a single vacuum calibration, and the persistent half-gap is $E_0\sqrt{W/C_{\mathrm{bg}}}\log W$. **Scope:** Complete timing is used; the time-dependent averaging formula applies to ungated rest-frame and background phases; a gated protocol is calibrated once using its own complete phase, with the actual force given in S9. The actual success column, flag capacity, and bright-mode frequency on the configuration side are given in S6.5; the common logarithmic rule and background squared matching are retained. | 2.5, Chapter 6, 9.3.2, 10.6.3, S6.3, S9.6. |

## S0.3 Participation and resonance response

Table S0-3 Participation weights and corrected couplings.

| Item | Assumptions, results, and identifications | Applicable results and main-text locations |
|---|---|---|
| Q07 Electroweak participation | **Assumed:** Equal-amplitude participation of four-layer fibers, unit weights, 21 roles within the same $P_4$, and 3/1/2 routing. **Obtained:** Dimension 2058; complete spectral control yields positive occupation B_W and the complementary remainder, $W_{\mathrm{vac}} = 2058 - \operatorname{Tr} B_W$; when matching preserves $P_r v = 0$, $N_W^{\mathrm{eff}} = \operatorname{Tr} B_W$. The complete residual purification column yields $\|v_{\mathrm{vac}}\|^2 = W_{\mathrm{vac}}/\Lambda_{\mathrm{col}}^2$ through the common capacity. **Conditions:** Coherent control of the address reference, spectral records, and inverse operations; the energy scale follows Q06. S6.1 extends matching preservation to the full 15-dimensional maximal subspace and arbitrary positive matching effects. | v and related masses, weak coupling; 6.1, 7.3, S6.3. |

| Item | Assumptions, results, and identifications | Applicable results and main-text locations |
|---|---|---|
| Q08 Metric weighting | **Assumed:** Layer correspondence, participation of all addresses, a spectral background of the same capacity, complete preparation, and simultaneous reflections; the common logarithmic response and calibration follow Q06. **Obtained:** Return weights from a common source and the actual collective frequency are given in S6; S6.4 gives comparison-configuration dimension 41, and S7, S10 give occupation from a common source, a common potential, and complete timing. The same zero-blocking reference, full gating, and equal windows imply 2mb=t_H. **Scope:** 41 is not the maximum depth of the entire background; S9 replenishment and the genuinely inverse outward and return paths in S10 each use their own complete protocol, while three-dimensional position and final strength matching remain in place. | Metric energy scale, actual potential, and strength; 6.2, 7.6, S6, S9, S6.4, S7, S10. |
| Q09 Nucleon spectral preparation | **Obtained:** The lowest nonempty singlet of a pure fundamental color representation uses three copies. **Identification:** The leading term of the variance is $1/(\beta-1)^2 = 3$, hence $\beta_B = 1 + 1/\sqrt{3}$ and $W_B = \zeta(\beta_B)$. **Scope:** Exact variance V($\beta$)=3 specifies a different spectral preparation. | $M_B$ and derived masses; 6.3. |
| Q10 Coupling corrections | **Assumed:** Background and routing weights, independent spectral auxiliary units, simultaneous arms, and virtual-end couplings; terminal squared transfer weights are identified with x and y. **Obtained:** An all-order static realization of $D = x^2/(x+y)$ and a complete conditional Born realization of $N = 6{+}6D$. **Conditions:** The connection from the original virtual-end action to this terminal identification remains unproved; the color direction retains a finite prescription. S8 proves that the actual exit effects satisfy $yE_+$=$xE_-$ if and only if normalization is preserved at every finite cutoff; no scalar-continuation assumption is needed, while the correspondence between the original static action and the actual exits remains to be established. | Corrected couplings; 7.3–7.4. See Q07 for the common-source realization of weak $N_W$. |
| Q11 Nuclear-response weighting | **Assumed:** 42 orthogonal complete outputs share a matching input and $R_1$, with $\lvert c_g\rvert^2 = 1/\mathcal{L}_7$ on each path. **Obtained:** An orthogonal output sum for the nuclear response. **Scope:** Weighting here is separate from Q08; event probabilities additionally contain instrument calibration and input shares. | $\alpha_{\text{nuclear}}$; 7.5, 9.5.2. |

## S0.4 Elementary-state mass spectrum

Table S0-4 Paths and readouts of the elementary families.

| Item | Assumptions, results, and identifications | Applicable results and main-text locations |
|---|---|---|
| Q13 Lepton paths | **Assumed:** Even-conjugate two-spectrum access, coherent binary nodes, within-group matching, and a reference from the same source. **Obtained:** The normalized fourth power, node $1/\sqrt{2}$, and $p_j/p_0$ enter the same persistent action. **Identification:** Lepton masses use the instrument normalization and s=1. | e, $\mu$, $\tau$; 2.4, 2.5.3, 8.2. |
| Q14 Common down-type return | **Adopted:** one odd source, one whole-group comparison, the original first return, color exchange and direction recording on continuation, the complete current link, and preparation maintenance. **Derived:** common kernel $\mathcal{F}_d$, the group 0 three-color linear rate, group 1 intensity rates, and $m_d/m_s = b_L = e(1+3\alpha_0)$. Every failure, formation record, current work, and actual inverse is retained. | b, s, d; Section 8.3, S16.2–S16.3. |
| Q15 Common up-type return | **Adopted:** the complete spectral-path preparation, source-group-preserving three-port competition, reentry controlled by six earlier corrected records, the current link and independent weak visit, common neighboring positions, and complete preparation maintenance. **Derived:** three rates $1, G_N, b_L a_W$ on the formation image, the complete history tail, common-source background, and joint motion. Populations and persistent frequencies are read separately. | c, t, u; Section 8.4, S12, S16.4–S16.6. |
| Q16 Neutrino scale | **Assumed:** A symmetric indirect entrance, two copies of even closure, two coherent nodes, address–background filtering in two stages on the same uniform preparation, neutral reference-group routing, and equal-amplitude port flags; complete return and simultaneous protection follow Q05, and common calibration follows Q06. **Obtained:** Two relays give $N^{-2}$, two nodes give $1/2$, and the ports give $\kappa_{\text{port}} = 1/2$; the common schedule gives $\eta_\nu = 1$, yielding the complete common mass coefficient. The neutral action contains no charged frequency shift. | Complete neutral round trip, survival records, and common scale; 2.5.3, 8.5, S13.2–S13.5. |

| Item | Assumptions, results, and identifications | Applicable results and main-text locations |
|---|---|---|
| Q17 Neutrino chain kernel | **Assumed:** Common transport in the three-dimensional odd sector, amplitude $n^{-1/2}$ for each unit, orthogonal multiplier outputs, and the cyclic connections of (8.38); the auxiliaries participate in the complete return. **Obtained:** Spectral exponent 3, $h_a = \zeta_a(3)\zeta_{7-a}(3)$, and the fully weighted three-chain Gram kernel; the actual persistent generator yields a mass reading linear in the eigenvalues, while the common formation angle changes only the eigenvectors. **Identification:** Normal ordering; mixing and oscillations are calculated separately. | Relative neutrino spectrum, squared-mass differences, and finite-truncation bounds; 8.5–8.6, S13.1, S13.3, S13.6. |

## S0.5 Collective states and physical comparisons

Table S0-5 Collective-state evaluations and comparison conventions.

| Item | Assumptions, results, and identifications | Applicable results and main-text locations |
|---|---|---|
| Q18 Nucleon mass difference | **Obtained:** Conjugate spectral phases $\pm\theta$ at $\beta_B$. **Assumed:** Phase prescription $-(M_B^2/v)\tan\Theta_N$ and electromagnetic difference $\alpha_{\mathrm{EM}}M_B/6$, with a shared energy reference and charge-preserving diagonal readout. **Obtained:** The nucleon mass difference; the undetermined common shift $\delta_0$ cancels. S14 distinguishes unwrapped phase from the tangent prescription and gives finite-depth shares for protected branched electromagnetic vertices; signs, scales, and the determination of double counting still require actual vertices. | $m_n - m_p$; 9.4. |
| Q19 Weak bosons | **Assumed:** Direct response, condensate matching, and the common entrance vertex are each calibrated according to their own action, retaining the complete bright carrier, purification auxiliaries, and inverse return. W and Z shares are both 1; the two entrance output columns for neutral Born mixing and $2\mathrm{w}_2$ routing are collinear. **Obtained:** A rank-one neutral mass matrix and W and Z mass readings. | $M_W$, $M_Z$; 10.1.3, 10.2.1, 10.3, 10.6.3 (common-entrance realization). |

| Item | Assumptions, results, and identifications | Applicable results and main-text locations |
|---|---|---|
| Q20 Common $\pi$ mass | **Assumed:** Product preparation of the collective and flavor sides, orthogonal addition of u and d, and three equal-strength color-rearrangement routes with mutually orthogonal complete outputs. **Identification:** The amplitude-level reading of the total response is mass. **Obtained:** $m_\pi^2 = 3M_B(m_u + m_d)$. S4.4 derives the bright half-gap from a common entrance and orthogonal exits that can be returned; permanent intermediate records cannot replace persistent coherent return, and the actual column norms and common-clock calibration remain conditions. | $\pi$ mass; 10.5.2. |
| Q21 Radial oscillation | **Assumed:** m=(p−1)/2 independent $I_2/2$ checks, purification auxiliaries, a coherent single entrance, inverse return, and a common joint action relative to unit coupling. **Obtained:** The all-identical $\pm$ outcomes each have share $2^{-m}$, and an effective drive $\nu$ is obtained with matter and background sharing the total elapsed duration. **Calibration:** Squared matching $\nu$=v gives $m_H = v/2$ at p=7. | $m_H$; 10.6. |
| Q22 Physical comparisons | **Comparison:** Couplings, quark masses, and W/Z masses are compared at their respective scales and under their respective definitions; the nuclear response uses the $\pi$NN vertex, and neutrinos use normal-ordering combinations. **Scope of identification:** Chapter 4 retains fixed-source passive response and connects it to active positive-source delivery, scalar thermal survival, end-of-cycle replenishment, and complete timing; S9 gives positive leading-order strength and explicit spectral matching conditions. The identification of $G_E$ with Newton's constant inherits three-dimensional space, common calibration, and the conditions on the actual schedule. | Data and deviations; 4.5, 7.4–7.6, 8.6, 10.3.3, 12.2, S9. |

Finite-return differences and correlations between comparisons are discussed in Chapter 8 and 12.2, respectively; this table does not replace the spectral-coverage and exclusivity tests in 11.5.

# S1 Cycle with fixed thermal preparation

This section retains the original cycle of fixed thermal preparation, coherent retrieval, and shell-average restoration. Its thermalization conclusion follows §4.4 of Ref. [2]. Coherent delivery and persistent protection in Chapter 2 of the main text do not automatically satisfy the restoration structure of this section; the isospectral temporary storage in 4.4.3 separately preserves and returns the carrier.

## S1.1 Complete cycle

Using the joint unitary action $\mathbb{U}$ and environment preparation $\eta_{\mathcal{E}}$ from 2.3.1, denote the projections onto the shell-average subspace $\mathcal{C}$ and the direction space $\mathcal{M}$ by $\Pi$ and P, respectively. The shell-preserving input control V is the identity on the shell-average subspace, preserves the direction space, and commutes with $H_K$. The fixed restoration preparation is

$$\sigma_g^{\mathrm{sh}} = \sum_{j=0}^{K} g_j |e_j\rangle\langle e_j|, \qquad g_j > 0, \quad \sum_j g_j = 1. \tag{S1}$$

Preparing the environment independently in each cycle and taking the partial trace of the complete output gives the cycle $\boldsymbol{\mathcal{F}}_{V,g}$ in the main text [7], [8]. The following calculation allows background–direction coherence in the initial state; decomposition according to the actual output is performed only after restoration.

## S1.2 Formation, decay, and retrieval

Take a direction $\nu$, background endpoint $e_a$, and their shell energies $E_\nu$ and $E_a$. We calculate below the coefficients of the exchange and restoration specified in 2.3.1.

In the specified channel, choose thermal environment states u and v such that the joint vectors $|\nu,\mathrm{u}\rangle$ and $|e_a, v\rangle$ have the same total energy and total unit-group charge. The exchange angle $\alpha_{a\nu}$ determines the rotation between them:

$$\begin{aligned} W_{a\nu}|\nu, u\rangle &= \cos\alpha_{a\nu}|\nu, u\rangle + \sin\alpha_{a\nu}|e_a, v\rangle, \\ W_{a\nu}|e_a, v\rangle &= \cos\alpha_{a\nu}|e_a, v\rangle - \sin\alpha_{a\nu}|\nu, u\rangle. \end{aligned} \tag{S2}$$

The component remaining in the direction space is then coherently retrieved through the zero element. Write z=$|0\rangle$. Prepare the retrieval auxiliary in g, and associate direction $\nu$ with auxiliary state $a_\nu$; g and all $a_\nu$ form an orthonormal set. The round trip temporarily stores the direction coefficients in the same auxiliary system and then returns them with phase $w_\nu$:

$$|\nu, g\rangle \longmapsto w_\nu|z, a_\nu\rangle \longmapsto w_\nu|\nu, g\rangle, \qquad |w_\nu| = 1. \tag{S3}$$

The same auxiliary is retained throughout the round trip and is restored to the common g at the endpoint. Including free evolution, denote the net cycle phase of each direction by $\phi_\nu$, represented in the reduced cycle by $U_\phi = \Pi + \sum_\nu e^{i\phi_\nu} P_\nu$. Relative phases between different directions are calculated from this complete action.

The component transferred into the shell-average subspace is restored according to the prescribed preparation. Introduce auxiliary labels $r_{ja}$ that are jointly

orthonormal in both indices and orthogonal to g and all $a_\nu$. For input $e_a$, the restoration acts as follows:

$$\begin{aligned} |e_a, g\rangle \longmapsto |Y_a\rangle &= \sum_{j=0}^{K} \sqrt{g_j}\, |e_j, r_{ja}\rangle, \\ \mathrm{Tr}_R\, |Y_a\rangle\langle Y_b| &= \delta_{ab}\sigma_g^{\mathrm{sh}}, \\ \mathrm{Tr}_R(|\nu, g\rangle\langle Y_a|) &= 0. \end{aligned} \tag{S4}$$

Orthogonality in both indices gives the latter two equations: a background input is restored to $\sigma_g^{\mathrm{sh}}$, background–direction cross terms vanish under the partial trace, and returned directions remain unchanged. The distinctions between the input and shell labels remain in the joint state; auxiliary state g and share $g_j$ are defined separately.

## S1.3 Formation shares and uninterrupted maintenance

Formation and decay shares in the same exchange are determined by the channel preparation and exchange angle. For the selected channel, let the preparation shares be $a_{a\nu} > 0$ and $0 < \alpha_{a\nu} < \pi/2$; the unexchanged preparation has share $a_0 = 1 - \sum a_{a\nu} > 0$. Put $\Delta_{a\nu} = E_a - E_\nu$. The thermal-block shares on u and v satisfy

$$t_u = \frac{1}{1 + e^{\beta\Delta_{a\nu}}}, \qquad t_v = \frac{e^{\beta\Delta_{a\nu}}}{1 + e^{\beta\Delta_{a\nu}}}, \qquad t_u + t_v = 1. \tag{S5}$$

Multiplying the preparation shares by the squared moduli of the corresponding transfer amplitudes gives exit share $\epsilon_{a\nu}$ and formation share $f_{\nu a}$:

$$\epsilon_{a\nu} = a_{a\nu} t_u \sin^2 \alpha_{a\nu}, \qquad f_{\nu a} = a_{a\nu} t_v \sin^2 \alpha_{a\nu}, \qquad \frac{\epsilon_{a\nu}}{f_{\nu a}} = e^{-\beta(E_a - E_\nu)}. \tag{S6}$$

The ratio formula applies to the selected positive exchange channels at finite $\beta$; both shares are set to zero for unselected channels. Thermal preparation fixes the ratio between forward and reverse shares, while the exchange angle and channel-preparation shares determine their absolute sizes. Summing for the same input, write $\epsilon_\nu = \sum_a \epsilon_{a\nu}$ and $k_a = 1 - \sum_\nu f_{\nu a}$. Without input control, the system outputs of the complete cycle for the two inputs are

$$\mathcal{F}_{I,g}(P_\nu) = (1 - \epsilon_\nu) P_\nu + \epsilon_\nu \sigma_g^{\mathrm{sh}}, \qquad \mathcal{F}_{I,g}(P_{e_a}) = \sum_\nu f_{\nu a} P_\nu + k_a \sigma_g^{\mathrm{sh}}. \tag{S7}$$

The two equations give survival of the original direction and new formation from the background, respectively; the system state obtained by summing over the formation channels is generally mixed.

For a direction input X=PXP, the non-exit subchannel $\mathcal{Q}_V(X) = P\boldsymbol{\mathcal{F}}_{V,g}(X)P$ is completely positive and trace-nonincreasing. Pattern preservation for normalized $\psi$, with $P_\psi = |\psi\rangle\langle\psi|$, requires

$$\mathcal{Q}_V(P_\psi) = s_\psi P_\psi, \qquad s_\psi = \mathrm{Tr}\, \mathcal{Q}_V(P_\psi) > 0. \tag{S8}$$

Starting from a normalized direction state $\rho_0$, $\mathrm{Tr}\, \mathcal{Q}_V^n(\rho_0)$ gives the share surviving n consecutive cycles without exit. In particular, a single uncontrolled direction satisfies

$$\mathcal{Q}_I^n(P_\nu) = (1-\epsilon_\nu)^n P_\nu, \qquad s_n = (1-\epsilon_\nu)^n. \tag{S9}$$

This formula allows pattern preservation with a finite lifetime; components newly formed after an exit are not counted as uninterrupted survival.

For a general initial state, after first-cycle restoration and the partial trace over the environment, the system state has the form $\rho_n = X_n + y_n \sigma_g^{\mathrm{sh}}$, where $X_n = PX_nP \geq 0$, $\mathrm{Tr}\, X_n + y_n = 1$, and $n \geq 1$. By equation (S4), direction–background cross terms disappear from the reduced system state because of distinct restoration labels, while correlations may remain in the joint state. Denote the direction state supplied by a unit background share by $B_g = \sum_\nu (\sum_a g_a f_{\nu a}) P_\nu$, with $f_\Sigma = \mathrm{Tr}\, B_g$. Subsequent cycles satisfy

$$X_{n+1} = \mathcal{Q}_V(X_n) + y_n B_g, \qquad y_{n+1} = \mathrm{Tr}[X_n - \mathcal{Q}_V(X_n)] + (1-f_\Sigma) y_n, \quad n \geq 1. \tag{S10}$$

If the initial state already has this form, the recurrence starts at n=0. The two equations preserve unit total trace; long-term thermalization additionally requires the exchange-sufficiency and Gibbs-restoration conditions in §4.4 of Ref. [2].

## S1.4 Coherent readings of the original cycle

Take V=I, complete cycle period $\tau$, and the same fresh environment preparation in every cycle, with readout only in the selected final cycle. Thermal exchange and restoration do not supply new coherence between directions $\nu$ and $\mu$ in the same shell; their existing coherence is continued by the returned component. Write

$$d_\nu = \sum_a a_{a\nu} t_u (1 - \cos \alpha_{a\nu}), \qquad \gamma_{\nu\mu} = 1 - d_\nu - d_\mu, \qquad a_0 \leq \gamma_{\nu\mu} \leq 1. \tag{S11}$$

The terms $t_u$ in the sum take their values from the respective thermal channels. S1 specifies $a_0>0$, so the above coherence factor is positive; it changes the amplitude without adding phase. Put $\Delta\phi_{\nu\mu} = \phi_\nu - \phi_\mu$. Iteration over cycles gives

$$c_n := \langle \nu | \rho_n | \mu \rangle = \gamma_{\nu\mu}^n e^{in\Delta\varphi_{\nu\mu}} c_0. \tag{S12}$$

When $c_0 \neq 0$, coherence remains after any finite number of cycles; nontrivial $\Delta\phi$ changes the relative phase from cycle to cycle, while $\gamma<1$ weakens the signal. Pattern preservation of a single direction does not imply that a superposition pattern remains unchanged. If V mixes directions or probes are added in each cycle, their actual action and duration must be included and the result recalculated.

Here $\gamma$ and the scalar survival coefficient $q_T$ of the persistent carrier use different joint preparations and are not interchangeable. The retrieval phase is still specified by the actual action, not fixed by the thermal shares.

## S1.5 Compatibility conditions for conserved energy

The original cycle in S1 uses shell energy $H_K$ and fixed environment energy $H_\mathcal{E}$. If another phase generator is also to serve as the conserved energy in the same exchange, it must satisfy the following restriction. Under the strict-contraction conditions of

§4.4 in Ref. [2], fix the same complete joint unitary action $\mathbb{U}$ and environment energy $H_{\mathcal{E}}$, using the same energy unit. If a self-adjoint quantity $\widetilde{H}$ on the full space and $H_K$ both satisfy the same joint energy-conservation equation, subtraction gives $\mathcal{F}^*(D) = D$ for $D = \widetilde{H} - H_K$. Since every initial state approaches the same stationary state, $\mathrm{Tr}(D\rho) = \mathrm{Tr}(D\rho_*)$ holds for every normalized state, so D can only be a common constant. Thus

$$[\mathbb{U}, \widetilde{H} \otimes I + I \otimes H_{\mathcal{E}}] = 0 \quad \Longrightarrow \quad \widetilde{H} = H_K + \kappa_E I, \qquad \kappa_E \in \mathbb{R}. \tag{S13}$$

If this energy also generates the cycle phase being read, its actual unwrapped value must satisfy $-\widetilde{\phi}_\nu/\tau = E_\nu + \kappa_E$; at the cycle endpoints, this requires $w_\nu = e^{-i\tau\kappa_E}$ to be the same for every direction. The common endpoint phase determines compatibility modulo $2\pi$, while the process specifies the actual unwrapped branch. Under the fixed complete exchange, environment energy, and strict-contraction conditions above, directions in the same shell have the same conserved energy. Distinct mass phases within a shell therefore require a separately established compatible joint dynamics if they are to represent exchange energy differences.

## S1.6 Thermal exchange and actual flux

Thermal equilibrium determines how the two directions of exchange are apportioned; the actual flux also contains the share of the selected thermal block and the exchange angle. Following S1.2–S1.3, write $q_{a\nu} = a_{a\nu} \sin^2 \alpha_{a\nu}$. Then

$$\epsilon_{a\nu} = \frac{q_{a\nu}}{1 + e^{\beta(E_a - E_\nu)}}, \qquad f_{\nu a} = \frac{q_{a\nu}}{1 + e^{\beta(E_\nu - E_a)}}, \qquad \epsilon_{a\nu} + f_{\nu a} = q_{a\nu}. \tag{S14}$$

The terminal thermal channel in the main text and S5.1 already use the corresponding $q_{K\chi} = \alpha_\chi^{(0)} = |R_\chi|^2/N_\chi$. Substituting $E_\chi = 0$ and $E_K = K \log p$ gives $\epsilon_{K\chi} = \alpha_\chi^{(0)}/(1 + p^{\beta K})$ and $f_{\chi K} = \alpha_\chi^{(0)} p^{\beta K}/(1 + p^{\beta K})$. This fixes the exchange product, without separately determining the angle and preparation share.

**Proposition S1.1 (Actual flux of the thermal stage).** Take Gibbs occupation probabilities $p_i^\beta = e^{-\beta E_i}/Z_{\mathrm{full}}$ at the same temperature. The equilibrium one-way flux along each selected edge and the net transfer for arbitrary diagonal occupation probabilities are

$$\begin{aligned} j^*_{a\nu} &= p_\nu^\beta \epsilon_{a\nu} = p_a^\beta f_{\nu a} = \frac{q_{a\nu}}{Z_{\mathrm{full}}(e^{\beta E_\nu} + e^{\beta E_a})}, \\ J_{\nu\to a} &= p_\nu \epsilon_{a\nu} - p_a f_{\nu a} = j^*_{a\nu}(h_\nu - h_a), \qquad p_i = p_i^\beta h_i. \end{aligned} \tag{S15}$$

**Proof.** Multiplying the two exchange shares by the Gibbs occupation probabilities and taking a common denominator gives the same $j^*$; substituting $p_i = p_i^\beta h_i$ gives the second line. Thus the two directional flows are equal at equilibrium and the net flow vanishes. Let each column of $B$ be an oriented edge $e_\nu - e_a$. Summing edge by edge gives

$$\begin{aligned} p' - p &= -B\,\mathrm{diag}(j^*)B^{\mathsf{T}}h, \\ \mathcal{D}_{\mathrm{th}}(h) &= -\frac{h^{\mathsf{T}}(p' - p)}{\tau} = \frac{1}{\tau}\sum_{a,\nu} j^*_{a\nu}(h_\nu - h_a)^2. \end{aligned} \tag{S16}$$

Here $\tau$ is the complete duration of one cycle executing this thermal stage; division by it gives the cycle-averaged rate of change. This quadratic form belongs only

to diagonal occupation-probability exchange in the thermal stage. It is not the entropy-production rate of the full cycle, which includes restoration and coherent operations, and does not turn the discrete channel into a continuous semigroup. Thermal edges connect directions to shell-average states; resource-graph edges in Chapter 4 connect resource units at different positions. The outputs, records, and timing of these operations must be identified separately; a terminal exit rate cannot directly replace a refreshing rate of the entire background resource unit.

## S2 Return of the formation phase

This section completes the return proof in 2.3.3, including ordered composition and actual duration. We retain the auxiliary preparation over the entire three-dimensional formation space, without assuming that the final pattern has already formed. Put $P = I_F \otimes |r_M\rangle\langle r_M|$. The two actual generators are $G_0 = Z_F \otimes A_0$ and $G_1 = -Z_F \otimes A_1$, where $A_0 = Q_1$ and $A_1 = Q_1 Q_2 B$, the two spectral projections are independent, $B$ is a coherent relay, and $[G_0, G_1] = 0$. The relay preparation satisfies $\langle B\rangle = 1/\sqrt{m}$.

For an arbitrary fixed self-adjoint $G$, define the block-diagonal part preserving the input subspace and its orthogonal complement:

$$\mathcal{B}_P(G) = PGP + (I-P)G(I-P), \qquad O(s) = e^{isG}, \qquad P_{\text{path}}(s) = O(s)PO(s)^\dagger. \tag{S17}$$

Parallel transport along the actual projection path has the explicit solution

$$K(s) = i[\partial_s P_{\text{path}}(s), P_{\text{path}}(s)] = O(s)\mathcal{B}_P(G)O(s)^\dagger - G, \qquad W(s) = O(s)e^{-is\mathcal{B}_P(G)}. \tag{S18}$$

Direct differentiation gives $i\partial_s W = K(s)W$ and $WPW^\dagger = P_{\text{path}}(s)$, with $P_{\text{path}}(s)K(s)P_{\text{path}}(s) = 0$. Retrieval along this path must therefore implement $W(s)^\dagger$. Implementing the inverse outward path $O(s)^\dagger$ also cancels the entire outward phase, giving a different return.

The actual outward path executes $O_0 = e^{isG_0}$ followed by $O_1 = e^{isG_1}$, with intermediate projection $P_1 = O_0 P O_0^\dagger$. Since $[G_0, G_1] = 0$, we have $\mathcal{B}_{P_1}(G_1) = O_0\mathcal{B}_P(G_1)O_0^\dagger$. Substitution of the single-stage $W$ gives $W_1W_0 = O_1O_0e^{-is\mathcal{B}_P(G_1)}e^{-is\mathcal{B}_P(G_0)}$; taking the adjoint and multiplying by the outward evolution yields

$$(W_1W_0)^\dagger O_1O_0 = e^{is\mathcal{B}_P(G_0)}e^{is\mathcal{B}_P(G_1)}. \tag{S19}$$

The two factors on the right need not commute on the full space; on the input isometry $\mathcal{J}_F\psi = \psi \otimes r_M$, however, they compress to $w_{\text{sp}}Z_F$ and $-w_{\text{sp}}^2 Z_F/\sqrt{m}$, respectively, so their product gives $\mathcal{J}_F e^{isaZ_F}$. This proves the actual formation return of 2.3.3, including background preservation, opposite phases for the two directions, and restoration of every auxiliary.

The control can be written as an explicit joint generator. Put $S_{\text{path}}(s) = 2P_{\text{path}}(s) - I$, so that $\mathcal{B}_{P_{\text{path}}(s)}(G_i) = (G_i + S_{\text{path}}(s)G_iS_{\text{path}}(s))/2$. The return parameter runs backward according to $ds/dt = -h_R$, hence $H_{\text{back},i} = -h_R K_i$, namely

$$H_{\text{back},i}(t) = \frac{h_R}{2}\big(G_i - S_{\text{path}}(s)G_iS_{\text{path}}(s)\big). \tag{S20}$$

The return therefore requires two simultaneous arms with a relative minus sign, together with preparation reflections along the path. The deterministic reference constructed above from a half exchange and direction phase is used to synthesize the final fixed reflection; here the entire formation subspace moves with the outward path. These serve different purposes.

Suppose that every elementary arm on the outward path has strength $\omega$, and every return arm also has strength $\omega$. Then $h_R = 2\omega$. Each of the two outward

stages takes $s/\omega$, and each of the two return stages takes $s/(2\omega)$, giving net round-trip duration $3s/\omega$. Reflection synthesis, readout, routing, and waiting are included separately. This formation clock cannot be replaced by the persistent-stage clock merely because the final logical unitaries coincide.

The unit-action identification remains a condition linking the process to spectral evaluation. Its kernel comparison is as follows: the Euclidean continuation of the actual return $e^{isaZ_F}$ gives $e^{-sa}$ on this branch, while the specified unit-spectral kernel is $e^{-a}$. For real $a \neq 0$, the kernels coincide if and only if $s = 1$. This is the correspondence condition between spectral evaluation and the process used; the return construction itself retains the continuous parameter s.

The actual return continuously feeds the mean of the input block into the phase; for an unprotected bare spectral auxiliary, the mean of a matrix exponential generally cannot be replaced by the exponential of the mean.

# S3 Convergence of the formation process

The attraction result in 2.3.7 of the main text requires a joint test of the three-dimensional source and unready auxiliaries. We use the implemented product-form isometric embedding, complete conjugated feedback, and independent auxiliary replacement below; the fixed restoration of the original thermal cycle does not directly yield this section's result.

## S3.1 Complete feedback and deficit contraction

We work in decoded relational coordinates, retaining the outer encoding and the same internal group reference in the actual steps. Take $\xi = (\psi_\theta - \psi_{\bar{\theta}})/\sqrt{2}$, with $b_s, b_a$ two orthogonal states of equal energy and charge among the original thermal labels. The complete participating space also contains a block outside the bright plane. The actual exchange and feedback sign are

$$\begin{aligned} &X_A = |e_0, b_s\rangle\langle\sigma_0, b_s| + |e_0, b_a\rangle\langle\xi, b_s| + \text{h.c.}, \\ &P_A = X_A^2, \qquad Z_A = I - 2P_{\sigma_0,b_s} - 2P_{\xi,b_s}, \\ &R_s = I \otimes P_{b_s}, \qquad R_a = I \otimes |b_s\rangle\langle b_a|. \end{aligned} \tag{S21}$$

Exchange vanishes outside the participating space, where $Z_A$ is the identity. The two exchanges and pointer protocol use $U_\alpha = e^{-i\alpha X_A}$ and $V_\alpha = e^{i\alpha P_A} U_\alpha$ from the main text, with pointer-outcome actions $L_0 = U_\alpha(I + V_\alpha)/2$ and $L_1 = U_\alpha Z_A(I - V_\alpha)/2$. Finally, $R_s, R_a$ brings the thermal labels to the common $b_s$, with separate orthogonal records for the two merging outcomes; complementary outcomes complete the operation outside the thermal-label subspace used, which this participating input cannot reach. Charge balancing must still include the original internal reference; equal charge of the two thermal labels does not replace a covariant realization of the exchange vertex.

Take $\alpha = \pi/2$. In the order $(L_0, R_s), (L_1, R_s), (L_0, R_a), (L_1, R_a)$, the four actual outcomes on $\mathrm{span}\{\Phi_+, \Phi_-, \xi\}$ are

$$\begin{aligned} &A_0 = -iP_+ - \tfrac{i}{2}P_\xi, \quad A_1 = -i|\Phi_+\rangle\langle\Phi_-| + \tfrac{i}{2}P_\xi, \\ &A_2 = -\tfrac{i}{2}|e_0\rangle\langle\xi|, \qquad A_3 = \tfrac{i}{2}|e_0\rangle\langle\xi|, \qquad \sum_r A_r^\dagger A_r = I. \end{aligned} \tag{S22}$$

The last two terms belong to different records and cannot cancel through their opposite signs. Put $\mathcal{T} = \sum_r \mathrm{Ad}_{A_r}$ and $Q_F = I - P_+$. Multiplying the four matrices term by term gives

$$\mathcal{T}(P_+) = P_+, \qquad \mathcal{T}^*(Q_F) = \tfrac{3}{4}P_\xi \le \tfrac{3}{4}Q_F. \tag{S23}$$

Conjugating the entire feedback by the actual $D_s$ and then replacing the old auxiliary with an independently prepared pure auxiliary gives $\mathcal{E}$ in the main text. A SWAP transfers the old auxiliary to the outgoing space, so information is not deleted. For $Q_s = I - P_{\Phi_s} \otimes P_r \otimes I$, we have $\mathcal{E}^*(Q_s) = \tfrac{3}{4}P_{D_s\xi} \otimes I \otimes I \le \tfrac{3}{4}Q_s$. Adding the global bypass for readiness failure yields $\kappa = 1 - q_R/4$.

## S3.2 Limit of the persistent orbit

This section uses the code-subspace definition of 2.3.5 in the main paper: the image of the complete carrier's isometric embedding.

Deficit contraction drives the transient state toward the persistent orbit. Let $\rho_n = \mathcal{F}^n(\rho)$ and $\varepsilon_n = \mathrm{Tr}(Q_s\rho_n) \le \kappa^n\varepsilon_0$. Since the channel on the code is a single unitary channel, the fine-history criterion in 2.3.4 allows each complete Kraus operator to be written in blocks on and outside the code as

$$K_r = \begin{pmatrix} c_r U & B_r \\ 0 & C_r \end{pmatrix}, \qquad \sum_r |c_r|^2 = 1, \qquad \sum_r \bar{c}_r B_r = 0. \tag{S24}$$

The last equation is the cross block of the trace-preservation condition. Write $A_n = \mathcal{J}_s^\dagger \rho_n \mathcal{J}_s$ and $D_n = Q_s\rho_n Q_s$; then $A_{n+1} = UA_nU^\dagger + \sum_r B_r D_n B_r^\dagger$. The increment is positive semidefinite and has trace $\varepsilon_n - \varepsilon_{n+1}$; the interaction-picture quantity $U^{\dagger n}A_nU^n$ therefore increases monotonically to a trace-one $\sigma_\infty$. The remaining trace is $\varepsilon_n$. Applying the purification projection estimate $\|\rho_n - P_s\rho_nP_s\|_1 \le 2\sqrt{\varepsilon_n}$ gives the orbital bound in the main text. Cancellation of cross terms after summing all records does not mean that each postselected history vanishes separately.

## S3.3 Survival envelope and spectral separation

Take the actual survival share $0 < q_T \le 1$. Thermal reconnection uses target transport and filtering that act as the identity on the source, auxiliaries, and logical degrees of freedom not participating in this action, and compensates the free unitary over the entire transient domain. Each successful fine branch is $\sqrt{r_\eta}I$, and the complementary output effects sum to $(1-q_T)I$. The survival subchannel is then exactly $\mathcal{L} = q_T\mathcal{F}$, with exit effect $(1-q_T)I$. Cycle-by-cycle composition gives the survival and first-exit formulas of the main text; source- or mode-dependent filtering does not satisfy this premise.

Formation transients can also be spectrally separated from coherence within the code. The interaction-picture limit above defines a completely positive trace-preserving map $\Pi$ satisfying $\Pi\mathcal{J} = \mathrm{Id}$ and $\Pi\mathcal{F} = \mathrm{Ad}_U\,\Pi$, where $\mathcal{J}(\sigma) = \mathcal{J}_s\sigma\mathcal{J}_s^\dagger$. Thus $\mathcal{P}_\infty = \mathcal{J}\Pi$ is a superoperator projection commuting with $\mathcal{F}$, and the orbital bound extends linearly in finite dimension to $\|\mathcal{F}^n(I-\mathcal{P}_\infty)\| = O(\kappa^{n/2})$. If $h|a\rangle = E_a|a\rangle$, then

$$\lambda_{ab}^{\mathrm{live}} = q_T e^{-i(E_a-E_b)t_H}, \qquad r(\mathcal{L}|_{\ker\mathcal{P}_\infty}) \le q_T\sqrt{\kappa} < q_T. \tag{S25}$$

This includes the degeneracy multiplicities of coherence within the code, while giving only a radius bound for the remaining spectrum. The map $\Pi$ is a mathematical asymptotic readout, not an additional measurement; the infinite-cycle result uses the reduced limit with a continuing supply of fresh auxiliaries and environments. When $q_T < 1$, the unnormalized survival share ultimately tends to zero, while the orbit conditioned on survival still satisfies the proved bound. Visible spectral lines also depend on input and readout overlaps; no specific decay products or lifetimes are obtained here.

The selected serial cycle period is $\tau = t_H + T_R + T_X$: $T_R$ includes readiness, auxiliary replacement, forward and inverse spectral operations, and feedback; $T_X$ includes thermal transport, filtering, compensation, and routing. Formation auxiliary units

restored to the same preparation along every fine record can be reused across cycles; retrieval auxiliaries are restored on survival and leave with the information-carrying output at first exit. Initial loading counts as startup; fresh thermal baths, pointers, and unrestored auxiliaries must still be supplied.

# S4 Records, nodes, and coherent return

This section supplies the record conditions required by the path rules in the main text and calculates nodes and whole-group projections at general depth and for a general background amplitude ratio.

## S4.1 Records and conditional outputs

Actual records must be distinguished by orthogonal pointers and remain in the complete output through retrieval, reconstruction, and the retention period under discussion. The zero-element round trip in S1 restores the auxiliary to the common state g, so the intermediately distinguishable $a_\nu$ are not terminal records. Distinguishing return, decay, formation, and background continuation also requires retention of the input class; a terminal label alone does not distinguish all these processes.

Fix the same final system state, with path amplitudes $A_\alpha$ and normalized final environment states $\epsilon_\alpha$. The squared norm of the complete output gives

$$w = \sum_{\alpha,\alpha'} A_\alpha \overline{A_{\alpha'}} \langle \varepsilon_{\alpha'} | \varepsilon_\alpha \rangle. \tag{S26}$$

For complete pointer projections $\{\Pi_r\}$, including the no-response outcome, write $v = \sum_\alpha A_\alpha \varepsilon_\alpha$. The shares $w_r = \|\Pi_r v\|^2$ then satisfy $\sum_r w_r = \|v\|^2 = w$. If complete final states of paths in the same class are $\varepsilon_\alpha = e^{i\phi_\alpha}\varepsilon_r$, with distinct r orthogonal, only after absorbing $A_\alpha \leftarrow A_\alpha e^{i\phi_\alpha}$ do we obtain $w = \sum_r |\sum_{\alpha\in r} A_\alpha|^2$. Merely merging pointer names does not change environment inner products; mixed preparations require a separate sum over preparation weights.

Under statistical conditions (F0)–(F4) [2], the joint final state $\Omega_{\text{out}}$ including detection gives $p_r = \mathrm{Tr}[(I \otimes \Pi_r)\Omega_{\text{out}}]$. When outcome r occurs and $p_r > 0$, the unnormalized conditional state must be divided by $p_r$; orthogonal records do not guarantee statistical independence of earlier and later stages, whose composition still uses the complete instrument of 2.4.2. Selected records may be separated by many cycles; no-response cycles must still be counted in both statistics and elapsed duration.

Internal-state identification in 9.1.2 is also tested by actual measurement. For normalized states $\sigma_A$ and $\sigma_C$, perfect single-shot discrimination within the unrestricted measurement class is possible if and only if their supports are orthogonal. If $0 \le E \le I$ has probabilities 1 and 0 on the two states, positivity forces their supports into the eigenvalue-1 space and kernel of E, respectively, so they are orthogonal; conversely, the support projection of $\sigma_A$ distinguishes two preparations with orthogonal supports. A restricted instrument must additionally implement this deterministic discrimination.

For a pure joint output $\sum_a c_a|a\rangle|e_a\rangle$, with orthonormal system basis $|a\rangle$ and normalized environment vectors $|e_a\rangle$, the reduced matrix entries are $\rho_{ab} = c_a\bar{c}_b\langle e_b|e_a\rangle$: occupations use squared moduli, while coherences also contain environment overlaps. Restoration in S1 removes background–direction coherence from the system; the associated phases must be read before restoration or through a joint readout of the still-accessible environment.

## S4.2 Node overlaps

We use the normalization rules of 2.4.3 and 3.3: an equal-amplitude orbit of h points has single-point amplitude $1/\sqrt{\mathrm{h}}$ and share 1/h, where h must be the actual orbit size. Multiplication of CRT overlaps requires product preparation and product readout. The following calculations distinguish a single point, a single deep-layer point pair, and a complete species group.

Write N=$\phi(\mathrm{p}^K)$ and $e_0 = N^{-1/2}\sum_{u\in S_0}|u\rangle$. For $D=\{u,-u\}$, put $d_D^{\pm} = (|u\rangle \pm |-u\rangle)/\sqrt{2}$ and let $Q_D$ be the projection onto this point pair. Then

$$\langle u|e_0\rangle = \frac{1}{\sqrt{N}}, \qquad \langle d_D^+|e_0\rangle = \sqrt{\frac{2}{N}}, \qquad \langle d_D^-|e_0\rangle = 0, \qquad \langle e_0|Q_D|e_0\rangle = \frac{2}{N}. \tag{S27}$$

The symmetric overlap of a single point pair equals $1/\sqrt{\mathrm{m}}$, with m=(p−1)/2, only at K=1. If the input is the complete restored background, the unit-shell preparation share must also be included:

$$\mathrm{Tr}(P_u\sigma_g^{\mathrm{sh}}) = \frac{g_0}{N}, \qquad \mathrm{Tr}(Q_D\sigma_g^{\mathrm{sh}}) = \frac{2g_0}{N} \quad (u\in S_0). \tag{S28}$$

The factor $\mathrm{g}_0$ cancels only in the distribution conditioned on the unit shell. The symmetric amplitude above depends on the coherence of $\mathrm{e}_0$; replacing it by the uniform mixture on the unit shell leaves point and point-pair occupations unchanged but changes the projection share onto the symmetric direction.

Let $\pi$ be reduction modulo p. Coherently lifting all deep-layer addresses of a species-layer point pair $\overline{D}=\{a,-a\}$ gives

$$|\bar{d}_{\bar{D}}\rangle = \frac{1}{\sqrt{2p^{K-1}}}\sum_{\pi(u)\in\bar{D}}|u\rangle, \qquad \langle\bar{d}_{\bar{D}}|e_0\rangle = \sqrt{\frac{2p^{K-1}}{N}} = \frac{1}{\sqrt{m}}, \quad m=\frac{p-1}{2}. \tag{S29}$$

The whole group contains $2\mathrm{p}^{K-1}$ points, so this $1/\sqrt{\mathrm{m}}$ holds for every K. It and $\sqrt{(2/\mathrm{N})}$ for a single deep-layer point pair refer to different endpoints.

A coherent relay contains both entrance and exit overlaps. For example, for u,v∈$\mathrm{S}_0$,

$$\langle v|P_{e_0}|u\rangle = \langle v|e_0\rangle\langle e_0|u\rangle = \frac{1}{N}, \qquad \Pr(P_{e_0}\ \text{success}, v \mid u) = \frac{1}{N^2}. \tag{S30}$$

Summing over v gives selection success probability 1/N; the single-point share conditioned on success is also 1/N. Repeating the same projection satisfies $P_{e_0}^2 = P_{e_0}$ and does not automatically add a new factor; a new factor must come from an actual intervening action or repreparation. Nontrivial directions are orthogonal to $\mathrm{e}_0$, so transfer into the background requires an actual exchange; uniformity alone cannot specify its matrix element.

For $N_r$ mutually exclusive outcomes, $w_{rj} = w_r/N_r$ holds only when the actual preparation and instrument give equal weights; when $w_r > 0$, $\Pr(j|r) = 1/N_r$. Sums over orthogonal outcomes and coherent amplitude sums into the same final state remain distinct as in S4.1.

## S4.3 General whole-group projections

Let m=(p−1)/2 and let g generate the unit group modulo p. At arbitrary depth K, the species-layer groups j=0,…,m−1 are

$$\mathcal{D}_j = \{u \in S_0 : \pi(u) \in \{g^j, -g^j\}\}, \qquad |\mathcal{D}_j| = 2p^{K-1}, \qquad |\bar{d}_j\rangle = \frac{1}{\sqrt{2p^{K-1}}} \sum_{u\in\mathcal{D}_j} |u\rangle. \quad \text{(S31)}$$

The groups are disjoint, and the $\bar{d}_j$ are mutually orthogonal. The whole-group projection $\bar{Q}_j = \sum_{u\in\mathcal{D}_j} |u\rangle\langle u|$ has rank $2\mathrm{p}^{K-1}$, with $\bar{d}_j$ one coherent direction within it. Whole-group readout preserves within-group coherence; pointwise recording followed by grouping can give the same group probabilities but a different conditional pattern.

First take p≥7 and m≥3. Choose a faithful character of the quotient group $G_{p,1}/\{\pm 1\}$ and pull it back to an even direction $\theta$ such that $\theta(g) = e^{2\pi i/m}$. Then $\mathrm{e}_0$, $\psi_\theta$, and $\psi_{\bar{\theta}}$ are orthogonal, and

$$|e_0\rangle = \frac{1}{\sqrt{m}} \sum_{j=0}^{m-1} |\bar{d}_j\rangle, \qquad |\psi_\theta\rangle = \frac{1}{\sqrt{m}} \sum_{j=0}^{m-1} e^{2\pi i j/m} |\bar{d}_j\rangle. \quad \text{(S32)}$$

The conjugate direction has the opposite phase. A symmetric entrance is chosen here; odd directions have zero amplitude at this entrance but still respond at an antisymmetric entrance.

Fix the common phase using $\mathrm{e}_0$, retain a general background amplitude ratio w≥0, and take

$$|\sigma_\delta\rangle = \frac{e^{i\delta}|\psi_\theta\rangle + e^{-i\delta}|\psi_{\bar{\theta}}\rangle}{\sqrt{2}}, \qquad |\Phi_{w,\delta}\rangle = \frac{w|e_0\rangle + |\sigma_\delta\rangle}{\sqrt{w^2+1}}. \quad \text{(S33)}$$

This preparation contains background–direction coherence; its persistence must be tested under the actual cycle. Expansion gives

$$|\Phi_{w,\delta}\rangle = \sum_{j=0}^{m-1} \frac{f_j}{\sqrt{m(w^2+1)}} |\bar{d}_j\rangle, \qquad f_j = w + \sqrt{2}\cos\left(\delta + \frac{2\pi j}{m}\right). \quad \text{(S34)}$$

For m≥3 the cosines sum to zero and their squares sum to m/2, hence

$$\begin{aligned} \sum_j f_j &= mw, \\ \sum_j f_j^2 &= m(w^2+1), \\ p_j = \langle\Phi_{w,\delta}|\bar{Q}_j|\Phi_{w,\delta}\rangle &= \frac{f_j^2}{m(w^2+1)}, \\ \sum_j p_j &= 1. \end{aligned} \quad \text{(S35)}$$

Each of the $\mathrm{p}^{K-1}$ deep-layer point pairs in a group has share $p_j/p^{K-1}$; the species-group ratios are independent of this repetition multiplicity. Setting w=1 recovers

the delivered output of 2.3.2; normalization itself does not determine w. For any reference group with $f_{j_*} \neq 0$,

$$\varrho_j := \frac{p_j}{p_{j_*}} = \left(\frac{f_j}{f_{j_*}}\right)^2, \qquad p_j = \frac{\varrho_j}{\sum_\ell \varrho_\ell}. \tag{S36}$$

The values $f_j$ may be positive, negative, zero, or degenerate; if $f_0$=0, another reference must be chosen, while the original distribution remains valid. The m groups count readings; whether these are nonzero and correspond to physical generations still depends on the pattern and dynamics.

Small primes require separate calculation: p=3 has no nontrivial even character; at p=5 the nontrivial even character is self-conjugate, and the two numerator terms combine into $we_0+\sqrt{2}\cos\delta\cdot\psi_\theta$, requiring normalization by the nonzero squared norm $w^2+2\cos^2\delta$. Other self-conjugate cases likewise use a single direction; nonfaithful characters require repeated phases to be treated according to their actual order.

If $L_\theta = L(1/2,\theta) \neq 0$, put $\delta_\theta = \arg L_\theta$. Conjugacy at real spectral points gives

$$\widetilde{\mathcal{B}}\sigma_0 = \frac{L_\theta\psi_\theta + \overline{L_\theta}\psi_{\bar\theta}}{\sqrt{2}} = |L_\theta|\,\sigma_{\delta_\theta}. \tag{S37}$$

If the same spectral transfer also acts on the background, its coefficient acquires an additional $L_0$; equal amplitude $\Phi_{1,\delta_\theta}$ cannot be inferred directly. If only background–direction cross terms are removed while coherence within the conjugate pair is retained, then

$$\rho_{\mathrm{dec}} = \frac{w^2 P_{e_0} + P_{\sigma_\delta}}{w^2+1}, \qquad p_j^{\mathrm{dec}} = \frac{w^2 + 2\cos^2(\delta + 2\pi j/m)}{m(w^2+1)}. \tag{S38}$$

The $2\sqrt{2}w\cos$ term disappears, showing that the preceding $p_j$ requires actual retention of background–direction coherence. Restoration in S1 also changes other weights, so this pure-state share cannot be carried over directly. Stored group records can be preserved by a subsequent trace-preserving channel that does not act on them, but this does not guarantee that the system pattern remains in its original group; persistence must still be tested for the corresponding carrier and actual process.

Finally, suppose the persistent process has yielded a common positive coefficient $M_j = Cf_j^2$, C>0. Take w>0, all m groups, and $s_j = \operatorname{sgn} f_j$. The two summation identities above give

$$\sum_{j=0}^{m-1} M_j = Cm(w^2+1), \qquad \sum_{j=0}^{m-1} s_j\sqrt{M_j} = \sqrt{C}\,mw, \qquad \frac{\sum_j M_j}{\left(\sum_j s_j\sqrt{M_j}\right)^2} = \frac{w^2+1}{mw^2}. \tag{S39}$$

At w=1 the right-hand side is 2/m; signs in front of the square roots may be omitted only when all $f_j \geq 0$. The formula does not apply directly to a subset of groups, self-conjugate directions, or unequal mass coefficients. It gives a spectral-shape relation; the common mass coefficient still comes from the actual generator and calibration.

## S4.4 Returnable routes and permanent records

Species directions, vacuum-counting addresses, and propagation wave numbers specify the formation pattern, participating preparation, and propagation eigendirections, respectively. Group-element and character descriptions of a factor can be interchanged only after establishing a concrete Fourier transform on the same group and representation. A species pattern at zero propagation wave number does not eliminate the matter's logical position register. All records below belong to the complete joint state; not reading a record does not erase it.

Suppose two paths give system vectors $a, b$, with complete final auxiliary states $R_a, R_b$. After tracing out the auxiliaries, the cross block is $|a\rangle\langle b|\langle R_b|R_a\rangle$. Orthogonal terminal records make it vanish; a common unitary inverse that restores auxiliaries to a shared ready state can preserve it. This criterion applies to addresses, paths, and exit times, but labels of these three distinct factors cannot simply be identified as one record.

**Proposition S4.4 (Bright-mode frequency of a common entrance).** Let the normalized entrance $i$ be orthogonal to every complete exit $v_{cf}$; the exits are mutually orthogonal but can be returned by the same reversible action. Write $V = \sum_{cf} v_{cf}$ and $S_V = \|V\|^2 > 0$. The actual persistent action and bright eigenvalues are

$$H = \nu\big(|V\rangle\langle i| + |i\rangle\langle V|\big), \qquad H\frac{i \pm V/\sqrt{S_V}}{\sqrt{2}} = \pm\nu\sqrt{S_V}\frac{i \pm V/\sqrt{S_V}}{\sqrt{2}}. \tag{S40}$$

Proof: the action preserves $\mathrm{span}\{i, V\}$ and, in the normalized basis $i, V/\sqrt{S_V}$, is a two-dimensional matrix with off-diagonal entry $\nu\sqrt{S_V}$; the remaining exit combinations are dark states. If $v_{cf} = |c\rangle \otimes a_B \otimes a_f$, the three color routes have equal strength, the $u, d$ flavor exits are orthogonal, and the actual columns satisfy $\|a_B\|^2 = b_B, \|a_f\|^2 = b_f$, then $S_V = 3b_B(b_u + b_d)$. If common-clock calibration further gives $\nu = E_0$, the $\pi$ mass relation in the main text follows. The product norms of the actual output columns and common-frequency calibration must still be provided by the vertices used; route counting alone cannot replace them.

A permanent intermediate record gives a different actual output. Write $V_c = \sum_f v_{cf}$. Dephase between the entrance block and the separate route-exit blocks, and let $z = \sin^2(\nu\sqrt{S_V}t)$ and $q_c = \|V_c\|^2/S_V$. The entrance probability after evolution, recording, and a second evolution, and the probability for fully coherent evolution, are respectively

$$P_i^{\mathrm{rec}}(2t) = (1-z)^2 + z^2\sum_c q_c^2, \qquad P_i^{\mathrm{coh}}(2t) = (1-2z)^2. \tag{S41}$$

Proof: the entrance weight after the first stage is $1 - z$, contributing $(1 - z)^2$ in the second stage; route $c$ has weight $zq_c$ and probability $zq_c$ of returning to the entrance, so summing the recorded branches gives the formula above. The coherent formula follows directly by doubling the evolution time. The two generally differ, so the return occupation probability after a permanent record cannot directly serve as the persistent half-gap above. For the neutral mass matrix with two inputs, each route contributes the Gram matrix of its output columns; if the two columns have the same fixed ratio on every route, the total Gram matrix remains rank one.

Orthogonal exit routes alone neither remove the $2w_2$ term in the W/Z construction of the main text nor destroy its proved zero mode.

**The same effect and different returns.** Continue with $U = e^{-itH}$ above, and denote the fine projections of the orthogonal exits by $\Pi_g$, satisfying $\Pi_g i = 0$ and $\sum_g \Pi_g V = V$; write $v_g = \Pi_g V$. The sets $C_s$ group the fine-route labels, with $\Pi_s = \sum_{g\in C_s} \Pi_g$. When generating records, one may retain each $g$ separately or retain only the group label $s$. The former merges labels when reporting them; the latter never retains the within-group distinction. The corresponding group-outcome instruments are

$$\begin{aligned} \mathcal{I}_s^{\mathrm{fine}}(\rho) = \sum_{g\in C_s} \Pi_g U\rho U^\dagger \Pi_g, \qquad \mathcal{I}_s^{\mathrm{coarse}}(\rho) = \Pi_s U\rho U^\dagger \Pi_s, \\ (\mathcal{I}_s^{\mathrm{fine}})^*(I) = (\mathcal{I}_s^{\mathrm{coarse}})^*(I) = U^\dagger \Pi_s U. \end{aligned} \tag{S42}$$

The entrance and remaining orthogonal outputs complete the other outcomes, so both instruments are complete. Equality of effects follows directly from $\Pi_g\Pi_h = \delta_{gh}\Pi_g$, but the within-group cross blocks retained by coarse records vanish under fine recording. For the same entrance $i$, put $p_g = \|v_g\|^2/S_V$; after recording, run the same $U$ again without accessing the record pointers. By the branchwise proof of (S41), the fine-record return is $(1-z)^2 + z^2 \sum_g p_g^2$ and the coarse-record return is $(1-z)^2 + z^2 \sum_s (\sum_{g\in C_s} p_g)^2$, generally different. This proves that the same event effect does not determine subsequent return; it does not permit restoration of coherence by reading fewer of the records already stored.

# S5 Spectral reference and normalization

This section checks the reference objects for analytic responses and the distinction between positive spectra and shell thermal spectra, and gives the finite-truncation bounds used in Chapter 6.

## S5.1 Analytic response and exchange normalization

At $s_0{=}1/2$, write $L_\chi = L(s_0, \chi)$ and $\zeta_0{=}\zeta(s_0)$. The unnormalized spectral transfer on directions, the full-space background, and the unit-shell average state satisfies, respectively,

$$\widetilde{\mathcal{B}}\psi_\chi = L_\chi \psi_\chi, \qquad \widetilde{\mathcal{B}} b = \zeta_0 b, \qquad \widetilde{\mathcal{B}} e_0 = L_0 e_0, \qquad L_0 = \zeta_0(1 - p^{-1/2}). \tag{S43}$$

Relative to b one obtains $R_\chi = L_\chi/\zeta_0$; using $e_0$ as reference instead gives $L_\chi/L_0$. The diagonal response of the zero element z is also $\zeta_0$, but its complete output contains components of other shell-average states:

$$\widetilde{\mathcal{B}} z = \zeta_0 \left( z + \sqrt{1 - p^{-1}} \sum_{j=0}^{K-1} e_j \right), \qquad \|\widetilde{\mathcal{B}} z\|^2 = |\zeta_0|^2 \left[ 1 + K(1 - p^{-1}) \right]. \tag{S44}$$

The latter equation follows from orthogonality of z to every $e_j$, so the diagonal response cannot replace the norm of the complete output. If the zero element serves only as a path-connecting endpoint and no spectral transfer acts on it, it adds no $\zeta_0$ factor. To turn response ratios into actual probabilities still requires a complete instrument; 7.1.3 gives the unit-spectral instrument used. For any normalized preparation $\sigma$, the matching response is $|R_\chi|^2 \operatorname{Tr}(\sigma P_{\mathrm{match}})$. Fix a trace-preserving preparation process, pull the effect of the actual $\chi$ event class back to the label space as $E_\chi^{\mathrm{act}}$, and fix the same $\kappa > 0$. Then

$$\begin{gathered} E_\chi^{\mathrm{act}} = \kappa^2 |R_\chi|^2 P_{\mathrm{match}} \\ \Longleftrightarrow \\ q_\chi^{\mathrm{act}}(\sigma) = \kappa^2 |R_\chi|^2 \operatorname{Tr}(\sigma P_{\mathrm{match}}) \\ \text{for every normalized preparation } \sigma. \end{gathered} \tag{S45}$$

The forward implication follows by taking the trace; the reverse holds for all pure states, so the quadratic form of the difference between the two self-adjoint effects vanishes identically, making the difference operator zero. Checking a single equal-weight preparation does not determine the effect. This criterion determines event shares; actual outputs and phases still come from the branch operators, and the joint realization must satisfy the corresponding conservation laws.

Chapter 3 additionally uses the original thermal exchange of S1 for event-rate comparisons. For selected channels from a unit-shell direction to the terminal shell, specify $a_{K\chi} \sin^2 \alpha_{K\chi} = |R_\chi|^2 / N_\chi$. Then

$$\epsilon_{K\chi} = t_u \frac{|R_\chi|^2}{N_\chi}, \qquad t_u = \frac{1}{1 + e^{\beta(E_K - E_\chi)}} = \frac{1}{1 + p^{\beta K}}. \tag{S46}$$

Here $E_\chi = 0$ and $E_K = K \log p$; the thermal factor belongs to the actual share and cannot be canceled by spectral normalization. If only this set of terminal channels is configured, then under $a_0 > 0$ and $0 < \alpha_{K\chi} < \pi/2$, a valid realization requires exactly $|R_\chi| > 0$ for every selected channel and $S = \sum_\chi |R_\chi|^2/N_\chi < 1$: necessity follows from $\sin^2\alpha < 1$ and $\sum a < 1$; sufficiency follows by choosing $S < \lambda < 1$, $\sin^2\alpha_{K\chi} = \lambda$, and $a_{K\chi} = |R_\chi|^2/(N_\chi\lambda)$. This short construction proves realizability of the specified correspondence, without making it a strength necessarily selected by thermal preparation.

## S5.2 Positive spectral weights and shell thermal preparation

Use the logarithmic energy spectrum indexed by positive integers $H_{\log,L}|n\rangle_{\log} = (\log n)|n\rangle_{\log}$ from 2.5 and 6.3, with 1≤n≤L. The summation cutoff L and address depth K are specified separately, and energy is zero at n=1. For $\beta$>0,

$$\Gamma_{\log,L}(\beta) = e^{-\beta H_{\log,L}} = \sum_{n=1}^{L} n^{-\beta} |n\rangle_{\log}\langle n|, \qquad Z_L(\beta) = \operatorname{Tr}\Gamma_{\log,L}(\beta) = \sum_{n=1}^{L} n^{-\beta}. \quad \text{(S47)}$$

The normalized shares are $n^{-\beta}/Z_L$. A uniform energy shift by a multiplies both $\Gamma$ and $Z_L$ by $e^{-\beta a}$, leaving shares unchanged but changing absolute participation; using the partition sum to calibrate energy requires fixing this zero-point convention.

The original cycle has $H_K$ equal to j log p on shell j, with Gibbs partition sum

$$Z^{\mathrm{sh}}_{p,K}(\beta) = \operatorname{Tr} e^{-\beta H_K} = \sum_{j=0}^{K} |S_j| p^{-\beta j}. \quad \text{(S48)}$$

It contains shell degeneracies and generally differs from $Z_L$. Even when L=$p^K$, every level of the logarithmic spectrum is simple, whereas the zero-energy unit shell of $H_K$ has multiplicity $p^{K-1}(p-1)\geq 2$; hence there is no generator-preserving unitary equivalence. Using integer representatives to label addresses also requires a separate energy assignment for the zero residue class; equal address counts do not identify the two spectra.

Since $T_n$b=b, the finite response $\widetilde{\mathcal{B}}_L(\beta) = \sum_{n=1}^{L} n^{-\beta} T_n$ satisfies $\widetilde{\mathcal{B}}_L(\beta)b = Z_L(\beta)b$. Different integers can induce the same finite address operation; their coefficients still add without creating independent directions. Although the trace on the logarithmic spectral space and the response coefficient of b are numerically equal, they act on different objects; $W_{\mathrm{em}} = Z_L$ is an adopted participation correspondence, not the rank of $P_b$ or the weight of an arbitrary one-dimensional projection.

## S5.3 Finite truncation and tail bounds

For $\beta$>1, monotonicity of $n^{-\beta}$ and integral comparison give

$$Z_L(\beta) \longrightarrow \zeta(\beta), \qquad 0 < \zeta(\beta) - Z_L(\beta) < \int_L^\infty x^{-\beta}\, dx = \frac{L^{1-\beta}}{\beta - 1} \quad (\beta > 1). \quad \text{(S49)}$$

For fixed L, $Z_L(1)$ is finite; the pole of $\zeta$ comes from infinite summation, and the tail bound must be rechecked near $\beta$=1. For $\beta \leq 1$ the infinite positive-weight

sum diverges; analytic continuation at regular spectral points may still serve as a specified response, but not as that positive-weight sum.

Nontrivial character sums contain phases and are not traces of positive operators. The full-space b accumulates all integer terms, whereas the trivial unit-group character excludes multiples of p, giving responses $\zeta(\beta)$ and $(1-\mathrm{p}^{-\beta})\zeta(\beta)$, respectively, in the convergence domain. Their physical connections to positive spectral participation all use the reference and matching conditions specified in the main text.

# S6 Common-source participation and common frequency

This section first obtains participation outputs from the same occupation process, then constructs the return columns and common frequency. Configuration formation uses this return construction, and its complete matching subsequently enters the collective column.

## S6.1 Common-source occupation and matching preservation

**Assumptions.** In a finite joint state space, the process participation projection is $P_4$ of rank $N_0$, and the orthogonal reference-background projection is $P_b^{\mathrm{ref}}$ of rank $M$. The reduced input address state is $(P_4+P_b^{\mathrm{ref}})/Z_0$, with $Z_0 = N_0+M$. If a finite thermal preparation approximates it, its preparation error must be retained separately; a finite number of cycles cannot automatically be called an exact stationary state. We use role encoding within the same $P_4$, compatible with joint energy and group charge: $P_B$ has rank 18 and $P_r \le P_B$ has rank 3; $P_2, P_3 \le P_4 - P_B$ have ranks 1 and 2, respectively, and are orthogonal. Identifying the neutral-register representation of the eighteen labels with these address roles is an adopted encoding condition.

The three fixed probes of the original complete spectral instrument give success weights $w_i \in [0,1]$, using the verified unit capacity. The initial occupation is $P_B$; $P_r$ is released when the first type of spectral probe succeeds, while $P_2, P_3$ are added on their respective spectral successes. Success controls only the occupation bit, without transporting addresses, and all original spectral-failure and auxiliary records are retained. Occupation is read after a single preparation; repeated attempts use actual repreparation, rather than repeatedly applying XOR to the old occupation bit.

**Theorem S6.1 (Common-source outputs and the maximal matching-preserving space).** The actual process above yields three unnormalized address outputs: occupation, process remainder, and reference background,

$$\begin{gathered} B_W = P_B - w_1 P_r + w_2 P_2 + w_3 P_3, \qquad 0 \le B_W \le P_4, \\ \boxed{\Gamma_W = B_W/Z_0, \quad \Gamma_{\mathrm{vac}} = (P_4 - B_W)/Z_0, \quad \Gamma_b = P_b^{\mathrm{ref}}/Z_0.} \end{gathered} \tag{S50}$$

Let $F$ be an actual positive matching effect acting within $\operatorname{Ran} P_B$, with $0 \le F \le P_B$. Its matching weight and signal probability satisfy

$$\begin{aligned} N_{\mathrm{occ}} &= \operatorname{Tr} B_W = 18 - 3w_1 + w_2 + 2w_3, \\ h_F &= \operatorname{Tr}(F B_W) = \operatorname{Tr} F - w_1 \operatorname{Tr}(F P_r), \\ p_{\mathrm{sig}} &= \kappa_W^2 w_3 h_F / Z_0, \qquad \alpha_W(F) = w_3 h_F / N_{\mathrm{occ}}. \end{aligned} \tag{S51}$$

Here subsequent spectral auxiliaries are independent or refreshed, and the actual signal effect is $\kappa_W^2 w_3 F \le I$; $\alpha_W(F)$ is the occupation-conditioned response with instrument calibration $\kappa_W^2$ removed. If $w_1 > 0$, then

$$\boxed{h_F = \operatorname{Tr} F \quad \Longleftrightarrow \quad \operatorname{supp} F \subset \operatorname{Ran}(P_B - P_r).} \tag{S52}$$

Thus the maximal matching-preserving space has dimension $18 - 3 = 15$. For any trace-one matching effect within it,

$$\boxed{N_W^{\mathrm{eff}} = N_{\mathrm{occ}}, \quad \alpha_W = \frac{w_3}{N_{\mathrm{occ}}}, \quad W_{\mathrm{vac}} = N_0 - N_{\mathrm{occ}}, \quad \alpha_W(N_0 - W_{\mathrm{vac}}) = w_3.} \tag{S53}$$

**Proof.** Calculate all outputs of the original controlled spectral instrument for each orthogonal address $a$. Addresses are not transported, and final auxiliary states retain the actual spectral-success, failure, and occupation records, with norm still equal to 1. The roles have final occupation probabilities $1, 1 - w_1, w_2, w_3, 0$, respectively. Reducing the specified uniform address input gives the address-by-address occupation output $B_W/Z_0$; the unoccupied process output is its complement, and the action on the reference background is the identity, proving (S50). Occupation in each of the four orthogonal blocks lies in $[0, 1]$, establishing positivity and the upper bound; the traces of the three output classes sum to 1. No artificially constructed square-root failure has replaced the original spectral failure.

Since $F = P_B F P_B$, the added blocks are orthogonal to $F$, and taking traces directly gives (S51). Moreover, $\mathrm{Tr}(FP_r) = \|P_r F^{1/2}\|_{\mathrm{HS}}^2$, which vanishes if and only if the support of $F$ is orthogonal to the released space. For $w_1 > 0$, this is equivalent to preserving the bare matching weight, proving the equivalence. For a trace-one effect in the maximal space, $h_F = 1$; dividing the total weight of the same output by its matching weight gives (S53). ■

**Physical scope.** The adopted flip block $P_r = I_u \otimes |j_0\rangle\langle j_0| \otimes |-o_0\rangle\langle -o_0|$ preserves matching for every $v = \xi \otimes |o_0\rangle$. This is a sufficient 9-dimensional subspace; the more general 15-dimensional space also includes modes of the other orientation that avoid $j_0$. If $w_1 = 0$, the entire 18-dimensional initial space preserves matching. For a general trace-one mixed matching effect $F$, put $\eta = \mathrm{Tr}(FP_r)$. Then exactly $h_F = 1 - w_1\eta$ and $N_W^{\mathrm{eff}} = N_{\mathrm{occ}}/h_F$ ($h_F > 0$). It is the matching test that is mixed here; the address source is not replaced.

If an entire $d$-dimensional subspace is accepted, the actual effect is $F = P_S$, not $P_S/d$; when matching is preserved, the response is correspondingly $dw_3/N_{\mathrm{occ}}$. For a fixed complete attempt duration $\tau$, the event rate is $p_{\mathrm{sig}}/\tau$. Spectral preparation, all failures, record readout, and reset all belong to $\tau$; retaining the preceding address while applying a new matching measurement requires a separate proof of that measurement's return. The choices $k = 4$ and the common energy-scale reading retain the specified conditions of Q06 and Q07.

## S6.2 Reference from a common source and return weights

Let the input address space have dimension $d$, let $U_\beta$ be an isometry from this space to the working code, and let $z$ be the background zero-element state. For positive integer $h, m$, use the same normalized preparation

$$C\psi = |u_h\rangle \otimes \frac{1}{\sqrt{m}} \sum_{\beta=1}^{m} |\beta\rangle \otimes U_\beta \psi \otimes z, \qquad |u_h\rangle = \frac{1}{\sqrt{h}} \sum_{\alpha=1}^{h} |\alpha\rangle, \qquad C^\dagger C = I_d. \tag{S54}$$

Fix two points $\alpha_0, \beta_0$ relative to the internal group reference, and let

$$
\begin{aligned}
Q_s &= P_{\alpha_0}^{(h)} \otimes I_m \otimes I, \qquad & Q_r &= I_h \otimes P_{\beta_0}^{(m)} \otimes I, \\
C^\dagger Q_s C &= \frac{1}{h} I_d, & C^\dagger Q_r C &= \frac{1}{m} I_d.
\end{aligned}
\tag{S55}
$$

The latter two equations follow directly from orthogonality of distinct paths and $U_\beta^\dagger U_\beta = I_d$. The common preparation remains normalized; the fixed points are implemented together with the internal reference and are not separately called a group-invariant selection.

First specify the source of the background denominator. In the complete shell-average space of S5, write $T = \widetilde{B}/\zeta(1/2)$, $u = p^{-1/2}$, and $z = e_K$. The normalized spectral block and its self-adjoint real part are

$$
\begin{aligned}
&Te_j = (1-u)\sum_{i=0}^{j} e_i \quad (j < K), \quad Tz = z + \sqrt{1-u^2}\sum_{i=0}^{K-1} e_i, \\
&\operatorname{Re} T = \begin{pmatrix} A & t \\ t^\dagger & 1 \end{pmatrix}, \qquad A = \frac{1-u}{2}(I_K + \mathbf{1}\mathbf{1}^\dagger), \qquad t = \frac{\sqrt{1-u^2}}{2}\mathbf{1}.
\end{aligned}
\tag{S56}
$$

Here $\mathbf{1}$ is the all-ones column of dimension $K$. Put $S_z = 2P_z - I$; simultaneous equal-strength reflection protection gives

$$
\Delta_0 = \frac{\operatorname{Re} T + S_z(\operatorname{Re} T)S_z}{2} = \begin{pmatrix} A & 0 \\ 0 & 1 \end{pmatrix} > 0, \qquad \Delta_0 z = z. \tag{S57}
$$

Since $0 < u < 1$, all eigenvalues of $A$ are positive, so the inverse background exists, with its action on the other shells fully retained. Write $Z = |\zeta(1/2)|$ and $\vartheta = \arg\zeta(1/2)$. The phase-corrected unnormalized spectral real part is $(e^{-i\vartheta}\widetilde{B} + e^{i\vartheta}\widetilde{B}^\dagger)/2 = Z \operatorname{Re} T$ and becomes $Z\Delta_0$ under the same protection. This step uses a bidirectional self-adjoint action and zero-element protection; the unprotected spectral transfer itself is not treated as a positive generator. Energy-balancing auxiliaries are retained coherently with the shells, and these finite self-adjoint actions are implemented in the joint code of 2.3.

The two branches use the same actual entrance coefficient $\gamma \neq 0$, spectral capacity, and background control $c_B > 0$. Take $B_s = c_B Z \Delta_0$ and $B_r = c_B \Delta_0$ and the identity on the remaining working auxiliary units. Since $B_i$ commutes with $Q_i$, implementing actual simultaneous reflection with $R_i = I_{\text{ext}} \oplus (2Q_i - I_{\text{virt}})$ gives

$$
\begin{aligned}
H_i^{\text{raw}} &= \begin{pmatrix} 0 & \bar{\gamma} C^\dagger \\ \gamma C & B_i \end{pmatrix}, \\
H_i &= \frac{H_i^{\text{raw}} + R_i H_i^{\text{raw}} R_i}{2} = \begin{pmatrix} 0 & F_i^\dagger \\ F_i & B_i \end{pmatrix}, \qquad F_i = \gamma Q_i C, \quad i = s, r.
\end{aligned}
\tag{S58}
$$

Unselected ports remain in the complete space, with their entrance terms canceled between the two arms. No branch has been measured and then renormalized, and no remaining working auxiliary units have been deleted.

**Theorem S6.2 (Participation weights of returns from a common source).** For the complete protocol above, taking the actual single-point reference response as the unit of participation gives

$$\mathcal{S}_s = F_s^\dagger B_s^{-1} F_s = \frac{|\gamma|^2}{hc_B Z} I_d,$$
$$\mathcal{S}_r = F_r^\dagger B_r^{-1} F_r = c_{\mathrm{ref}} I_d, \qquad c_{\mathrm{ref}} = \frac{|\gamma|^2}{mc_B}, \tag{S59}$$
$$\Gamma = \frac{\mathcal{S}_s}{c_{\mathrm{ref}}} = \frac{m}{hZ} I_d, \qquad W = \operatorname{Tr}\Gamma = \frac{dm}{hZ}.$$

**Proof.** Zero-element protection gives $B_s F_s = c_B Z F_s$ and $B_r F_r = c_B F_r$. The inverse of each background acts on the corresponding coupling image, dividing by $c_B Z$ and $c_B$, respectively. Substitution of the two entrance Gram operators gives the two responses. Their ratio cancels the common entrance and background control, and taking the trace gives total participation. The proof neither requires the values $\gamma, c_B, c_{\mathrm{ref}}$ to be equal nor adds $m$ unit-strength entrances.

The reference can be read independently from the actual schedule. When a general block satisfies $F^\dagger F = JI_d$, $BF = bF$, and $b > 0$, its coupled bright block is $\left(\begin{smallmatrix} 0 & \sqrt{J} \\ \sqrt{J} & b \end{smallmatrix}\right)$. Put $s = \sqrt{b^2 + 4J}$. The transfer probability starting from the outer endpoint is

$$p_{\mathrm{virt}}(t) = \frac{4J}{s^2} \sin^2 \frac{st}{2}, \qquad \mathcal{A} = \frac{4J}{s^2}, \qquad t_{\mathrm{ret}} = \frac{2\pi}{s}, \qquad \frac{J}{b} = \frac{\mathcal{A}s}{4\sqrt{1-\mathcal{A}}}. \tag{S60}$$

Thus, with the two branches on the same clock and each using its own return,

$$c_{\mathrm{ref}} = \frac{\mathcal{A}_r s_r}{4\sqrt{1-\mathcal{A}_r}}, \qquad \frac{\mathcal{S}_s}{c_{\mathrm{ref}}} = \frac{t_r}{t_s} \frac{\mathcal{A}_s}{\mathcal{A}_r} \sqrt{\frac{1-\mathcal{A}_r}{1-\mathcal{A}_s}}\, I_d. \tag{S61}$$

This is an exact inversion at finite drive, requiring no weak-coupling approximation. The actual reference is neither defined by inverting the return frequency nor equal to an arbitrarily chosen old response unit. If $c_{\mathrm{unit}}$ is additionally fixed, the participation operator is $(c_{\mathrm{ref}}/c_{\mathrm{unit}}) m I_d/(hZ)$. The same reference and common control are fixed across layers; if the entrance covers only a proper subspace, the conclusion holds only on the common support, without filling missing addresses by zero-response ratios. Taking $h = 2, m = 3, d = 7^K$ gives $\Gamma_{\mathrm{met}}$ and $W_{\mathrm{met}}$ of 6.2 under these conditions.

## S6.3 Return amplitudes and common frequency

We now retain the actual return above and examine how it becomes an input to the persistent mass action. Let the reference generator be

$$H = \begin{pmatrix} 0 & F^\dagger \\ F & B \end{pmatrix}, \qquad F^\dagger F = JI_d, \quad BF = bF, \quad B > 0, \qquad W = \frac{dJ}{bc_{\mathrm{ref}}} > 1. \tag{S62}$$

Here $F$ has not yet been multiplied by the drive $\log W$. The exact exponential of the coupled bright block gives, at $s = \sqrt{b^2 + 4J}$ and $t_* = \pi/s$,

$$P_{\mathrm{virt}} e^{-iHt_*} P_{\mathrm{ext}} = -ie^{-ib\pi/(2s)} \frac{2}{s} F. \tag{S63}$$

Compensate this known common phase while retaining the outer-end residual. Prepare the normalized vector $\Phi_d = d^{-1/2}\sum_a |a\rangle|\bar{a}\rangle$ on the internal address and its conjugate reference, and apply the same $H$ to the internal-address factor. The virtual-end success column is

$$u_W = \frac{2}{s\sqrt{d}}\operatorname{vec} F, \qquad \operatorname{vec} F = \sum_a F|a\rangle \otimes |\bar{a}\rangle, \qquad \|u_W\|^2 = \frac{4J}{s^2}. \tag{S64}$$

The group representation and energy compensation of the actual address reference are included in the preparation; the norm of a normalized state has not been changed to $\sqrt{d}$. For the finitely many response endpoints used, choose a common capacity

$$C_d = \frac{s\sqrt{d}}{2\sqrt{bc_{\mathrm{ref}}}}, \qquad \Lambda_{\mathrm{col}} \geq \max_{\text{ports used}} C_d, \qquad C_d^2 = W + \frac{db}{4c_{\mathrm{ref}}} > W. \tag{S65}$$

At every virtual endpoint, implement a flag-retaining unitary rotation with acceptance amplitude $C_d/\Lambda_{\mathrm{col}} \leq 1$. The complete success column is then

$$\boxed{v_W = \frac{\operatorname{vec} F}{\Lambda_{\mathrm{col}}\sqrt{bc_{\mathrm{ref}}}}, \qquad \|v_W\|^2 = \frac{W}{\Lambda_{\mathrm{col}}^2} \leq 1.} \tag{S66}$$

This follows directly by multiplying $u_W$ by the actual acceptance amplitude; the outer-end residual, rejection flag, and other records are all retained. The coefficient is determined by the reference action $F, b$, address dimension, and fixed reference, not inferred backward from a target mass. Common capacity is part of the adopted actual protocol, whose control cost cannot be omitted.

Let $U$ be the complete unitary for this preparation, half-return, phase compensation, and flag rotation; let $a_0$ be the ready vector and $\Pi$ the success projection. Then $\Pi U a_0 = v_W$. Add orthogonal entrance and exit ports, and define

$$\begin{aligned} A_0 &= |1\rangle\langle 0| \otimes U + |0\rangle\langle 1| \otimes U^\dagger, \\ z_0 &= |0, a_0\rangle, \qquad P = P_{z_0} + |1\rangle\langle 1| \otimes \Pi, \qquad S = 2P - I, \\ \widehat{A} &= \frac{A_0 + SA_0S}{2} = PA_0P + (I-P)A_0(I-P). \end{aligned} \tag{S67}$$

**Theorem S6.3 (Common carrier of actual return columns).** If the complete preparation $U$ and its inverse can be coherently controlled, and $\widehat{A}$ is actually deployed using the simultaneous equal-strength arms of 2.3, the entire $P$ code is invariant. Applying the common coefficient $g_{\mathrm{col}} \log W$ on this code gives its bright carrier the rest-frame half-gap

$$\widehat{A}|_{P\mathcal{H}} = X_{v_W} = |1, v_W\rangle\langle z_0| + |z_0\rangle\langle 1, v_W|, \qquad \boxed{m_W = \frac{g_{\mathrm{col}}}{\Lambda_{\mathrm{col}}}\sqrt{W}\log W.} \tag{S68}$$

**Proof.** Reflection averaging exactly cancels coupling between $P$ and the exterior of the code, retaining the entire exterior block. The entrance satisfies $PA_0z_0 = |1, v_W\rangle$. Every $w \in \operatorname{Ran}\Pi$ satisfies $PA_0|1, w\rangle = z_0\langle v_W, w\rangle$, so the restriction on the entire successful exit space is the expression above; this verifies more than a one-column compression. The bright states $(z_0 \pm |1, v_W\rangle/\|v_W\|)/\sqrt{2}$ have eigenvalues

$\pm\|v_W\|$, and the other successful directions are dark states. Multiplying by the actual drive and substituting the success-column norm gives the half-gap.

Using the shared motion port of 2.4, connect the same bright carrier to the orientation and motion actions. At constant drive its dispersion is $E_\pm(k) = \pm\sqrt{m_W^2 + 4\omega^2 \sin^2(k\ell_x)}$, so the rest-frame half-gap and inertia at the origin use the same mass parameter. Auxiliary oscillations alone, or the trace of several degenerate frequencies, do not provide this connection to motion.

Simultaneous action is an actual premise of the theorem. Put $D = |0\rangle\langle 0| \otimes I + |1\rangle\langle 1| \otimes U$. Although $A_0 = D(\sigma_x \otimes I)D^\dagger$, the ability to execute $D, S$ as successive gates does not mean that both conjugate actions have been switched on persistently. With only piecewise control, the complete time-ordered propagation must be used; the phase rate of the simultaneous generator does not apply directly.

The common vacuum-end column also has an actual output from the same source. The complete occupation protocol of 6.1.2 gives reduced residual-branch output $(P_4 - B_W)/Z_0$. Retaining the address reference and all spectral records, its purification column $u_{\mathrm{vac}}$ has squared norm $W_{\mathrm{vac}}/Z_0$. Require the same capacity also to satisfy $\Lambda_{\mathrm{col}} \geq \sqrt{Z_0}$; a flag rotation with acceptance amplitude $\sqrt{Z_0}/\Lambda_{\mathrm{col}}$ gives

$$\|v_{\mathrm{vac}}\|^2 = \frac{W_{\mathrm{vac}}}{\Lambda_{\mathrm{col}}^2}, \qquad H_{\mathrm{ref}} = g_{\mathrm{col}}(\log W_{\mathrm{vac}})\widehat{A}_{\mathrm{vac}} \otimes G. \tag{S69}$$

This uses an actually controllable complete unitary and its inverse; a reduced mixed state alone does not satisfy the requirement. Following the background-gradient action of 2.5, suppose the bright-mode coefficients of the background gradient satisfy $\sum_r \kappa_r^2 = C_{\mathrm{bg}}$. On the preparation of the positive-eigenvalue vacuum bright state (the positive bright preparation), the background frequencies are $\omega_r = (g_{\mathrm{col}}/\Lambda_{\mathrm{col}})\sqrt{W_{\mathrm{vac}}}\log W_{\mathrm{vac}}\,\kappa_r$. Applying the original squared matching once, only at the vacuum endpoint, gives

$$\begin{aligned} \frac{\sum_r \omega_r^2}{E_0^2} &= W_{\mathrm{vac}} \log^2 W_{\mathrm{vac}}, \\ \frac{g_{\mathrm{col}}}{\Lambda_{\mathrm{col}}} &= \frac{E_0}{\sqrt{C_{\mathrm{bg}}}}, \qquad m_W = E_0\sqrt{\frac{W}{C_{\mathrm{bg}}}}\log W. \end{aligned} \tag{S70}$$

The finite capacity therefore cancels in the common calibration, without separate family-by-family compensation; bare-control resource costs remain in the actual $g_{\mathrm{col}}$. The complete-background limit uses the gradient-squared tail bound of 2.5.2, with $C_{\mathrm{bg}} = C^*$ used numerically. Together with Theorem S6.2, this yields the metric energy scale of 6.2 under the adopted layer correspondence and all-address conditions. After resource gating, S9 performs a new, single background calibration for that complete operating process; its phase mass is the effective accumulation divided by twice the total elapsed duration, and cannot be combined with the bare calibration of the ungated half-gap. S9.6 obtains absolute strength from the actual source–background–test response and states the calibration condition for equality with the spectral scale.

The total preparation duration of the metric return column is

$$T_U = T_\Phi + \frac{\pi}{s} + T_{\mathrm{phase}} + T_{\mathrm{att}} + T_{\mathrm{route}}, \qquad T_{\mathrm{att}} = \frac{\arccos(C_d/\Lambda_{\mathrm{col}})}{g_a}, \tag{S71}$$

The flag acceptance amplitude here uses a cosine rotation at angular rate $g_a$. Loading the positive bright state additionally requires $\pi/(4m_W)$ and phase operations. At the vacuum endpoint, the actual durations of address preparation, the occupation spectral protocol, flags, and routing are counted separately. A one-time loading is not deducted anew each cycle; actual repeated controls are timed on every use. End-of-cycle replenishment in S9 is an actual repeated preparation, whose bridging, failure padding, and thermal transport enter that protocol's complete cycle period each time. When both endpoints switch together along the same history and inactive stages act as the identity on the logical carrier or are compensated, the rest-frame branches and complete-cycle background phases use $\bar{g}_{\mathrm{eff}} = T^{-1}\int_0^T g_{\mathrm{col}}(t)/\Lambda_{\mathrm{col}}\,dt$; a single matching fixes this average effective strength. General motion has a noncommuting schedule, so average strength cannot simply be inserted into the constant-drive dispersion.

## S6.4 Comparison using a common source and configuration formation

### S6.4.1 Comparison space and reachable directions

Take an odd prime $p$, $G = \mathrm{Aff}(\mathbb{F}_p)$, $n = p(p-1)$, and address-configuration space $A = \mathbb{F}_p^G$. Translation $x \mapsto x+1$ and multiplication $x \mapsto ax$, with $a$ a primitive element, generate $G$. Denote these two transports by $s_1, s_2$ and the common-phase vector by $\mathbf{j}$. Actual comparison reads only differences, so the phase labels and dual writing directions are respectively

$$Q = A/\mathbb{F}_p\mathbf{j}, \qquad I = \{v \in A : \sum_g v_g = 0\}, \qquad N_C = |Q| = |I| = p^{n-1}. \tag{S72}$$

**Proposition S6.4.** All transport-edge differences $w_{sg} = e_{s^{-1}g} - e_g$ span $I$, and comparison removes exactly the common-phase direction.

Proof: the two transports generate the full group, so their undirected Cayley graph is connected. Choose a spanning tree; summing edge differences from its root to any vertex gives the coordinate difference between that vertex and the root. These $n-1$ differences form a basis of $I$. A vector $c$ pairs to zero with every edge difference if and only if it takes equal values across every edge; connectivity makes it constant. The standard nondegenerate pairing therefore induces the perfect pairing $Q \times I \to \mathbb{F}_p$. Taking $p = 7$ gives $n - 1 = 41$. This determines the dimension of the actual comparison configuration, without proving that the entire background must be truncated at 41.

### S6.4.2 Cycle-by-cycle formation under the actual echo channel

Let $\zeta = e^{2\pi i/p}$ and $Z_w|c\rangle = \zeta^{\langle c,w\rangle}|c\rangle$. The actual comparison echo applies this phase at probe address $g$. Use the complete half-return probe of S6.3, with $F_0^\dagger F_0 = JI_d, BF_0 = bF_0$, and take $d > n, J > 0$. The virtual-branch weight is $\eta = 4J/(b^2 + 4J)$; $n$ probe directions perform the comparison and the other directions do not participate in the comparison. A new probe is prepared independently each cycle; the old probe and its records are isolated and traced out at the complete output endpoint. Averaging over the two transports gives the actual configuration channel

$$\mathcal{T}(\rho) = \ell\rho + \frac{\eta}{2d} \sum_{s\in\{s_1,s_2\}} \sum_{g\in G} Z_{w_{sg}} \rho Z_{w_{sg}}^\dagger, \qquad \ell = 1 - \frac{\eta n}{d} > 0. \tag{S73}$$

Proof: both the half-return weight $1-\eta$ that does not enter the virtual branch and the weight of directions not participating in the comparison $\eta(d-n)/d$ within that branch give identity branches; every remaining address has weight $\eta/d$, split equally between the two transports. The coefficients sum to 1 and all failures have been counted.

Take the initial state as the uniform coherent vector on $A$, namely state $v = 0$ in the Fourier basis. The action $Z_w$ shifts the Fourier index by $w$, so evolution remains in sector $I$ and becomes the finite random walk with the edge-difference steps above. Each coefficient in the spanning-tree basis can be reached with at most $p-1$ forward steps; reversing an edge of the spanning tree realizes a negative step in $p-1$ steps. Full support is therefore reached by step $(p-1)(n-1)$ at the latest, with idle branches padding the cycle count. Full support here does not mean uniformity after finitely many cycles.

This random walk is irreducible on the index set and has a positive holding probability, so its unique stationary distribution is uniform. Its Fourier eigenvalues and the purity for the specified initial state are

$$\lambda_{[c]} = \ell + \frac{\eta}{2d} \sum_{s,g} \zeta^{\langle c, w_{sg}\rangle}, \qquad \operatorname{Tr} \rho_L^2 = \frac{1}{N_C} \sum_{[c]\in Q} |\lambda_{[c]}|^{2L} \longrightarrow \frac{1}{N_C}. \tag{S74}$$

Proof: characters diagonalize translation convolution on the finite abelian group $I$. If an eigenvalue has modulus 1, positive idle weight forces every step phase to be 1; Proposition S6.4 makes this character trivial, and all other eigenvalues have modulus strictly below 1. Parseval's identity gives the purity formula. It applies to the initial state and actual channel used here, not as a finite-cycle matching formula for arbitrary nonuniform or correlated preparations. Other total-sum sectors are not attracted into $I$ by this channel. If all probes and comparison actions are completely reversed, the initial state is restored; the reduced formation above cannot simultaneously be claimed to remain.

Likewise, for a fixed observable $X$ of a random-unitary channel with positive weights, strict convexity of the Hilbert–Schmidt norm gives $[X, Z_{w_{sg}}] = 0$: equality of norms requires all conjugated $X$ to coincide, and the idle branch fixes them to $X$. This criterion identifies which structure remains preserved by actual observation after the partial trace.

### S6.4.3 Matching probability of the complete comparison

Read $n-1$ independent differences along a spanning tree, encoding each with a Fourier phase query on one $p$-dimensional working auxiliary unit. The perfect pairing ensures that distinct $q \in Q$ yield orthogonal encodings $|F_q\rangle$. The receiver applies the inverse query using its own $q'$ and checks that all working auxiliary units return to their ready states, accepting if and only if $q = q'$. Thus the actual acceptance effect and its independent uniform compression are

$$P_{\mathrm{eq}} = \sum_{q\in Q} |q,q\rangle\langle q,q|, \qquad \operatorname{Tr}_S \left[ \left( \frac{I_S}{N_C} \otimes I_R \right) P_{\mathrm{eq}} \right] = \frac{I_R}{N_C}. \tag{S75}$$

The formula is exact whenever one endpoint is exactly uniform and the endpoints are independent; if both marginals are uniform but perfectly correlated, success probability can be 1. Only the diagonal entries in the protocol's $Q$ basis need be uniform here. The phase channel in S6.4.2 preserves these entries; the specified uniform coherent preparation satisfies this from the first cycle. Nonuniform occupation in the Fourier–$I$ basis does not obstruct exact matching here. A corresponding deviation is retained only if a nonuniform preparation in the protocol basis is substituted. Formation supplies the configuration set, complete comparison gives the event effect, and the actual collective column of S6.5 gives the frequency; each step uses its own explicit preparation and complete operation.

## S6.5 Configuration capacity and actual collective columns

Under independent uniform preparation, the complete comparison of S6.4 gives compression $I/N_C$ of the matching projection $P_{\mathrm{eq}}$. We retain the complete output generating this response below, rather than taking the norm of a normalized mixed state as total participation. Let the common-source response per configuration be $\kappa > 0$, with $W_C = N_C\kappa > 1$. Retain the original logarithmic action rule $\Omega_C = \log W_C$. Let the actual half-return column $F_\theta$ satisfy

$$F_\theta^\dagger F_\theta = \frac{q_s}{Z_0}\big(N_v I - \delta n P_{\mathrm{eq}}\big), \qquad q_s = q_r\kappa, \quad b_0 = N_v/Z_0, \quad 0 \le \delta n \le N_v. \tag{S76}$$

Here $q_r > 0, N_v > 0, b_0 > 0$, and $F_\theta$ is a contraction column already delivered by the specified spectral instrument, whose inner capacity ensures $F_\theta^\dagger F_\theta \le I$; the outer $\Lambda_C$ controls only the subsequent flag and cannot replace this inner normalization. All failure outputs are retained. The controllable purification preparation $N_C^{-1/2}\sum_q |q\rangle|\bar{q}\rangle$ and the unchanged receiving input define the isometric embedding $J_R$. For $T_\theta = (F_\theta \otimes I)J_R$, matching compression gives $T_\theta^\dagger T_\theta = q_s b_0(1-\delta n/(N_C N_v))I$. Using common capacity $\Lambda_C^2 \ge \max\{N_C/(q_r b_0), Z_v\}$, execute the actual flag acceptance amplitude $a_C = \sqrt{N_C}/(\Lambda_C\sqrt{q_r b_0}) \le 1$. Then

$$v_\theta = a_C T_\theta, \qquad v_\theta^\dagger v_\theta = \frac{W_{\mathrm{eff}}}{\Lambda_C^2} I, \qquad W_{\mathrm{eff}} = W_C - \frac{\kappa\delta n}{N_v}. \tag{S77}$$

**Theorem S6.5.** Constructing the common entrance, successful exit, and simultaneous reflection protection from this complete preparation and its inverse as in S6.3 gives bright-mode frequency $g\sqrt{W_{\mathrm{eff}}}\log W_C/\Lambda_C$.

Proof: for $W_{\mathrm{eff}} > 0$, the column $Q_\theta = \Lambda_C v_\theta/\sqrt{W_{\mathrm{eff}}}$ is isometric. The matrix on the complete protected code has off-diagonal block $v_\theta, v_\theta^\dagger$; the two isometric carriers $(J_R, \pm Q_\theta)/\sqrt{2}$ have eigenvalues $\pm\sqrt{W_{\mathrm{eff}}}/\Lambda_C$. Zero-weight endpoints follow by continuity. The vacuum residual column is accepted by a flag of the same capacity and has squared norm $W_{\mathrm{vac}}/\Lambda_C^2$. The same background squared calibration fixes the complete-cycle average $\bar{g}/\Lambda_C = E_0/\sqrt{C_{\mathrm{bg}}}$, so at the unperturbed endpoint

$$E_C = E_0\sqrt{\frac{W_C}{C_{\mathrm{bg}}}}\log W_C, \qquad \frac{E_C(\theta)}{E_C(0)} = \frac{\bar{g}(\theta)}{\bar{g}(0)}\sqrt{1 - \frac{\delta n}{N_C N_v}}. \tag{S78}$$

During local perturbation the selected $\log W_C$ rule remains unchanged. This formula shows both how preparation capacity cancels and how the complete holding time remains; counting configuration degeneracy alone does not enforce the

logarithmic drive, common squared calibration, or physical identification of the per-configuration weight. If the complete unitary and address reference have become inaccessible permanent records, the premise of this persistent exchange is not satisfied.

**Occupation-controlled bright-channel coupling.** Retain the complete comparison of S6.4 and the actual coherent entrance of S6.3. Let the fresh configuration preparation be independent of the incoming resource packet, and let its occupation projection $O$ commute with label projection $P_q^L$; the incoming resource packet may still be correlated with the source and old records. For $N_C > 1$, take

$$\begin{aligned} v_C &= \frac{\sqrt{\kappa}}{\Lambda_C} \sum_{q=1}^{N_C} |q, \bar{q}, f_q\rangle, \qquad \beta_C^2 = \frac{N_C \kappa}{\Lambda_C^2} \le 1, \\ P_{\mathrm{eq}} &= \sum_q P_q^C \otimes P_q^L, \qquad F = (I - O P_{\mathrm{eq}})(v_C \otimes I). \end{aligned} \tag{S79}$$

The complete configuration columns are mutually orthogonal, with $\sum_q P_q^L = I$, hence $(v_C \otimes I)^\dagger P_{\mathrm{eq}} (v_C \otimes I) = \beta_C^2 I / N_C$. Since $O P_{\mathrm{eq}}$ is also a projection,

$$\boxed{\begin{aligned} F^\dagger F &= \beta_C^2 (I - O/N_C), \\ B_C = \sqrt{F^\dagger F} &= \beta_C (I - \delta_C O), \qquad \delta_C = 1 - \sqrt{1 - 1/N_C}. \end{aligned}} \tag{S80}$$

This is an operator identity on unknown occupation. Replacing $O$ by mean occupation before taking the square root generally changes the output; the matching share $1/N_C$ is already included in the Gram operator and must not multiply the persistent frequency again. Here $\delta_C$ and the flag amplitude $a_C$ above are defined separately.

Bright-channel coupling must enter the maintenance frequency through actual loading. Let $\widehat{A}_C$ be the two-end exchange implemented in S6.3 by complete preparation, inverse preparation, and reflection, and let $J$ be its ready entrance. On occupation branch $o = 0, 1$, the bright-channel coupling amplitude is $\beta_o = \beta_C \sqrt{1 - o/N_C} > 0$. Take actual strength $g_F > 0$, use the same control to hold branch $o$ for $T_o = \pi / (4 g_F \beta_o)$, and then add exit phase $i$; this sends the entrance into the positive bright state. The shorter branch pauses until the common final time $T_1$; control remains coherent, producing no irreversible occupation or stopping-time record. The resulting actual loading $\mathcal{L}$ satisfies

$$\mathcal{L} J = \mathcal{J}_+ = \frac{1}{\sqrt{2}} \begin{pmatrix} J \\ F B_C^{-1} \end{pmatrix}, \qquad \widehat{A}_C \mathcal{J}_+ = \mathcal{J}_+ B_C. \tag{S81}$$

The proof is a quarter rotation in the two-dimensional space of the entrance and normalized exit on each occupation branch; both branches reach the same positive bright form, and linearity preserves unknown occupation and its entanglement with subsystems not participating in this action. Actual preparation, pausing or phase compensation, and inverse loading all enter the total duration.

Let $K_Y$ be the original complete self-adjoint common action of matter and the background gradient, acting on logical factors independent of the incoming resource packet. Common maintenance on the two-port code above gives off-diagonal block $(\log W_C) K_Y B_C$ and its adjoint, hence yields $(\log W_C) B_C K_Y$ on the positive bright code. Swap the incoming resource packet in under position control; denote the swap

unitary by $C$, the entire loading by $\mathcal{T} = \mathcal{L}C$, and the original displacement term by $D_{\mathrm{mov}}$. When motion edges and the maintenance term are transported together by $\mathcal{T}$, the complete ready code $J_0$ satisfies

$$\begin{aligned} H_C &= \zeta_C K_Y [I - \delta_C Q(X)] + D_{\mathrm{mov}}, \\ \zeta_C = \beta_C \log W_C, &\qquad Q(X) = \sum_x P_x O_x, \\ H_{\mathrm{work}} \mathcal{T} J_0 = \mathcal{T} J_0 H_C, &\qquad \mathcal{T}^\dagger e^{-ibH_{\mathrm{work}}} \mathcal{T} J_0 = J_0 e^{-ibH_C}. \end{aligned} \tag{S82}$$

Here $P_x$ are the actual position projections and $O_x$ the occupation projections of the corresponding incoming resource packets; $K_Y$ commutes with $Q(X)$. Maintenance intertwining follows from the block matrix above, and displacement intertwining from transport by the entire loading; adding them and expanding in powers gives the exponential identity. It proves auxiliary restoration while retaining the common intermediate action; correlations among the source, incoming resource packet, and target remain.

This collective reception replaces the original incoming-resource availability projection $I - O$, while the matter's own matching and occupation remain in $K_Y$. Multiplying the two occupation-control factors instead gives $(I-O)(I-\delta_C O) = I-O$, a different action. Collective bright-channel coupling changes the source phase, background calibration, and receiving response, while additional processing also changes the complete cycle duration; this proposition therefore cannot simply be an extra multiplier in the old strength formula. It proves how collective participation enters the same maintenance action; absolute strength must still be calculated from the complete operating protocol.

# S7 Common carrier of matter and background

## S7.1 Common transport of the original exchange with occupation

On the finite participating space, let the original formation state be $\Phi$ and the background-direction projection be $P_0$, with $q_0 = \langle\Phi|P_0|\Phi\rangle$. The matter isometric column is $J_m$, and the complete matching projection is $Q_m$, where $J_m = \Phi \otimes J_R$; its compression $E = J_m^\dagger Q_m J_m$ satisfies $0 \le E \le I$. The original equal-strength reflection exchange yields $H_s = g_M E/4$ on this code. The actual background zero-element action uses the same $\Phi$, finite capacity $\Lambda$, and full gradient $G = G^\dagger$, with compression coefficient $c_B = g_B q_0/\Lambda$. Take $q_0, g_M, g_B > 0$ below. These quantities come from matrix elements of the original ports; $G$ must be the already defined self-adjoint gradient, not an arbitrary replacement operator chosen for a target strength.

Specifically, the original tail-removed background module is $\bar{X}_B = P_0^S Z_\Lambda U_G$, with capacity preparation satisfying $\langle r_\Lambda|Z_\Lambda|r_\Lambda\rangle = 1/\Lambda$ and gradient isometric column satisfying $j_G^\dagger U_G j_G = G$. On $j_B = \Phi \otimes r_\Lambda \otimes j_G$, take preparation reflection $\bar{S}_B = 2j_B j_B^\dagger - I$; then $\bar{H}_B = g_B(\bar{X}_B + \bar{S}_B \bar{X}_B \bar{S}_B)/2$ yields exactly $\bar{H}_B j_B = j_B c_B G$. The background action on the same common port is $\sigma_x \otimes \bar{H}_B$; no independent binary tail stage is added for it. Actual matter matching takes $Q_m = P_0^S \otimes Q_R$, hence $E = q_0 J_R^\dagger Q_R J_R$. The two actions intertwine separately on the same joint preparation, without requiring commutativity outside the code.

Denote the original complete counting auxiliary state by $\beta$ and its success projection by $\Pi_B$, with $s = \langle\beta|\Pi_B|\beta\rangle$. The new occupation bit starts in $|0_d\rangle$, and the actual writing is the controlled flip

$$\begin{aligned} U_{\mathrm{occ}} &= (I - Q_m \otimes \Pi_B) \otimes I_d + (Q_m \otimes \Pi_B) \otimes X_d, \\ O &= I_M \otimes \Pi_B \otimes P_{1d}, \\ A_{\mathrm{av}} &= I_M \otimes \Pi_B \otimes P_{0d}. \end{aligned} \tag{S83}$$

Here the background reference capacity $\Lambda$, matter spectral capacity, and counting capacity $\Lambda_B$ belong to distinct actual factors; the same $\beta$ is prepared only once, and both counting success and failure are retained. Branchwise calculation of this controlled flip gives $W^\dagger O W = sE$ and $W^\dagger A_{\mathrm{av}} W = s(I - E)$.

Jointly prepare the source, gradient, and occupation auxiliaries as $J_0$, using only one source formation state. The actual occupation-writing unitary $U_{\mathrm{occ}}$ is controlled by $Q_m$ and the background projection; write $W = U_{\mathrm{occ}} J_0$. The reflections for the original matter and background actions are transported by the same writing, giving $H'_M = U_{\mathrm{occ}} H_M U_{\mathrm{occ}}^\dagger$ and $H'_B = U_{\mathrm{occ}} H_B U_{\mathrm{occ}}^\dagger$. The original intertwining identities then directly give

$$H'_M W = W(H_s \otimes I), \qquad H'_B W = W(I \otimes c_B G), \qquad W^\dagger A_{\mathrm{av}} W = s(I - E) \otimes I. \tag{S84}$$

Here $A_{\mathrm{av}}$ is the available-background projection left by this writing, and $0 < s \le 1$ is the original acceptance share. The last identity follows branchwise for matching and nonmatching: the matching part writes occupation, and its complement leaves available resources. Put $F = A_{\mathrm{av}} W$ and $B = \sqrt{s(I - E)}$, and on the support of $B$

define the polar-decomposition isometric column $V = FB^{-1}$. The zero-availability subspace does not participate in this exchange.

**Complete effects on the formation input.** Let the actual formation-acceptance column $K_5 : \mathcal{H}_{\mathrm{pre}} \to \mathcal{H}_{\mathrm{post}}$ be a contraction. For the construction in which formation output feeds subsequent matching, write

$$\begin{aligned} D_F &= K_5^\dagger K_5, \qquad & E_{\mathrm{RO}} &= K_5 K_5^\dagger, \\ E &= q_0 E_{\mathrm{RO}}, \qquad & K_5^\dagger E_{\mathrm{RO}} K_5 &= D_F^2. \end{aligned} \tag{S85}$$

The operator $D_F$ acts on the preformation input, and $E_{\mathrm{RO}} = J_R^\dagger Q_R J_R$ on the post-formation logical space; the equality here refers to the actual formation–matching representation used. They are defined separately from the complete effect $E$, which includes background share $q_0$. Composing the actual occupation, availability, and old-failure output columns with $K_5$, while retaining the unformed outcome, gives four effects on the preformation input:

$$\begin{gathered} T_{\mathrm{av}} = s(D_F - q_0 D_F^2), \qquad T_{\mathrm{occ}} = s q_0 D_F^2, \\ T_{\mathrm{oldfail}} = (1-s) D_F, \qquad T_{\mathrm{unformed}} = I - D_F, \qquad T_{\mathrm{av}} + T_{\mathrm{occ}} + T_{\mathrm{oldfail}} + T_{\mathrm{unformed}} = I. \end{gathered} \tag{S86}$$

Proof: sandwiching $s(I-E), sE, (1-s)I$ between $K_5^\dagger, K_5$ gives the first three equations; actual formation failure gives $I - D_F$. Since $0 \le D_F \le I$ and $0 \le q_0, s \le 1$, all four terms are positive and sum to the identity. The original output columns and records are retained; an arbitrary square root of the same effect does not replace the actual post-state. After positive bright loading, the equal-weight two-port structure below contributes a further one-half to occupation, giving preformation-input effect $s q_0 D_F^2/2$. This conclusion applies only to support that has actually been loaded; actual loading of different spectral branches must be checked separately, and independent preparation costs of source and background remain included.

**Theorem S7.1 (Common restricted exchange).** Put $L = H_s \otimes I + I \otimes c_B G$ and define on the two outer ports

$$\begin{aligned} Y &= \sigma_x \otimes (H'_M + H'_B), \\ P &= \mathrm{diag}(WW^\dagger, A_{\mathrm{av}}), \\ R_P &= 2P - I, \\ H_{\mathrm{com}} &= \frac{\gamma}{4}(Y + R_P Y R_P). \end{aligned} \tag{S87}$$

If this simultaneous action is actually switched on and $\gamma = 4$ inherits the original matrix element of each elementary arm, then the positive bright carrier $\mathcal{J}_+ = (W, V)/\sqrt{2}$ satisfies

$$H_{\mathrm{com}} \mathcal{J}_+ = \mathcal{J}_+ \, 2BL, \qquad \mathcal{J}_+^\dagger O \mathcal{J}_+ = \frac{sE}{2}. \tag{S88}$$

Proof: $B$ is a function of $E$ and therefore commutes with $L$. The intertwining identities above give $A_{\mathrm{av}}(H'_M + H'_B)W = FL = VBL$, while self-adjointness gives reverse block $LB$. Protection averaging cancels the exterior-code blocks, so the restriction in $\mathrm{diag}(W, V)$ is $\gamma \left(\begin{smallmatrix} 0 & LB \\ BL & 0 \end{smallmatrix}\right)/2$, immediately proving positive-bright intertwining. The occupation projection $O$ compresses to $sE$ on branch $W$ and vanishes

on the accepted branch $V$; orthogonality of the two outer ports makes bright-state occupation half of this.

These outer ports can be obtained by reconnecting the complete entrance of S6.3. Put $P_{\mathrm{in}} = J_0 J_0^\dagger$ and $D = \mathrm{diag}(U_{\mathrm{occ}}, I)$; then $D\left(\begin{smallmatrix} 0 & U_{\mathrm{occ}}^\dagger \\ U_{\mathrm{occ}} & 0 \end{smallmatrix}\right) D^\dagger = \sigma_x \otimes I$, and the entrance–success reflection simultaneously becomes $R_P$. This is unitary reconnection on the full space, retaining failures and the complement; the polar part $V$ cannot be treated as a cost-free preparation gate. If each normalized elementary arm inherits original matrix element $g_M/2$, the coefficient $\gamma g_M/8$ of the expanded common matter arm equals it, hence $\gamma = 4$. Serial operation synthesis, additional attenuation, or different background matrix elements require their own actual actions.

Arbitrary amplitude adjustment cannot replace the roles of the two arms: for the same $Y, P$, one has $(I-P)(aY+bR_PYR_P)P = (a-b)(I-P)YP$. If a nonzero leakage block exists, leakage cancellation requires $a = b$. This is the algebraic constraint for the original simultaneous equal-strength protection; the common overall strength is still determined by the actual elementary action and calibration.

## S7.2 Resource occupation, mass phase, and background gradient

Take eigenbranch $0 \le e < 1$ of $E$, with $r_e = \sqrt{s(1-e)}$. Let the actual common holding integral per cycle be $t_H$ and the complete cycle period be $\tau$, and write $f = t_H/\tau$. Denote the effective capacity of the occupation preparation by $N = s\Lambda_B^2$. Converting bright-state occupation probability to resource occupation using capacity $\Lambda_B^2$, the same bright carrier gives

$$M_e = \frac{f r_e g_M e}{2}, \qquad V_e = 2 f r_e c_B, \qquad n_e = \frac{Ne}{2}, \qquad \boxed{n_e = \frac{2Nc_B}{g_M}\frac{M_e}{V_e}}. \tag{S89}$$

Proof: the generator in Theorem S7.1 is $2r_e(g_M e/4 + c_B G)$; multiplying the coefficients of its matter constant term and entire gradient term by the same $f$ gives the first two equations. Resource occupation is the bright-state occupation probability $se/2$ times $\Lambda_B^2$. Their ratio cancels the same holding ratio and availability amplitude, giving the last equation. Here $V_e$ is the coefficient of the entire gradient and cannot also be treated as an event rate; effective capacity $N$ is not automatically equal to configuration count $N_C$ in S6.4.

Thus $1/V_e$ has an explicit origin in the same exchange. This does not independently fix full-background capacity, spatial geometry, or every schedule; cancellation holds only when comparing the same carrier with the same complete-cycle timing. The ratio is defined for nonzero holding and $c_B > 0$; the zero-availability endpoint $e = 1$ does not use this division. Finite occupation also gives $r_e^2/r_0^2 = 1 - e = 1 - 2n_e/N$. This explains how the same matching enters both matter phase and background availability, without assigning them separate response functions.

## S7.3 Complete return and residuals

If reversible outward and return transport applies the same writing and its inverse before and after holding, the action is $U_{\mathrm{occ}}^\dagger e^{-itH'} U_{\mathrm{occ}} = e^{-itH}$. This retrieves aux-

iliary writing while retaining the phase accumulated in between; the entire initial state is restored only if the holding action is also reversed. With isospectral temporary storage and a complete survival instrument, surviving histories take the form $K_\eta \mathcal{J}_+ = c_\eta \mathcal{J}_+ e^{-i2t_H BL}$, while other exit histories remain in orthogonal outputs. This requires a scalar survival effect on the selected code and actual execution of storage and retrieval according to the original complete instrument. One cannot discard failures, renormalize, and then treat the conditional phase as an unconditional full output.

Holding itself can leave a nonzero local background deformation. Let $f_0 = (I - \Pi_B)\beta$ and $b_0 = \Pi_B \beta$, and define $\beta_0 = (f_0 + b_0)|0_d\rangle$, $\beta_1 = f_0|0_d\rangle + b_0|1_d\rangle$, and $\widehat{b}_0 = b_0|0_d\rangle/\sqrt{s}$. Fix source branch $e$ and trace out source, common bright port, and gradient logic, giving

$$\sigma_e = \frac{1}{2}\left[(1-e)P_{\beta_0} + eP_{\beta_1} + P_{\widehat{b}_0}\right], \qquad \sigma_e - \sigma_0 = \frac{e}{2}(P_{\beta_1} - P_{\beta_0}), \quad \mathrm{Tr}(O\sigma_e) = \frac{se}{2}. \tag{S90}$$

Proof: branch $W$ decomposes over matching and nonmatching matter projections, with traced weights $e, 1-e$; branch $V$ carries the normalized resource auxiliary vector $\widehat{b}_0$. The two outer ports each contribute one-half, giving the formula. Theorem S7.1 preserves the bright carrier at fixed $e$, and the remaining evolution acts only on the traced-out gradient, so the reduced state of this resource subsystem is unchanged during holding. A survival cycle may therefore add zero net resource occupation while the local deformation maintained by the stable state remains nonzero; whether it becomes a persistent distant distribution depends on actual transport and return.

Genuinely inverse transport without readout retrieves distant occupied resources, while the intermediate residence integral remains nonzero; a sustained multi-cycle distribution must arise from an actual in-transit queue, residuals after return, or the explicit replenishment process of S9. End-of-cycle refreshing in S9 and this reversible return are separately complete deployments; stationary conditions of the former cannot silently be imported into the latter. S10 calculates one complete return with readout and explicitly retains feedback on the source, so the source mass used is the prereadout accumulated phase divided by the complete cycle duration; spectra under repeated long-term readout still require a separate calculation of the complete repeated channel.

## S8 Matching normalization of the output state

**Fixed auxiliary preparation and actual exits.** Use the existing seven-dimensional output-label space: $P$ has rank 6, $R_d \le P$ has rank 2, and $P_b$ has rank 1 and is orthogonal to $P$; the normalized matching direction $v$ satisfies $Pv = v$ and $R_d v = 0$. Take $w = w_2 \in (0, 1]$ and $V = w_3 > 2w$, so $x = (V - 2w)/6 > 0$; following 7.3.2 of the main text, take $y = (w_2/6)(2w_3/18 + w_2/6) > 0$ and $A = x + y$. Let

$$\begin{aligned} &\Omega = (P + VP_b)/(6 + V), \qquad B = VP_b - wR_d, \quad \operatorname{Tr} B = 6x, \quad Bv = 0, \\ &K_+ = \sqrt{P - wR_d} + P_b, \quad L_+ = \sqrt{wR_d}, \qquad K_- = P, \quad L_- = P_b. \end{aligned} \tag{S91}$$

Each pair $K_\pm, L_\pm$ is a complete instrument, and the retained outputs of the two routes are $(P + B)/(6 + V)$ and $P/(6 + V)$, respectively.

Denote the controller's finite-dimensional reachable space by $\mathcal{K}$. Each actual complete single-cycle instrument contains a common continuation subchannel $\mathcal{C}$, effects $E_+, E_-$ for two terminal records, and the remaining exit effect $E_\times$. With $T = \mathcal{C}^*$,

$$E_+, E_-, E_\times \ge 0, \qquad T(I) + E_+ + E_- + E_\times = I. \tag{S92}$$

Here $E_\pm$ count the probabilities of controller selection of the two routes, before auxiliary-system filtering. The input is $\rho \otimes \Omega$, with controller and output auxiliary system independently prepared. Each cycle uses the same instrument and fresh ready auxiliaries, while old history records are retained and isolated; the controller includes internal degrees of freedom that will be revisited. Continuation does not act on the output auxiliary system, and $\mathcal{K}$ is invariant under continuation; the first terminal $+$ or $-$ triggers the corresponding auxiliary-system filter once, with rejected auxiliary-filter outcomes exiting directly. All terminal, continuation, exit, and cutoff histories retain orthogonal records and are not coherently recombined at the end. If cross-history coherence exists, all cross terms must be retained and this theorem does not apply.

**Theorem S8.1 (Finite-schedule normalization and a single-cycle criterion).** For at most $L \ge 1$ cycles, define

$$G_{\pm,L} = \sum_{n=0}^{L-1} T^n(E_\pm), \qquad a_L = \operatorname{Tr}(\rho G_{+,L}), \quad b_L = \operatorname{Tr}(\rho G_{-,L}), \quad t_L = a_L + b_L. \tag{S93}$$

The actual unnormalized output state and matching denominator are

$$\boxed{\Gamma_L = \frac{t_L P + a_L B}{6 + V}, \qquad N_L = \frac{\operatorname{Tr}\Gamma_L}{\langle v|\Gamma_L|v\rangle} = 6 + 6x\frac{a_L}{t_L} \quad (t_L > 0).} \tag{S94}$$

For the given identical single-cycle instrument, the following three statements are equivalent:

1. The two single-cycle terminal effects satisfy $yE_+ = xE_-$.
2. For some fixed finite $L \ge 1$, every input state with $t_L > 0$ gives $N_L = 6 + 6x^2/A$.
3. For every finite $L \ge 1$, every input state with $t_L > 0$ gives the same denominator above.

Thus, whenever the actual single-cycle effects balance,

$$\boxed{N_L = 6 + \frac{6x^2}{x+y}, \qquad \alpha_L = \frac{w_2}{N_L} \quad \text{for every finite } L \text{ with nonempty delivery.}} \tag{S95}$$

The continuation subchannel may change the internal state, need not commute with the original $H_J$, and need not be scalar survival or restore the same controller state each cycle. This scope is bounded by (S92) and the fixed-auxiliary-preparation conditions and does not require changing the original static generator.

**Proof: output from complete histories.** The probabilities of the first terminal outcome in cycle $n+1$ are $\mathrm{Tr}[E_\pm \mathcal{C}^n(\rho)]$. Summing orthogonal histories and pulling them back to the common entrance gives (S93). The output auxiliary system remains in the prepared state $\Omega$ and independent of the controller until termination, so delivery is a weighted sum of two positive outputs, proving (S94). The matching weight is $t_L/(6+V)$, and the total weight is $(6t_L + 6xa_L)/(6+V)$.

**Proof: sufficiency.** Put $\Delta = yE_+ - xE_-$. If $\Delta = 0$, linearity of $T$ directly gives $yG_{+,L} - xG_{-,L} = \sum_{n=0}^{L-1} T^n(\Delta) = 0$. Thus $ya_L = xb_L$ and $a_L/t_L = x/A$; substituting into (S94) gives (S95). No reduction of $T$ to a survival probability is needed.

**Proof: necessity.** If condition 2 holds, (S94) implies, for every input with nonzero delivery, $\mathrm{Tr}\,\rho \sum_{n=0}^{L-1} T^n(\Delta) = 0$. For zero-delivery inputs, both $a_L$ and $b_L$ vanish, so the equality also holds. Testing every pure state yields the operator identity $\sum_{n=0}^{L-1} T^n(\Delta) = 0$. Multiplication by $I - T$ gives $\Delta = T^L(\Delta)$.

The key is the positivity constraint from (S92). Write $R = I - T(I) = E_+ + E_- + E_\times$. Then

$$-xR \le \Delta \le yR, \qquad T^n(R) = T^n(I) - T^{n+1}(I) \longrightarrow 0. \tag{S96}$$

Because $T$ is positive and subunital, $T^n(I)$ decreases monotonically with lower bound zero; in finite dimension it converges in norm, so successive differences tend to zero. Applying the positive map $T^n$ to the order bounds gives $T^n(\Delta) \to 0$. On the other hand, $\Delta = T^{kL}(\Delta)$ for every $k$, hence $\Delta = 0$. This also covers dark subspaces that continue forever; they have no terminal effect and cannot conceal exit imbalance. Condition 3 plainly implies condition 2, completing the equivalence.

**All probabilities and durations.** Let $c_L = \mathrm{Tr}[\rho T^L(I)]$ be the cutoff tail and $d_L = \sum_{n=0}^{L-1} \mathrm{Tr}[\rho T^n(E_\times)]$ the other exits. Cycle-by-cycle completeness gives

$$\begin{gathered} t_L + d_L + c_L = 1, \qquad p_{\text{pool drop},L} = \frac{2wa_L + Vb_L}{6+V}, \\ \mathrm{Tr}\,\Gamma_L + p_{\text{pool drop},L} + d_L + c_L = 1, \\ \mathbb{E}\mathcal{T}_L = \mathcal{T}_{\text{prep}} + \tau \sum_{n=0}^{L-1} \mathrm{Tr}[\rho T^n(I)]. \end{gathered} \tag{S97}$$

The last equation uses a fixed complete-cycle duration $\tau$, including that cycle's preparation, continuation or exit processing, actual auxiliary filtering, and readout, with untriggered actions padded by bypasses. It does not apply directly to arbitrary branch durations. Nor does the finite-cycle theorem automatically exclude a nonterminating infinite tail or infinite mean duration. If the actual signal effect is $\kappa^2 w_2 |v\rangle\langle v|$, the signal probability of one complete attempt is $\kappa^2 w_2 t_L/(6+V)$; only conditioning on delivery gives $\kappa^2 \alpha_L$.

## S8.1 Actual terminal action and stability

Use neutral, equal-energy three-state terminal records, with the complete self-adjoint action

$$C_R = \sqrt{x}|+\rangle\langle 0| + \sqrt{y}|-\rangle\langle 0|, \qquad H_R = ig_R(C_R - C_R^\dagger) \tag{S98}$$

coupling $|0\rangle$ to the bright direction $(\sqrt{x}|+\rangle + \sqrt{y}|-\rangle)/\sqrt{A}$; the generator vanishes on the orthogonal complement of the preparation code subspace. Starting from $|0\rangle$, the actual exponential gives $M_\pm = \sqrt{\{x,y\}/A}\sin(g_R t_R\sqrt{A})I_\mathcal{K}$. Thus the single-cycle effects are $E_+ = s_R x I/A$ and $E_- = s_R y I/A$, where $s_R = \sin^2(g_R t_R\sqrt{A}) > 0$, directly satisfying $\Delta = 0$. This follows from the explicit Hamiltonian rather than taking target terminal probabilities as dynamical premises.

The proof also relaxes the previous thermal conditions. If both terminal routes accept the same actual thermal-success effect $F_T$ while retaining route records and the fixed auxiliary preparation, their joint effects are $E_+ = s_R x F_T/A$ and $E_- = s_R y F_T/A$, and remain balanced; common thermal success may depend on the controller state. Including continuation and thermal failure in (S92) through the actual instrument preserves (S93)–(S97). This requires the actual pulled-back effects of both routes to be the same $F_T$; if different states are written on the routes before thermal processing, use of the same thermal apparatus alone does not establish equality.

More generally, define $S = E_+ + E_-$ and $\Delta_L = \sum_{n=0}^{L-1} T^n(\Delta)$. If the actual action satisfies only the relative order bound $-\epsilon S \le \Delta \le \epsilon S$, order preservation by the positive map and summation over histories give $|\operatorname{Tr}(\rho\Delta_L)| \le \epsilon t_L$. By (S94),

$$\boxed{|N_L - (6 + 6x^2/A)| = \frac{6x|\operatorname{Tr}(\rho\Delta_L)|}{At_L} \le \frac{6x\epsilon}{A} \quad (t_L > 0).} \tag{S99}$$

This bound does not grow with the number of retries and does not require a uniform positive lower bound on the delivery probability. It controls relative exit imbalance; an absolute error bound unrelated to $S$ alone cannot give this guarantee at low delivery probabilities.

The terminal Hamiltonian above gives a complete conditional realization of balanced exits. The static generator of 7.3.2 in the main text gives $D = x^2/(x+y)$ but does not itself specify these two terminal projections. To connect the results as the same original process, one must still calculate $E_\pm = J_{\text{in}}^\dagger U^\dagger \Pi_\pm U J_{\text{in}}$ using its actual joint unitary $U$, input isometric column $J_{\text{in}}$, and actual exits $\Pi_\pm$, then test $yE_+ = xE_-$ and the connection conditions for the fixed auxiliary preparation and its filtering. Average balance for just one preparation and one cutoff does not prove the operator identity for all inputs. The theorem already includes any common continuation subchannel required for normalization preservation, without additionally requiring scalar restoration each cycle; the correspondence between the original virtual process and the actual exits still has to be established.

# S9 Matter sources, background propagation, and metric strength

This section first establishes delivery of the positive source, then distinguishes position propagation in the original thermal cycle from persistent replenishment of resource units. Each is calculated from its own complete procedure; the subsequent metric strength uses the explicitly specified resource return and timing.

## S9.1 Formation and delivery of the positive source

Write $r(W) = \sqrt{W}\log W$. The complete bright block of S6, on working entrance $z_0$, normalized bright exit $e$, and logical orientation $Q = Q^\dagger, Q^2 = I$, is $F_W = \mu_W \sigma_x \otimes Q$, where $\mu_W = (g_{\mathrm{col}}/\Lambda) r(W) > 0$. The quantity $\Lambda$ is the common capacity for all columns; $D_i$ multiplies the exit by phase $i$ and acts as the identity on the entrance.

**Proposition S9.1 (Delivery of the complete positive-source code subspace).** After resource gating is isolated, an actual quarter pulse and exit phase give

$$\begin{aligned} U_+ &= D_i e^{-i\pi F_W/(4\mu_W)}, \qquad \mathcal{J}_+\psi = \frac{z_0\otimes\psi + e\otimes Q\psi}{\sqrt{2}}, \\ U_+(z_0\otimes\psi) &= \mathcal{J}_+\psi, \qquad \mathcal{J}_+^\dagger\mathcal{J}_+ = I, \quad F_W\mathcal{J}_+ = \mu_W\mathcal{J}_+. \end{aligned} \tag{S100}$$

**Proof.** The complete bright block satisfies $(F_W/\mu_W)^2 = I$. The exponential gives $(z_0\otimes\psi - ie\otimes Q\psi)/\sqrt{2}$, and the exit phase removes $-i$. Orthogonality of the endpoints and $Q^2 = I$ establish the isometry and positive-eigenvalue relations, respectively. Thus arbitrary logical states and entanglement with subsystems not participating in this action are fully preserved, without additional selection of a target orientation.

Every accepted history in 2.3.4 of the main text satisfies $K_{h,d}\mathcal{J}_{\mathrm{in}} = -ib_h\mathcal{J}_{F,s}/\sqrt{2}$, where the deterministic preparation $\mathcal{J}_{F,s} = U_s\mathcal{J}_{\mathrm{in}}$ has an actual inverse protocol. Let the actual layout conversion be $V_0\mathcal{J}_{\mathrm{in}} = \mathcal{J}_0$, with $\mathcal{J}_0\psi = z_0\otimes\psi$; for the same ready layout one may take $V_0 = I$. First decode back to the entrance and then load, giving

$$L_{h,d} = U_+V_0U_s^\dagger K_{h,d}, \qquad L_{h,d}\mathcal{J}_{\mathrm{in}} = -\frac{ib_h}{\sqrt{2}}\mathcal{J}_+, \qquad \sum_{h,d}\mathcal{J}_{\mathrm{in}}^\dagger L_{h,d}^\dagger L_{h,d}\mathcal{J}_{\mathrm{in}} = Q_M I. \tag{S101}$$

The inverse here is that of the deterministic reference preparation, not of the random thermal channel. Isometric postprocessing leaves the acceptance effect unchanged, so neither a positive-branch share nor a return-column norm is multiplied in again. The actual bridging duration is

$$T_{\mathrm{bridge},+} = T_{U_s^\dagger} + T_{V_0} + \frac{\pi\Lambda}{4g_{\mathrm{col}}r(W)} + T_{\mathrm{phase},+}. \tag{S102}$$

Failure histories execute a window of equal duration; existing matter and resources must actually be paused or preserved by the specified transport, with these durations likewise included. If the entrance is not ready, $V_0$ and all accepted histories must be replaced by those of the actual protocol.

During operation, source writing also preserves the positive-source code subspace history by history. Under the common maintenance-phase convention, the two orig-

inal source-outcome actions are

$$k_{\pm}\mathcal{J}_{+} = c_{\pm}\mathcal{J}_{+}, \qquad c_{\pm} = \frac{e^{-iz_s/2}}{\sqrt{2}}\left(\cos\frac{z_s}{2} \pm \sin\frac{z_s}{2}\right), \qquad |c_{\pm}|^2 = \frac{1 \pm \sin z_s}{2}. \quad \text{(S103)}$$

The two effects sum to the identity, and the resource-writing bias is $b_s = \sin z_s/2$. All outcomes in this equation are scalar on the same positive-source code subspace, so they preserve the logic and its correlations. The separate filtering protocol needed to protect coherence across positive and negative branches is not executed here, and its exit rate cannot be appended to the positive source. Paused displacement defines the scope of this proposition; an anticommuting displacement orientation generally connects positive and negative codes, so moving sources require the complete motion generator.

## S9.2 Position propagation in the original cycle

Position response must retain all thermal inputs and direction coherences of the original cycle. Following S1.2–S1.3, take $\sigma_g = \sum_a g_a P_{e_a}$ and $Y = V\rho V^{\dagger}$, where $V$ preserves the direction space and acts as the identity on the shell-average space. Define $\epsilon_\nu = \sum_a \epsilon_{a\nu}$, $k_a = 1 - \sum_\nu f_{\nu a}$, $d_\nu = \sum_a a_{a\nu} t_{u,a\nu}(1 - \cos\alpha_{a\nu})$, and $\Gamma_{\nu\nu} = 1 - \epsilon_\nu$, $\Gamma_{\nu\mu} = 1 - d_\nu - d_\mu$ $(\nu \neq \mu)$. Here $t_{u,a\nu}$ is the environment occupation probability on the exit side of the corresponding thermal block, and $U_\varphi$ is the original retrieval phase. Retaining each thermal outcome and then executing shell restoration gives the complete channel

$$\begin{aligned}\mathcal{F}_{V,g}(\rho) = U_\varphi \left[\Gamma \circ (PYP) + \sum_\nu \left(\sum_a f_{\nu a}\rho_{aa}\right) P_\nu\right] U_\varphi^{\dagger} \\ + \sigma_g \left[\sum_\nu \epsilon_\nu Y_{\nu\nu} + \sum_a k_a \rho_{aa}\right].\end{aligned} \quad \text{(S104)}$$

Here $\circ$ denotes entrywise multiplication, and $\rho_{aa} = \langle e_a|\rho|e_a\rangle$. The two non-transfer thermal branches give $\Gamma$, while reverse exchange gives the direction diagonal entries; restoration replaces all background outputs by their total trace times $\sigma_g$ and removes direction–background and internal background cross coherences. This is the complete channel, with no selection of survival outcomes.

Take $V = I$. In the complete spectral basis $\{q_i\} = \{e_a\} \cup \{\nu\}$, the occupation-probability column evolves under the stochastic matrix $\mathsf{P}$, while direction coherences are independently multiplied:

$$\begin{gathered}\mathsf{P}_{\mu\nu} = (1 - \epsilon_\nu)\delta_{\mu\nu}, \quad \mathsf{P}_{a\nu} = g_a\epsilon_\nu, \quad \mathsf{P}_{\nu b} = f_{\nu b}, \quad \mathsf{P}_{ab} = g_a k_b, \\ \lambda_{\nu\mu} = (1 - d_\nu - d_\mu)e^{i(\varphi_\nu - \varphi_\mu)}, \qquad \nu \neq \mu.\end{gathered} \quad \text{(S105)}$$

Each column of $\mathsf{P}$ sums to one. Let the unitary transformation between the point and spectral bases be $A_{xi} = \langle x|q_i\rangle$, and write $v_y = (|A_{yi}|^2)_i$ and $w_x = (|A_{xi}|^2)_i$.

**Theorem S9.2 (Complete position kernel).** For every $n \geq 1$, the final-cycle position probability of the original cycle satisfies

$$\boxed{T_n(x,y) = w_x^{\mathsf{T}}\mathsf{P}^n v_y + \sum_{\nu\neq\mu} A_{x\nu}\overline{A_{x\mu}A_{y\nu}}A_{y\mu}\lambda_{\nu\mu}^n.} \quad \text{(S106)}$$

**Proof.** Point input $|y\rangle\langle y|$ has spectral occupation probabilities $v_y$ and direction coherences $\overline{A_{y\nu}}A_{y\mu}$. The first cycle removes all other coherences; subsequent cycles iterate occupation probabilities by $\mathsf{P}$ and multiply direction coherences by $\lambda$. Transforming back to the point basis and taking diagonal entries proves the formula. Only the complete sum of the two terms is a probability; $T_0(x,y)=\delta_{xy}$ is specified separately.

The finite arithmetic basis further determines the spatial shape. Let $N_j=|S_j|$ and $\mathcal{I}_j=\{e_j\}\cup\mathcal{D}_j$, let $\mathcal{D}_j$ be the directions of shell $j$, and let $C_n(j,b)=\mathbf{1}_{\mathcal{I}_j}^{\mathsf{T}}\mathsf{P}^n\mathbf{1}_{\mathcal{I}_b}/N_b$. For $x\in S_j, y\in S_b$, the moduli and multiplicativity of characters within a shell give

$$T_n(x,y)=\frac{C_n(j,b)}{N_j}+\frac{\delta_{jb}}{N_j^2}\sum_{\substack{\nu,\mu\in\mathcal{D}_j\\ \nu\neq\mu}}\lambda_{\nu\mu}^n\chi_\nu(u_xu_y^{-1})\overline{\chi_\mu(u_xu_y^{-1})}. \tag{S107}$$

The unit ratio is evaluated only within the same shell, $u_x=x/p^j$; the zero-element shell has an empty direction set. Thus inter-shell transfer is uniform within the output shell, while direction coherence resolves relative positions within a shell. This position relation on the unit group is not yet the distance on the three-dimensional resource graph of Chapter 4.

## S9.3 Coherent memory and integrated response

At finite temperature, with all $g_a>0$, non-exchange share $a_0>0$, and at least one positive exchange edge for each direction, directions can reach the background, and the restored background can reach every direction, with self-retention also present. Hence $\mathsf{P}$ is primitive; its unique stationary occupation probabilities follow directly from flux balance:

$$\pi_{e_a}=y_*g_a,\quad \pi_\nu=y_*\frac{f_\nu(g)}{\epsilon_\nu},\quad y_*^{-1}=1+\sum_\nu\frac{f_\nu(g)}{\epsilon_\nu},\quad f_\nu(g)=\sum_a g_af_{\nu a}. \tag{S108}$$

Within the strict-contraction range of the original partial-exchange angles, direction coherences satisfy $|\lambda_{\nu\mu}|<1$. Put $\Pi_*=\pi\mathbf{1}^{\mathsf{T}}$ and $p_*(x)=w_x^{\mathsf{T}}\pi$. Summing on the zero-sum subspace of diagonal components and separately on each coherence component gives

$$\begin{aligned}\mathscr{R}_{xy}&:=\tau\sum_{n=1}^{\infty}[T_n(x,y)-p_*(x)]\\ &=\tau w_x^{\mathsf{T}}[(I-\mathsf{P}+\Pi_*)^{-1}-I]v_y\\ &\quad+\tau\sum_{\nu\neq\mu}A_{x\nu}\overline{A_{x\mu}A_{y\nu}}A_{y\mu}\frac{\lambda_{\nu\mu}}{1-\lambda_{\nu\mu}}.\end{aligned} \tag{S109}$$

The identity used is $\sum_{n\geq1}(\mathsf{P}^n-\Pi_*)=(I-\mathsf{P}+\Pi_*)^{-1}-I$. With the trace-distance contraction constant $c_*=\min\{a_0,\min_\nu\epsilon_\nu\}>0$ from §4.4 of Ref. [2], truncation after $N$ cycles has pointwise tail error at most $\tau(1-c_*)^{N+1}/c_*$. Every column of the integrated kernel sums to zero; its entries may be negative, representing relaxation of a local preparation.

Position occupation probabilities need not evolve autonomously. Select only the terminal thermal edges from directions to the zero element. A unit-shell direction $\chi$ and the shell-average state $e_0$ have identical initial point probabilities, but their

first-cycle zero-element probabilities are $\epsilon_{K\chi} g_K$ and $g_K$, respectively. Thus $T_1$ cannot be repeatedly multiplied as a cycle-by-cycle position transition matrix; measuring each cycle and discarding coherence changes the process.

All coherent feedback can still be retained under a general control $V$. Let $\mathcal{D}$ denote the projection onto point-basis diagonals and let $\mathcal{Q} = I - \mathcal{D}$, as a mathematical decomposition without an actual measurement. Set $\mathsf{A} = \mathcal{D}\mathcal{F}\mathcal{D}$, $\mathsf{B} = \mathcal{D}\mathcal{F}\mathcal{Q}$, $\mathsf{C} = \mathcal{Q}\mathcal{F}\mathcal{D}$, and $\mathsf{D} = \mathcal{Q}\mathcal{F}\mathcal{Q}$. Then

$$p_{n+1} = \mathsf{A}p_n + \mathsf{B}\mathsf{D}^n h_0 + \sum_{k=0}^{n-1} \mathsf{B}\mathsf{D}^{n-1-k}\mathsf{C}p_k, \quad p_n = \mathcal{D}\rho_n, \quad h_n = \mathcal{Q}\rho_n. \tag{S110}$$

**Proof.** Iterate $h_{n+1} = \mathsf{C}p_n + \mathsf{D}h_n$ and substitute into $p_{n+1} = \mathsf{A}p_n + \mathsf{B}h_n$. The second term retains initial coherence, and the third retains the history of diagonal components entering coherence and then returning. Occupation-probability evolution is strictly autonomous for every input if and only if $\mathsf{B} = 0$; even when restricted to diagonal initial states, $\mathsf{B}\mathsf{D}^k\mathsf{C}$ must be checked. This memory comes from system coherence and does not require the old environment to participate again.

This gives complete results for propagation and relaxation in the original thermal cycle. To use it for metric-background propagation, the change in the joint state induced by a matter source and the test maintenance phase must additionally be connected to the same channel, with a correspondence for spatial scales. The resource-return coefficient of S9 comes from its explicit source-writing and testing protocol and cannot be replaced by $\mathscr{R}$ without this step.

## S9.4 Thermal survival and persistent replenishment

Let $s, t$ denote the distinct resource units associated with the source and test, respectively. Each component of resource configuration $\mathbf{n}$ takes value 0 or 1; $E_t$ refreshes the test resource unit to an unbiased preparation, while $E_s(b)$ replaces the source resource unit by a preparation of occupation probability $1/2 + b$, retaining all other resource units. The actual transport and unbiased boundary update is $\mathcal{R}$, with first moment $I - cL$, where $L > 0$ is the Laplacian of the finite Dirichlet resource graph. Use the edgewise exchange rule of the main text so that weights are nonnegative; for example, each edge has probability $c$ and the total edge weight is at most $1/2$. Each cycle selects no contact, test contact, or source contact with probabilities $1-2\varepsilon, \varepsilon, \varepsilon$, where $0 < \varepsilon < 1/2$. The complete configuration coherence block is

$$\begin{aligned} D_t(z) &= \operatorname{diag}_{\mathbf{n}} e^{-2iz(1-n_t)}, \\ T_b(z) &= \mathcal{R}\big[(1-2\varepsilon)D_t(z) + \varepsilon E_t D_t(z) E_t + \varepsilon E_s(b) D_t(z)\big], \\ T_0(z) &= T_b(z)\big|_{b=0}. \end{aligned} \tag{S111}$$

Records of the original resource units are not ignored; they are summed within the complete refreshing channel. When the source is absent, source contact is still performed using an unbiased preparation; turning off that contact requires recalculating the return kernel.

After source activity, execute the thermal instrument transported from 2.3.5 of the main text to the positive-source code subspace, with survival and exit effects $q_T I$ and $(1-q_T)I$. Exit outputs are moved out and retained, and the vacancy is restored to the same loading entrance; replenishment follows S9.1 at cycle end. Thermal and

preparation windows act as the identity on resources, the test, and their correlations, and the complete delivery effect is $fI$. With a ready entrance and all steps included, $f = Q_M$. Multiple attempts alone do not justify $f = 1 - (1-q)^M$; that formula additionally requires independent restoration after failure and complete preservation after success.

**Proposition S9.3 (Complete source-absent/source-present process).** In the order "source-absent, source-present," its coherence block and occupation recurrence are

$$\mathbb{T}(z) = \begin{pmatrix} (1-f)T_0 & (1-f)(1-q_T)T_b \\ fT_0 & [q_T + f(1-q_T)]T_b \end{pmatrix}, \tag{S112}$$
$$\lambda = (1-f)q_T < 1, \qquad \pi_{j+1} = f + \lambda\pi_j, \qquad \pi_j = \pi_* + (\pi_0 - \pi_*)\lambda^j, \quad \pi_* = \frac{f}{1-\lambda}.$$

**Proof.** The source-absent branch passes through $T_0$ and is then either filled or remains source-absent. The source-present branch passes through $T_b$; its surviving part remains, while its exited part is split according to $f$, giving the four blocks. The effects are scalar, so this decomposition remains valid even when the source-presence flag and resources are correlated. Summing the probabilities of the blocks at $z = 0$ gives the occupation recurrence; its affine fixed point and iteration give the last two formulas. When $f = 0, q_T = 1$, occupation stays at its initial value and is treated separately.

An existing individual still exits according to thermal survival; equilibrium occupation does not imply that the individual survives indefinitely. If the common reading gives individual phase mass $E_s = z_s/(2\tau)$, the stationary counting mass is $M_s = \pi_* E_s$. This is the average mass of individuals actually present; it does not define the coherence frequency of the source-absent/source-present mixture as $M_s$. The nonzero source-strength comparisons below take $\pi_* > 0$.

## S9.5 Source and test return response

Take an unbiased resource preparation, and let $y_j$ be the resource unit bias after contact and before the next transport. Write $S_0 = I - \varepsilon P_{\{s,t\}}$ and $\mathsf{A} = S_0(I - cL)$. Refreshing replaces the mean occupation of the original resource unit by the specified new mean; its dependence on the source-presence flag is affine, so correlated inputs still satisfy

$$y_{j+1} = \mathsf{A}y_j + \varepsilon b_s \pi_j e_s, \qquad y_N = \varepsilon b_s \sum_{j=0}^{N-1} \mathsf{A}^{N-1-j} \pi_j e_s, \quad y_0 = 0. \tag{S113}$$

No additional product of the source-presence flag and mean occupation of the original resource unit appears, because source contact replaces the old resource unit in both the source-absent and source-present cases. Differentiating the complete test signal at $z_t = 0$ makes each cycle's phase insertion give its mean occupation of the test resource unit. Substituting the formula above into the time average and

interchanging finite sums gives

$$\begin{aligned}\Phi_N^{\mathrm{weak}} &= -2b_s H_N, \qquad H_N = \pi_* K_N + (\pi_0 - \pi_*) J_N(\lambda),\\ K_N &= \frac{\varepsilon}{N}\sum_{l=0}^{N-1}(N-l)(\mathsf{A}^l)_{ts},\\ J_N(\lambda) &= \frac{\varepsilon}{N(1-\lambda)}\sum_{l=0}^{N-1}(1-\lambda^{N-l})(\mathsf{A}^l)_{ts}.\end{aligned} \tag{S114}$$

This is the actual finite buildup process. For the kernel $K_N$ at graph distance $d$, $N \le d$ permits no arrival yet; positive edge weights make it strictly positive after $N \ge d+1$. An initially absent source must also build up occupation, and the complete response $H_N$ retains this delay. Direct subtraction using nonnegativity gives $K_{N+1} - K_N = \varepsilon \sum_{l=1}^{N} l(\mathsf{A}^l)_{ts}/[N(N+1)] \ge 0$.

**Proposition S9.4 (Stationary two-contact return).** Let $\mathcal{G} = L^{-1}$, $u = \mathcal{G}_{ss}$, $v = \mathcal{G}_{tt}$, and $w = \mathcal{G}_{st} > 0$, and write $\alpha = c(1-\varepsilon)/\varepsilon$ and $D = (\alpha+u)(\alpha+v)$. Then

$$\boxed{K_\infty = \varepsilon[(I-\mathsf{A})^{-1}]_{ts} = \frac{\alpha w}{D-w^2}, \qquad F_{\mathrm{ret}} = \frac{\alpha}{D}.} \tag{S115}$$

**Proof.** Multiplying the stationary equation by $S_0^{-1}$ gives $cLy + \varepsilon P_{\{s,t\}} y/(1-\varepsilon) = \varepsilon b_s \pi_* e_s/(1-\varepsilon)$. Its restriction to the two endpoints is

$$\begin{pmatrix} \alpha+u & w \\ w & \alpha+v \end{pmatrix}\begin{pmatrix} y_s \\ y_t \end{pmatrix} = b_s \pi_* \begin{pmatrix} u \\ w \end{pmatrix}. \tag{S116}$$

The determinant is $D - w^2 > 0$, giving solution $y_t = b_s \pi_* \alpha w/(D-w^2)$. The stationary geometric series gives the first expression, in agreement. The finite $K_N$ increases monotonically toward it; this kernel belongs to the active two-contact process and does not replace the passive motion kernel of 4.3–4.4 in the main text.

The stationary weak-test potential is therefore $\Phi_\infty = -\pi_* K_\infty \sin z_s = -2\tau K_\infty M_s \operatorname{sinc} z_s$. Actual occupation appears in both response and counting mass and cancels from the weak-source coefficient per unit mass; the actual durations of replenishment and preparation do not cancel.

**Corollary S9.4a (Availability and two background responses).** Fix the resource unit availability share $H_{t|s} = 1/2 - y_t$ after test contact. In a single-window weak-angle readout, the normalized coherence signal and two squared responses are

$$\begin{aligned} C_1(z) &= 1 - H_{t|s} + H_{t|s} e^{-2iz}, \qquad -\tfrac{1}{2}\partial_z \arg C_1(0) = H_{t|s},\\ C_{\mathrm{ex}} &= C_M^* H_{t|s}, \qquad C_{\mathrm{ph}} = C_M^* H_{t|s}^2, \qquad y_t = \pi_* K_\infty b_s. \end{aligned} \tag{S117}$$

**Proof.** An unavailable resource unit is bypassed, while an available resource unit accumulates phase $-2z$; summing over the two occupation probabilities gives $C_1$. Differentiation at zero angle gives mean phase coefficient $H_{t|s}$. For gradient block $\sqrt{b_r}\sigma_x$, the short-time exchange probability is $H_{t|s}\sin^2(\sqrt{b_r}z)$; summing the coefficients of the quadratic term gives $C_{\mathrm{ex}}$, while summing squared phase rates gives $C_{\mathrm{ph}}$. Expansion about the unbiased reference gives

$$\Delta C_{\mathrm{ex}} = -C_M^* y_t, \qquad \Delta C_{\mathrm{ph}} = -C_M^* y_t + C_M^* y_t^2. \tag{S118}$$

Their first-order changes coincide, but their baselines differ. Multi-cycle weak-angle readings time-average $H$ over the same complete history. At finite angles, the complete signal of S9.7 is still used; it cannot be replaced by a single phase factor formed from the mean availability. This corollary connects actual availability and phase response; it does not separately fix the return kernel, clock, or absolute strength.

## S9.6 Complete cycle period and strength matching

Resource gating changes the operating phase, requiring a single background calibration on this complete protocol. Suppose all ports under the unbiased preparation satisfy phase reading $p = z/2$ within the window used; S9.7 provides a direct sufficient condition. Put $C = C_M^*$ and $\Omega = 2E_0/\sqrt{C}$, with gated accumulation $a_B = \int g_{\mathrm{col}}/\Lambda \, dt$. Common background squared matching gives $a_B = \Omega\tau$. The bare coefficient of the ungated half-gap in 2.5.3 of the main text is not used in addition to this formula.

Use a serial schedule with a common bare rate. Let $A_M$ be the dimensionless cost of all inactive steps, including the actual inverse formation reference, positive-source loading, thermal transport, replenishment, transport, and failure padding; denote independent waiting by $t_w$. Then

$$\begin{aligned} \tau &= \frac{\Lambda a_B + A_M}{g_{\mathrm{col}}} + t_w, \qquad & \rho_{\mathrm{clk}} &= \frac{g_{\mathrm{col}}}{\Omega}, \quad \delta = \Omega t_w, \\ a_B &= \frac{A_M + \rho_{\mathrm{clk}}\delta}{\rho_{\mathrm{clk}} - \Lambda}, \qquad & \tau &= \frac{a_B}{\Omega}, \quad \rho_{\mathrm{clk}} > \Lambda. \end{aligned} \tag{S119}$$

The proof substitutes the common calibration into the first line and rearranges. Positive cost requires $\rho_{\mathrm{clk}} > \Lambda$. The explicit solution requires that costs not depend on the accumulation or rate being solved for; otherwise the equation remains implicit. Different parallel protocols must be recalculated using actual elapsed duration; failure windows cannot be shortened by the success probability.

Using the common spectral ports of S6, write $r_s = r(W_s)$, $r_t = r(W_t)$, $r_v = r(W_v)$, and $r_m = r(W_{\mathrm{met}})$, so that $z_s = a_B r_s$ and $z_t = a_B r_t$. Under the old normalization $a = r_v a_B$ of the main text, $f_s = r_s/r_v$ and $f_t = r_t/r_v$; a general family uses its own actual phase coefficient. The common reading gives $E_s = \Omega r_s/2$ and $E_{\mathrm{met}} = \Omega r_m/2$, and the spatial correspondence of the main text gives $\ell_x = 2/(\Omega r_v)$. In a three-dimensional bulk region and a controlled far-field regime, $w \simeq \ell_x/(4\pi r)$; taking also the leading term in return ratio $\eta = w^2/D \ll 1$, the stationary weak-source, weak-test potential is $-2\tau M_s F_{\mathrm{ret}} w$. The finite-return denominator is retained in S9.7.

**Theorem S9.5 (Leading-order strength of the complete schedule).** Within the common process and reading regime above,

$$\boxed{\begin{aligned} G^{(0)}_{\mathrm{response}} &= \frac{\tau \ell_x F_{\mathrm{ret}}}{2\pi}, \qquad & \kappa_0 &= \frac{r_m^2 F_{\mathrm{ret}}}{4\pi r_v}, \\ \mathfrak{C}_0 &:= G^{(0)}_{\mathrm{response}} E_{\mathrm{met}}^2 = \kappa_0 \frac{A_M + \rho_{\mathrm{clk}}\delta}{\rho_{\mathrm{clk}} - \Lambda}. \end{aligned}} \tag{S120}$$

**Proof.** Writing the potential above as $-G^{(0)}_{\mathrm{response}} M_s/r$ gives the first equation. Substitution of the spatial scale, spectral mass, and complete cycle period gives the

second line. Source occupation has canceled through the same counting mass, while loading and replenishment costs remain in $A_M$. Changing the actual preparation schedule may therefore change the strength; the old clock cannot continue to be used after occupation cancels.

**Corollary S9.6 (Matching the spectral scale).** The same process satisfies the original inverse-square spectral calibration if and only if

$$\boxed{G^{(0)}_{\mathrm{response}} = E_{\mathrm{met}}^{-2} \quad \Longleftrightarrow \quad \rho_{\mathrm{clk}} - \Lambda = \kappa_0(A_M + \rho_{\mathrm{clk}}\delta).} \tag{S121}$$

This follows by direct rearrangement of the theorem. If the bare rate is not yet fixed, the candidate $\rho_{\mathrm{clk}} = (\Lambda + \kappa_0 A_M)/(1 - \kappa_0\delta)$ must also satisfy positivity and the common reading window; this is an explicit physical calibration. If the inputs have been independently fixed, the relation is tested and the actual calculated value retained. Uniform rescaling of energy units does not change $\rho_{\mathrm{clk}}, \delta, \mathfrak{C}_0$ and cannot achieve matching.

Here $\tau E_v = 1$ is not assumed. A separate proof that the original elementary return period equals this section's complete service period would give $a_B r_v = 2$, and strength matching would further require $r_m^2 F_{\mathrm{ret}} = 2\pi r_v^2$; this must still be checked against the actual process, together with the corresponding finite phases. A common background clock alone does not turn these relations into identities.

## S9.7 Finite processes and error bounds

Finite source angle, test angle, and observation duration are all calculated from the same joint block of S9.4. Let $v_0 \geq 0$ be the joint occupation-probability column of resource configurations and source-presence flags, normalized by $\mathbf{1}^{\mathsf{T}} v_0 = 1$, allowing correlations between them; take initial test coherence as their common factor. Define

$$\begin{gathered} C_N(z_t; b_s) = \mathbf{1}^{\mathsf{T}} \mathbb{T}(z_t; b_s)^N v_0, \\ \varphi_N = \mathrm{Arg}_{\mathrm{unfold}} \frac{C_N(z_t; b_s)}{C_N(z_t; 0)}, \qquad \Phi_N = -\frac{\varphi_N}{N z_t}, \qquad b_s = \frac{\sin z_s}{2}. \end{gathered} \tag{S122}$$

The phase is unwrapped along the actual continuous branch, and both signals must be nonzero; zero angle uses the derivative limit. The finite-test-angle result generally depends on the test port and does not automatically define a test-independent potential.

**Proposition S9.7 (Common finite window).** For an unbiased resource preparation, if $N|z| < \pi/2$, then $|C_N(z; b)| \geq \cos(N|z|) > 0$. On the unbiased baseline $b = 0$, the phase-mass reading is exactly $-\mathrm{Arg}_{\mathrm{unfold}}\, C_N/(2N) = z/2$.

**Proof.** Each complete configuration path contains exactly one $D_t$ phase per cycle, with nonnegative weights for all other records and unit total weight after removing phases. Thus $S_N = e^{iNz} C_N$ is a convex combination of phases in $[-N|z|, N|z|]$ and has real part at least $\cos(N|z|)$. On the unbiased baseline, $T_b = T_0$, and summing over source-presence flags gives exactly $T_0$, without requiring flags and resources to be independent; resource-flip symmetry makes $S_N$ real, and positivity of its real part then determines the unwrapped phase.

If $L_{\max}$ covers all angle coefficients of the common background, source, test, and metric endpoints, then $N a_B L_{\max} < \arccos V_{\min}$ guarantees the reading at every port under the unbiased preparation and the visibility lower bound $V_{\min}$ for the signal

with a source. This sufficient condition guarantees neither field arrival nor arbitrarily long windows; outside it, the signals are calculated directly.

Finite-buildup errors can be controlled independently. The geometric series of a nonnegative matrix gives

$$\begin{aligned} K_\infty - K_N &= \frac{\varepsilon}{N}[\mathsf{A}(I-\mathsf{A}^N)(I-\mathsf{A})^{-2}]_{ts} \geq 0, \\ |H_N - \pi_* K_\infty| &\leq \pi_*(K_\infty - K_N) + \frac{|\pi_0 - \pi_*| K_\infty}{N(1-\lambda)}. \end{aligned} \tag{S123}$$

The first equation follows by subtracting the finite weighted sum from $\sum_{l\geq 0}\mathsf{A}^l$; the second uses $0 \leq J_N \leq K_\infty/[N(1-\lambda)]$. When the exchange rule of the main text makes $I - cL$ symmetric and nonnegative with controlled spectrum, $q_A = \|S_0^{1/2}(I-cL)S_0^{1/2}\| < 1$; the identical $S_0$ factors at the two contacts cancel, and the first line is also bounded by $\varepsilon q_A(1-q_A^N)/[N(1-q_A)^2]$. Buildup of source occupation from vacancy and buildup through resource propagation are thus retained separately.

If potential coefficients are compared using $M_s = \pi_{\mathrm{ref}} z_s/(2\tau)$, the reference $\pi_{\mathrm{ref}} > 0$ may explicitly be the stationary $\pi_*$ or finite-window average occupation $\bar{\pi}_N = N^{-1}\sum_{j=0}^{N-1}\pi_j$. Under the same three-dimensional spatial correspondence, the finite potential coefficient is

$$\mathfrak{R}_N := G_N E_{\mathrm{met}}^2 = \mathfrak{C}_0 \frac{\varphi_N}{N z_t z_s \pi_{\mathrm{ref}} F_{\mathrm{ret}} w}. \tag{S124}$$

The proof writes $\Phi_N$ as $-G_N M_s/r$ and uses $w \simeq \ell_x/(4\pi r)$ together with the calibration in S9.6. Spatial-approximation error is counted separately. Taking a weak test, stationary occupation, and an established return gives

$$\mathfrak{R}_\infty = \mathfrak{C}_0 \frac{\operatorname{sinc} z_s}{1-\eta}, \qquad \eta = \frac{w^2}{D} < 1, \quad D = (\alpha+u)(\alpha+v). \tag{S125}$$

This is the long-time limit of the response kernel; if mass is also reread over that long window, the common phase identity must still be checked.

Finally, the force follows from a position difference or derivative of the actual potential. Keep source mass, source angle, cycle period, and contact parameters independent of test position; $V = E_t \Phi_\infty$ gives $\mathcal{F}_r = E_t \pi_* \sin z_s\, \partial_r K_\infty$. In a general geometry,

$$\partial_r K_\infty = \alpha \frac{(D+w^2)\partial_r w - w \partial_r D}{(D-w^2)^2}, \qquad \partial_r D = (\alpha+v)\partial_r u + (\alpha+u)\partial_r v. \tag{S126}$$

In a bulk region where $u, v$ is approximately constant, $w \simeq \ell_x/(4\pi r)$, and the spatial approximation also controls derivatives, substitution using the counting mass gives

$$\boxed{\mathcal{F}_r = -\frac{G^{(0)}_{\mathrm{response}} M_s E_t}{r^2} \operatorname{sinc} z_s \frac{1+\eta}{(1-\eta)^2}.} \tag{S127}$$

On a lattice, use position differences of the potential directly. The potential's return factor $(1-\eta)^{-1}$ differs from the force factor; finite tests, field buildup, and general boundaries continue to be calculated from the complete block. These are complete conditional results for the specified common process, without identifying every original thermal cycle with this controlled protocol or relabeling finite-window errors as experimental precision.

## S9.8 Averaging limit of passive maintenance

This section proves the averaging window used in 4.4.2 of the main paper for the passive resource–carrier channel of 4.3; resource transfer is independent of the test state and cannot be replaced by the active two-contact process. All constants below refer to the same fixed finite mixing resource process, whose mixing rate is not changed when taking the small-accumulation limit.

Fix a finite mixing resource process and its stationary distribution $\pi_b$, and put $K_{\mathbf{n}} = \eta B_{\mathbf{n}}$, where $B_{\mathbf{n}}$ remains bounded and $\eta$ is the small accumulation per cycle. Write $a_H = \eta A_H$ and $a_x = \eta A_x$. Starting from the stationary product entrance $\rho_{\mathbf{n}} = \pi_b(\mathbf{n})\rho$, resource updates preserve this entrance distribution. Expanding the active unitary of the main paper's resource–carrier channel to first order gives

$$\begin{aligned} \sum_{\mathbf{n}} \pi_b(\mathbf{n}) e^{-i\eta B_{\mathbf{n}}} \rho e^{i\eta B_{\mathbf{n}}} &= \rho - i\eta[\bar{B}, \rho] + O(\eta^2), \\ \bar{B} = \sum_{\mathbf{n}} \pi_b(\mathbf{n}) B_{\mathbf{n}} &= A_H \mathsf{F}_t \otimes \sum_i \bar{h}_i P_i + A_x Z \otimes D_x. \end{aligned} \tag{S128}$$

Here the average response comes from the first-order term of the same channel. The next cycle's entrance is generally correlated and cannot repeatedly be assumed to remain a product. Let $\rho^{(m)} = \sum_{\mathbf{n}} \rho_{\mathbf{n}}^{(m)}$ after cycle $m$, and decompose $\rho_{\mathbf{n}}^{(m)} = \pi_b(\mathbf{n})\rho^{(m)} + \delta_{\mathbf{n}}^{(m)}$, where $\sum_{\mathbf{n}} \delta_{\mathbf{n}}^{(m)} = 0$.

Denote the action of resource transfer on these matrix blocks by $\mathscr{P}$. Finite mixing ensures $\|\mathscr{P}^q\| \le C r^q$ on zero-sum blocks, where $0 < r < 1$. The first-order expansion of the active stage generates only $O(\eta)$ correlations per cycle, so starting from $\delta^{(0)} = 0$ gives

$$\begin{aligned} \delta^{(m+1)} = \mathscr{P}\delta^{(m)} + O(\eta), \qquad \|\delta^{(m)}\|_{\oplus,1} \le C'\eta \sum_{q=0}^{m-1} r^q = O(\eta), \\ \|\delta\|_{\oplus,1} := \sum_{\mathbf{n}} \|\delta_{\mathbf{n}}\|_1. \end{aligned} \tag{S129}$$

After tracing over resources, the zeroth-order correlation term cancels; its contribution to carrier motion in the next cycle acquires another factor $\eta$ and is therefore $O(\eta^2)$. The reduced action differs from the average unitary by $O(\eta^2)$ per cycle; summing cycle by cycle using trace-norm invariance under unitary evolution gives the averaging-error bound in equation (4.14) of the main paper. The constants depend on the mixing rate of the fixed resource process and uniform bounds on the norms of the operators used; the actual elapsed duration is $N\tau$. A nonstationary resource entrance additionally has an initial relaxation, and arbitrarily long-time coherent readings still require the complete return channel. Section 4.4.2 of the main paper therefore converts the average generator using the complete cycle period.

# S10 Spatial outward and return paths and common response

## S10.1 Reception and impulse for complete histories

Let matter position be logical register $X$. The source writes occupation at vertex $x_0 = s_0$ with probability $q_s = se/2$. At each step the propagation protocol uses fresh, ready, identically distributed actual edge/idle labels, giving the same Markov transition while retaining orthogonal histories; the outward path reaches $x_1, \ldots, x_m$ and the return follows the same labels in reverse order to $x_{m-1}, \ldots, x_0$. At the end of each step, target logical position controls local swapping in, readout, and sending back. The read generator is the target's complete $K_t = K_t^\dagger$ multiplied by the incoming resource packet's $I - O_B$. The incoming resource packet, target preparation, and target resource unit are distinct factors; the no-added-occupation baseline here is $I - O_B$, not the successfully available projection $A_{\mathrm{av}}$ of S7. The holding envelope in each readout window has time integral $b$; position motion and the source's own holding are frozen along the path, and the target is initially independent of the source.

For history $\omega$, the number of visits to position $x$ is $N_x(\omega) = \delta_{x,x_0} + \delta_{x,x_m} + 2\sum_{j=1}^{m-1} \delta_{x,x_j}$. Let $h$ be the time integral of any additional ungated target-holding envelope. For two target eigenstates with eigenvalue difference $\Delta\kappa$, their exact reduced coherence multiplier is

$$C_x(\Delta\kappa) = e^{-i(h+2mb)\Delta\kappa} \left[1 - q_s + q_s \mathbb{E}_\omega e^{ib\Delta\kappa N_x(\omega)}\right]. \tag{S130}$$

Proof: the unoccupied branch experiences the full $K_t$ in every window; the occupied branch skips that action in windows visiting $x$. Labels follow actual Born weights and complete terminal histories remain orthogonal, so tracing records sums probabilities; the same inverse return does not generate a second independent history. A matrix expression also gives the finite distribution directly: write $D_x(z) = I + (e^{iz} - 1)|x\rangle\langle x|$ and let the one-step probability matrix be $R$; the characteristic function is $\mathbf{1}^T D_x(z) R[D_x(2z)R]^{m-1} D_x(z) \boldsymbol{e}_{s_0}$. It counts each endpoint once and every interior point twice, agreeing with the complete-path calculation.

Take cumulative weak readout $2mb\|K_t\| \ll 1$. Expanding each history's exponential while retaining position control before the partial trace gives

$$\begin{aligned} H_t^{(0)} &= \frac{h + 2mb}{\tau} K_t, \\ \Delta H_t &= H_t^{(0)} \Phi_m(X), \\ \Phi_m(x) &= -\frac{q_s b}{h + 2mb} u_{m,x}, \\ u_{m,x} &= \mathbb{E} N_x. \end{aligned} \tag{S131}$$

This is the same multiplicative change of the entire target generator. The corresponding single-cycle impulse is

$$\Delta\langle P\rangle^{(1)} = -iq_s b \,\langle K_t[u_m(X), P]\rangle. \tag{S132}$$

Proof: after removing free background action, the first-order relative action is $I+iq_sbK_tu_m(X)$. Applying its adjoint action to $P$ gives the commutator formula. In a continuous-position window, $[u(X),P]=i\partial_x u$, giving the phase-gradient impulse; on a finite graph the commutator form remains. A symmetric isolated source has no directed net self-acceleration. Identifying the inertia parameter with this full-target coefficient still requires the motion port of the same protocol; a local phase alone does not establish all kinematic identifications.

A finite-angle reading is not a single potential. For a fixed history and target eigenvalue $\kappa$, the compression onto the original source code contains $I+a_s(e^{ibN_x\kappa}-1)$ and $a_s = sE/2$; its exterior-code leakage effect is $4a_s(I-a_s)\sin^2(bN_x\kappa/2)$. This follows directly from norm preservation of the complete unitary and the squared modulus of the compressed multiplier. It must enter the repeated round-trip channel; the source cannot be assumed to reset automatically to its original state after each cycle.

## S10.2 Arrival kernel, residence, and spatial limit

Let the graph be finite, connected, and undirected, with an actual one-step deployment giving $R = I - cL$, where $L$ is the combinatorial Laplacian and $c > 0$ makes $R$ nonnegative and aperiodic. Let the vertex count be $J_g$, let $\pi = \mathbf{1}/J_g$ and $\Pi = \mathbf{1}\mathbf{1}^T/J_g$, and let $L^+$ be the inverse on the zero-mean subspace. Summing in outward-and-return visit order gives exactly

$$u_m = (I+R)\sum_{j=0}^{m-1} R^j \boldsymbol{e}_{s_0}, \qquad \mathbf{1}^T u_m = 2m, \tag{S133}$$

$$u_m - 2m\pi = \left[\frac{2}{c}L^+ - (I-\Pi)\right](I-R^m)\boldsymbol{e}_{s_0}. \tag{S134}$$

Proof: the original path's expected distribution is $R^j\boldsymbol{e}_{s_0}$; counting the endpoints once and interior points twice gives the first equation. On the zero-mean subspace, $I-R=cL$ is invertible and $I+R=2I-cL$; applying the finite geometric series gives the second equation. If $Q_R = R-\Pi$, omitting the finite tail has error norm at most $(2\|L^+\|/c+1)\|Q_R\|^m$. The uniform term cancels only in spatial differences or under an explicit common-background subtraction.

For two nonsource positions, the local $I-\Pi$ term cancels in the spatial difference, leaving the actual graph's Green-function difference and the finite tail above. The cycle graph has exact kernel $L^+_{xs_0} = (J_g^2-1)/(12J_g) - d(J_g-d)/(2J_g)$, a piecewise quadratic shape; address count alone does not turn it into three-dimensional $1/r$. A Newtonian far field follows only after separately proving a three-dimensional window and length conversion for the actual position graph such that $L^+_{xs_0} \sim \ell_x/(4\pi r)$. Cumulative weak readout must also hold; taking $m\to\infty$ at fixed $b$ does not preserve the first-order potential.

Resource residence and visit readings are timed separately. If the first transport pulse in an original transport step lasts $a$ and the second auxiliary-reset pulse lasts $d$, with no extra endpoint waiting, the resource occupation-time kernel for outward and return transport without readout is

$$\ell_m = a\boldsymbol{e}_{s_0} + 2(a+d)\sum_{j=1}^{m-1} R^j \boldsymbol{e}_{s_0} + (a+2d)R^m \boldsymbol{e}_{s_0}, \qquad \mathbf{1}^T \ell_m = 2m(a+d). \quad \text{(S135)}$$

If local swapping in and sending back together with read holding take $t_F$, the additional residence in the entire local workspace is $t_F u_m$ and the total duration increases by $2mt_F$. Resources may leave the original network resource unit during transfer or temporary storage; counting only original network resource units misses this contribution. A complete genuine inverse can make external resource occupation zero at cycle end while these integrals remain positive, so resource occupation, residence integrals, and stationary net flux are not interchangeable.

## S10.3 Common calibration, complete-cycle timing, and strength

Use the same source, capacity, elementary matrix element, and zero-blocking background reference. Put $r_0 = \sqrt{s}$; the actual common background coefficient is $V_0$. A single calibration shared by source and target requires

$$V_0 = 2r_0 c_B \frac{t_H}{\tau} = 2r_0 c_B \frac{h + 2mb}{\tau}. \quad \text{(S136)}$$

When all target maintenance is supplied by gated reception along the path, $h = 0$; equal read windows in the same module further give $2mb = t_H$. This holding ratio follows from common calibration and the stated fully gated deployment, without using a target-strength equality. Nonuniform read windows use the sum of the time integrals of their holding envelopes, while ungated maintenance requires retaining $h$; the identity does not apply to different zero-blocking references.

The absence of a residual endpoint phase in the original transport follows directly from its two pulses. For label-controlled SWAP or idle operation $F = F^\dagger = F^{-1}$, define $L_F = F \otimes |1\rangle\langle 0|$, $L_I = I \otimes |1\rangle\langle 0|$, and $U_\theta(L) = e^{\theta(L - L^\dagger)}$. Acting on auxiliary $|0\rangle, |1\rangle$ gives $U_{-\pi/2}(L_I)U_{\pi/2}(L_F) = F \otimes I$. Thus even an arbitrary correlated auxiliary is fully restored; if both stages have angular rate $g$, the total duration is $\pi/g$. Each local swap in and send back is one such complete module.

Source holding and target readout below both use a normalized holding envelope identically equal to one, so $t_H, b$ give both the corresponding actual holding durations and envelope integrals; an envelope not identically equal to one requires separately calculating its time integral and the elapsed duration, and equal integrals do not imply equal durations.

At the same elementary matrix element $g = g_M/2$, a complete transport step takes $2\pi/g_M$ and a local swap-in/send-back pair takes $4\pi/g_M$. The outward and return paths contain $2m$ steps in total; source holding and target holding are serial in the present protocol, and all other actual preparation, temporary storage, control, and reset durations sum to $T_{\text{extra}}$. Hence

$$C_m = \frac{12m\pi}{g_M} + T_{\text{extra}}, \qquad \tau = 2t_H + C_m, \qquad \alpha = \frac{V_0}{2r_0 c_B} = \frac{t_H}{\tau}. \quad \text{(S137)}$$

When the actual drive is fixed and $C_m > 0$ is independent of the holding amount being solved for, positive finite holding exists if and only if $0 < \alpha < 1/2$, with solution

$$t_H = \frac{\alpha C_m}{1 - 2\alpha}, \qquad b = \frac{t_H}{2m}, \qquad \tau = \frac{C_m}{1 - 2\alpha}. \tag{S138}$$

Proof: substitute the total serial duration into $\alpha = t_H/\tau$ and solve the scalar linear equation. If the drive is itself to be determined by the same calibration, or control costs depend on holding duration, the formula is a compatibility relation or implicit equation; it does not independently determine all drives and durations at once. Actual loading is counted only when executed, without repeatedly charging a one-time load or removing controls actually executed each cycle.

**Theorem S10.3 (Calibrated finite potential difference).** Under the common fully gated equal-window deployment above and the cumulative weak-readout regime of S10.1, denote the source's prereadout phase mass by $M_s = r_e g_M e\, t_H/(2\tau)$. Then, for $0 < e < 1$,

$$\Phi_m(x) = -\frac{q_s}{2m} u_{m,x}, \qquad \frac{\Phi_m(x) - \Phi_m(y)}{M_s} = -\frac{s r_0 c_B}{m r_e g_M V_0}(u_{m,x} - u_{m,y}). \tag{S139}$$

Proof: first substitute $2mb = t_H$ into the potential of S10.1, then use $q_s = se/2$ and common calibration $\alpha = V_0/(2r_0 c_B)$ to eliminate $e, t_H/\tau$. This already holds at finite $m$, without a Green-function approximation. The holding ratio and additional processing duration are therefore no longer independent factors in this calibrated ratio of first-order potential difference to source phase mass; they still control actual operating duration, residence integrals, and finite-angle outputs.

For two nonsource positions, S10.2 further gives the positive coefficient of the Green-function term and its weak-matching limit:

$$\mathcal{A}_G = \frac{2 s r_0 c_B}{m c\, r_e g_M V_0}, \qquad \mathcal{A}_G^{(0)} = \frac{2 s q_0}{m c \Lambda V_0} \quad (e \to 0,\ g_B = g_M). \tag{S140}$$

Only in a separately established three-dimensional spatial window can the leading potential difference be identified as $-G^{(0)} M_s (1/r_x - 1/r_y)$, with $G^{(0)} = \mathcal{A}_G^{(0)} \ell_x/(4\pi)$. The present actual $m, c$, position graph and length scale, inertial reading of the motion port, and correspondence between the participating spectral scale and this common $V_0$ each retain their own conditions. These results alone do not imply $G^{(0)} E_{\mathrm{met}}^2 = 1$ or alter the exact conditional identity obtained in S9 for its specified replenishment protocol.

# S11 Approximation regime of nucleon spectral preparation

Section 6.3 of the main paper uses the leading-term prescription $\beta$_B=1+1/$\sqrt{3}$ and complete spectral participation. The comparisons below distinguish prescription choice, spectral cutoff, and the lowest scale within the allowed family.

$$\begin{aligned} V(\beta_B) &= 2.860214657918\ldots, \\ V(\beta_B) - 3 &= -0.139785342081\ldots, \\ \frac{V(\beta_B) - 3}{3} &\simeq -4.6595\%. \end{aligned} \tag{S141}$$

This difference cannot be treated as a mass error bar already supplied by the calculation. Replacing the $\beta$-selection rule by the exact condition V($\beta$)=3 requires solving for the root and evaluating again, giving $\beta$ ≈1.564273980920, W≈2.388897704196, and M≈962.7840 MeV. The value 936.6693 MeV corresponds to the leading-term condition, and 962.7840 MeV to the exact-variance condition; these are different spectral-preparation prescriptions.

The spectral cutoff is likewise a preparation condition. The main value in 6.3 uses the complete $\zeta$ spectrum; finite participation must instead use $Z_L$, with errors controlled by the tail bound in S5.3. For example, changing only the participation cutoff to $L = 2400$ while retaining $\beta_B$, $C^*$, and $E_0 = 1$ GeV gives $Z_L \simeq 2.330274457513$ and energy scale 923.7669 MeV, below the main value by 12.9024 MeV. This illustrates the cutoff effect without imposing the electroweak depth on nucleons; if the background is also truncated, its squared quantity must be replaced simultaneously. The number of operation terms carried by a finite state space also differs from its number of independent states.

Finally, the formal energy scale supplied by the prescribed rule increases strictly with the number n of fundamental color-representation factors. Indeed, $\beta$(n) decreases, $\zeta$ decreases for $\beta$>1, and

$$\frac{d}{dW}\left(\sqrt{\frac{W}{C^*}}\log W\right) = \frac{\log W + 2}{2\sqrt{C^* W}} > 0 \qquad (W > 1), \tag{S142}$$

Thus, for pure three-dimensional fundamental color representations, the above $\beta$ values, and the same energy-scale conditions, n=3 gives the lowest scale in this allowed family. This does not prove exclusion of every decay channel to arbitrarily lower masses, nor turn formal values at n=4,5 into an allowed color-singlet spectrum. Actual excitation, formation, and stability remain determined by the complete process.

# S12 Finite loading and persistent-return enhancement

This section proves the finite frequency enhancement of 8.4.2 in the main paper. Adopt $\alpha \in (0,1)$ and $q = 1-\alpha$, and denote the c-branch single-attempt success effect by $E_c$, with $0 \le E_c < \alpha I$. The complete three-outcome instrument for each attempt satisfies

$$K_s^\dagger K_s = E_c, \quad K_f^\dagger K_f = \alpha I - E_c, \quad K_c = \sqrt{q}\, I, \qquad E_c + (\alpha I - E_c) + qI = I. \tag{S143}$$

Execute at most $n+1$ attempts in succession, with only the continuation outcome triggering the next attempt; all outcomes, attempt labels, and the unfinished tail are retained in controllable workspace. Let the isometry after complete unitary loading be $J_t = U_n J_0$, and let $\Pi_s, \Pi_f$ mark success on some attempt and termination in failure without success, respectively. The success column of attempt $k+1$ is $K_s q^{k/2}$, where $k = 0, \dots, n$; different attempt outputs are orthogonal, hence

$$\begin{aligned} J_t^\dagger \Pi_s J_t &= \mathcal{R}_n E_c, \qquad J_t^\dagger \Pi_f J_t = \mathcal{R}_n(\alpha I - E_c), \\ J_t^\dagger (I - \Pi_s - \Pi_f) J_t &= q^{n+1} I, \qquad \mathcal{R}_n = \frac{1-q^{n+1}}{\alpha}. \end{aligned} \tag{S144}$$

The three effects sum to the identity because $\alpha \mathcal{R}_n + q^{n+1} = 1$. Enhancement comes from summing the success effects over all orthogonal attempts; the success flag is not measured and the tail branch is not discarded.

Keep the remaining spectral paths, nodes, and frequency shift of the c branch unchanged, and denote its dimensionless compression by $F_c$. On the fully loaded code, use the same self-adjoint action $X_t$ of the success projection and these original factors, with compression $J_t^\dagger X_t J_t = \mathcal{R}_n F_c$. Take $S_t = 2J_t J_t^\dagger - I$ and let the self-adjoint orientation $Y$ carry the phase. Simultaneous arms then give

$$\widehat{H}_t = \frac{g}{2}(X_t + S_t X_t S_t) \otimes Y, \qquad \widehat{H}_t (J_t \otimes I) = (J_t \otimes I)(g \mathcal{R}_n F_c \otimes Y). \tag{S145}$$

The proof uses only $S_t J_t = J_t$ and $(I + S_t) X_t J_t = 2 J_t J_t^\dagger X_t J_t$. Thus the entire working code is preserved, and the phase coefficient of continuous evolution is enhanced by a factor $\mathcal{R}_n$.

Loading is executed only once. Subsequent survival cycles share drive $g$ and active window $t_H$, under a schedule in which the additional working registers do not increase per-cycle repair and thermal-processing durations; all complete cycle periods are $\tau = t_H + T_{\mathrm{proc}}$. For the same nonzero c branch,

$$\frac{m_t^{(n)}}{m_c} = \mathcal{R}_n, \qquad \frac{1}{\alpha} - \frac{m_t^{(n)}}{m_c} = \frac{q^{n+1}}{\alpha}. \tag{S146}$$

The common active fraction $t_H/\tau$ cancels, leaving the exact tail term. The common period is an explicitly adopted operating condition; initial loading belongs to the formation history, and its duration must be included separately in averages from startup to a finite endpoint. The finite-pool implementation in Section 8.4.2 and S16.4 uses $\alpha = \alpha_8$ and $N = n+1$, so $\mathcal{R}_n = G_N$; $\alpha_L \to \alpha_{\mathrm{EM}}$ only as the pool cutoff tends to infinity. The continuation share $q$ of this competition is defined separately from the thermal exit probability.

# S13 Complete neutrino round trip and mass matrix

This section connects the indirect channel of 8.5 in the main paper to the persistent action of 2.3–2.5. Using the same neutral entrance, cyclic connections, and shared motion port, we calculate the two spectral transport stages, two relays, nodes, inverse return, and background response in the complete auxiliary space. Spectral and relay flags remain reversible; actual thermal exit occurs after one cycle's inverse connection. The common source has already been delivered, while S3 still specifies formation from unready inputs.

## S13.1 Complete carrier and spectral transport

**Transported object.** On species layer $p = 7$, take the odd sector under negation to be $U_- = \mathrm{span}(\chi_1, \chi_3, \chi_5)$. The same multiplier transports these three directions together, with each unit carrying amplitude $n^{-1/2}$. Since $T_n\chi_\ell = \chi_\ell(n)\chi_\ell$ and $\chi_1\chi_3\chi_5 = \chi_3$, its action on the top exterior power is

$$\begin{aligned} \det(n^{-1/2}T_n|_{U_-}) &= \chi_3(n)n^{-3/2}, \\ w_b &= \sum_{n\equiv b\ (7)} \chi_3(n)n^{-3/2}|n\rangle, \\ z_b &:= \|w_b\|^2 = \zeta_b(3). \end{aligned} \tag{S147}$$

Only nonzero residue classes $b = 1, \dots, 6$ are used here. Multiplier labels $|n\rangle$ are mutually orthogonal, and the series converges in norm; exponent 3 comes from the squared modulus of the amplitude. The quantity $\zeta_b(s)$ in (8.36) of the main paper can also represent a coherent matrix element, but this section uses the squared norm of the transfer column above; these two operations cannot be interchanged. Transport of the entire odd sector and orthogonal multiplier outputs are inputs of this neutral process; the identity of $\chi_3$ alone does not select them.

Let $\mathcal{H}_G = \mathbb{C}^3$ be the family space, use $C = C_\nu(\delta_*) = (c_1, c_2, c_3)$ from (8.38) of the main paper, and let $\|c_a\|^2 = 6$. On the six-dimensional spectral working auxiliary unit, take the actual spectral operator $S_L$ and write $A = S_L^\dagger S_L \geq 0$, $A\chi_2 = |L_2|^2\chi_2$, and $s_A = \|A\| > 0$. This spectral operator inherits the analytic-response correspondence of 2.4.1 and S5. Define

$$\begin{aligned} Z = \sum_{a=1}^{3} z_a c_a c_a^\dagger, \qquad \mathcal{M} = \sum_{a=1}^{3} z_a z_{7-a} c_a c_a^\dagger, \\ \Lambda \geq \max\{1, s_A^4 z_4 \|Z\|\}, \qquad b_2 = s_A^2 z_4, \quad b_1 = \Lambda/b_2. \end{aligned} \tag{S148}$$

Here the chain weights are $h_a = z_a z_{7-a}$. The capacity $\Lambda$ is actually shared by matter and background and simultaneously satisfies feasibility of the other modules used; it is not solved backward from neutrino masses. Both $w_b$ and $c_a$ are transfer columns, whose absolute sizes are not erased by normalizing them to unit states.

**Proposition S13.1 (Complete two-stage instrument).** The first transport stage outputs the chain label and first multiplier; the second reads the partner address while retaining these outputs:

$$V_1 = \frac{A}{\sqrt{b_1}} \otimes \sum_{a=1}^{3} |a\rangle|w_a\rangle\langle c_a|,$$
$$V_2 : \ |a,n\rangle \longmapsto \frac{A}{\sqrt{b_2}} \otimes |a,n\rangle|w_{7-a}\rangle. \tag{S149}$$

The operator $A$ in the second line acts on the original spectral working auxiliary unit. Both maps are contractions, and their double-success effect is exactly $A^4 \otimes \mathcal{M}/\Lambda$.

**Proof.** The positive series gives $z_4 \geq z_5 \geq z_6 > 0$, hence $\|V_2\|^2 = s_A^2 z_4/b_2 = 1$. The first transport stage satisfies $V_1^\dagger V_1 = A^2 \otimes Z/b_1$, so $\|V_1\|^2 = s_A^4 z_4 \|Z\|/\Lambda \leq 1$. For any contraction $V : \mathcal{H} \to \mathcal{K}$, use the complete unitary block on $\mathcal{H} \oplus \mathcal{K}$

$$U_V = \begin{pmatrix} \sqrt{I - V^\dagger V} & -V^\dagger \\ V & \sqrt{I - VV^\dagger} \end{pmatrix}. \tag{S150}$$

The identity $Vf(V^\dagger V) = f(VV^\dagger)V$ cancels the cross blocks, and both diagonal blocks equal the identity. First-stage failure follows a bypass, while only the remaining outputs enter the second transport stage. Thus the complete isometric column and success effect are

$$\begin{pmatrix} \sqrt{I - V_1^\dagger V_1} \\ \sqrt{I - V_2^\dagger V_2}\, V_1 \\ V_2 V_1 \end{pmatrix}^\dagger \begin{pmatrix} \sqrt{I - V_1^\dagger V_1} \\ \sqrt{I - V_2^\dagger V_2}\, V_1 \\ V_2 V_1 \end{pmatrix} = I,$$
$$V_2 V_1 = \frac{A^2}{\sqrt{\Lambda}} \otimes \sum_a |a, w_a, w_{7-a}\rangle\langle c_a|, \qquad (V_2V_1)^\dagger (V_2V_1) = \frac{A^4}{\Lambda} \otimes \mathcal{M}. \tag{S151}$$

The failure-output subspaces of both transport stages are retained; denote the complete unitary dilation by $U_L$. On $\chi_2$, $A^4$ gives $|L_2|^8$. The second transport stage acts according to the first transport stage's label, producing a product of partner weights rather than drawing a chain anew. The construction contains no three-chain averaging, columnwise normalization, or two independent normalized spectral samples, and therefore produces no additional 1/3, 1/6, or spectral partition denominator. ■

## S13.2 Relays, nodes, and ports

Prepare the relay register as $e = N^{-1/2} \sum_x |x\rangle$ in the $N$-dimensional participating space of $P_4$, and take $P_x = |x_0\rangle\langle x_0|$ and $P_e = |e\rangle\langle e|$. Both projections are implemented with complete success/failure flags, without the original shell thermal restoration between them. Direct multiplication gives

$$R = P_e P_x, \qquad Re = \frac{e}{N}, \qquad p(r_1) = \frac{1}{N}, \quad p(r_2 \mid r_1) = \frac{1}{N}, \quad \langle e, R^\dagger Re\rangle = \frac{1}{N^2}. \tag{S152}$$

After the first success the state is $|x_0\rangle$, and after the second success it is restored to $e$. The complete two-stage instrument is not statistically independent; the calculation multiplies conditional probabilities along the double-success history. The two

$P_x$ factors in $R^\dagger R = P_x P_e P_x$ are the same outward amplitude and its conjugate, not a third matching.

Denote the complete unitary for the two filtering stages by $U_R$ and its double-success projection by $\Pi_{R,ss}$. The corresponding two-stage spectral projection is $\Pi_{L,ss}$. The self-adjoint entrance projections are $Q_R = U_R^\dagger \Pi_{R,ss} U_R$ and $Q_L = U_L^\dagger \Pi_{L,ss} U_L$. They act on different working factors; their entrance compressions are the above $N^{-2}$ and the two-stage spectral effect, respectively. Subsequent inverse connection uses the complete flags and does not perform another independent acceptance test.

Take the two coherent nodes and independent port flag as

$$\begin{aligned} |n_i\rangle = \cos(\pi/8)|0\rangle + \sin(\pi/8)|1\rangle, \quad B_i = \sigma_x, \quad \langle n_i|B_i|n_i\rangle = \frac{1}{\sqrt{2}}, \\ P_q = |0\rangle\langle 0|, \quad |q\rangle = |+\rangle, \quad \kappa_{\mathrm{port}} = \langle q|P_q|q\rangle = \frac{1}{2}. \end{aligned} \tag{S153}$$

The two nodes give the product of amplitude matrix elements $1/2$, while the port projection additionally gives $\kappa_{\mathrm{port}}$. Orientation partner $Y$ has symmetric eigenvalues $\pm 1$, and the half-gap reading introduces no further factor of 2. The equal-amplitude port preparation is the physical normalization used and cannot be replaced by a real character or a notational factor in a quadratic form.

## S13.3 Persistent action and complete inverse connection

Let $J$ embed $\mathcal{H}_G \otimes \mathbb{C}^2_Y$ in the complete space together with common source $\Phi$, spectral preparation $\chi_2$, relay $e$, two nodes, the port, and ready flags. Take the background logical space separately as $\mathcal{H}_B$, with its local auxiliary prepared as in 2.5.3. The source reference group is $P_0^S$, with $p_0 = \langle \Phi | P_0^S | \Phi \rangle > 0$. The neutral endpoint also actually passes through this reference group. Write

$$\begin{aligned} D_\nu = \frac{\kappa_{\mathrm{port}} |L_2|^8}{2N^2} \mathcal{M}, \qquad c = \frac{p_0}{\Lambda}, \qquad U = U_L U_R, \\ D_{\mathrm{out}} = P_0^S \Pi_{L,ss} \Pi_{R,ss} B_1 B_2 P_q Y, \quad J_o = UJ, \quad S_o = 2J_o J_o^\dagger - I. \end{aligned} \tag{S154}$$

From here on, every $J, J_o$ includes the background auxiliary and the corresponding isometric embedding of the background logic. The projections and nodes act on their specified tensor factors, and $D_{\mathrm{out}}$ is a self-adjoint contraction. Factorwise calculation using S13.1 and S13.2 gives $J_o^\dagger D_{\mathrm{out}} J_o = c D_\nu \otimes Y$, with the unwritten background logic acted on by the identity.

**Theorem S13.2 (Persistent phase of the complete round trip).** Actually switching on the two equal-strength protection arms simultaneously on the output code gives

$$\begin{aligned} H_{\nu,o}(t) = \frac{g(t)}{2}(D_{\mathrm{out}} + S_o D_{\mathrm{out}} S_o), \\ H_{\nu,o}(t) J_o = J_o \, g(t) c (D_\nu \otimes Y), \\ \boxed{U^\dagger \mathcal{T} e^{-i \int H_{\nu,o}(t) dt} U J = J e^{-ic\theta D_\nu \otimes Y}, \quad \theta = \int g(t) dt.} \end{aligned} \tag{S155}$$

**Proof.** Since $S_o J_o = J_o$ and $I + S_o = 2 J_o J_o^\dagger$, the sum of the two arms equals $2 J_o J_o^\dagger D_{\mathrm{out}} J_o$ on $J_o$. This gives intertwining at every time; the fixed logical generator

makes the time-ordered exponential depend only on accumulation $\theta$. Left multiplication by the complete $U^\dagger$ gives the return identity. All failures remain within the domains of the unitary action and reflection. ■

The eigenvalues of $\mathcal{M}$ therefore enter the mass frequency linearly. Directly replacing this by bidirectional exchange with $\left(\begin{smallmatrix} 0 & V^\dagger \\ V & 0 \end{smallmatrix}\right)$ would give singular values; that is a different actual vertex. The paths here execute sequentially, while the protection arms in the active window still act simultaneously and cannot be replaced by serial finite pulses.

**Local extension of the background.** Let $j_B$ denote the preparation isometry for the common source and background-local auxiliaries, including the logical gradient spaces of both endpoints, and put $P_B = j_B j_B^\dagger \otimes I_{\nu,\mathrm{work,exit}}$ and $S_B = 2P_B - I$. Take the bare background action $X_B = P_0^S Z_\Lambda U_t U_G$ from (2.75) of the main paper, with local compression $cG$. Set

$$\begin{aligned} H_B(t) &= \frac{g(t)}{2}(X_B + S_B X_B S_B), \qquad [U, H_B(t)] = 0, \\ (H_{\nu,o} + H_B)J_o &= J_o\, g(t) c(D_\nu \otimes Y \otimes I_B + I_\nu \otimes G). \end{aligned} \tag{S156}$$

Commutativity follows because $U_L, U_R$ does not act on the source or background-local workspace; neutral protection and background protection each preserve the common code, so their actions can be added on it. They need not commute everywhere outside the code. The background reflection does not require the matter workspace to be ready, so the same background action remains on the exit domain.

## S13.4 Common cycle period and mass coefficient

Following the common processing protocol of 2.3, a cycle consists of successive nonoverlapping stages: forward execution $U$, output-end maintenance, complete $U^\dagger$, repair, thermal processing, and padded waiting. Matter and background switch on together in the maintenance window and switch off together in the other stages, or complete the specified free-phase compensation. Denote the complete cycle duration and accumulated drives at both endpoints by

$$\begin{gathered} T = T_U + t_H + T_{U^\dagger} + T_{\mathrm{repair}} + T_{\mathrm{thermal}} + T_{\mathrm{pad}}, \\ \Theta_\nu = \int_0^T g(t) a_\nu(t) dt, \qquad \Theta_B = \int_0^T g(t) a_B(t) dt > 0, \qquad \eta_\nu = \frac{\Theta_\nu}{\Theta_B}. \end{gathered} \tag{S157}$$

Simultaneous operations are timed only once; configurations rebuilt every cycle enter the corresponding stages, while one-time setup of a fixed apparatus and initial loading are counted separately. Processing for different thermal outcomes is padded to the same fixed-duration processing interval. Switching these operations is actual control; identical gate types do not automatically imply equal durations.

**Corollary S13.3 (Common coefficient).** The common gating above gives $a_\nu = a_B$ and $\Theta_\nu = \Theta_B = \theta$, hence $\eta_\nu = 1$. By Theorem S13.2 and the same background readout,

$$
\begin{aligned}
&v_B^{\mathrm{obs}} = \frac{p_0 \Theta_B}{\Lambda T}, \qquad M_\nu^{\mathrm{obs}} = \frac{p_0 \Theta_\nu}{\Lambda T} D_\nu, \qquad M_\nu^{\mathrm{obs}} = v_B^{\mathrm{obs}} \eta_\nu D_\nu, \\
&v_B^{\mathrm{obs}} = v_E \quad \Longrightarrow \quad \boxed{M_\nu^{\mathrm{obs}} = \frac{v_E |L_2|^8}{4N^2} \mathcal{M} \quad (\eta_\nu = 1,\ \kappa_{\mathrm{port}} = 1/2).}
\end{aligned}
\tag{S158}
$$

Finally apply the background squared calibration of (2.77) in the main paper. The factors $p_0$ and $\Lambda$ cancel in the actual common-source ratio; a neutral endpoint without $P_0^S$ would leave an additional $1/p_0$. The common capacity must still satisfy feasibility in S13.1 and the other modules; changing it changes actual preparation and bare frequencies. The mass/background ratio remains unchanged only when both endpoints use the same actual capacity and repeat the common calibration. Increasing processing duration at fixed drive and active window also lowers the actual absolute frequency.

Dividing the same round trip into two calendar cycles introduces no factor of one-half. Only when the accumulated neutral activity is half the background activity does $\eta_\nu = 1/2$ hold; if equal-amplitude ports are still used, both half factors remain. Ports and timing are separately specified in the present common protocol and cannot give two explanations for the same 1/2. After a one-time loading of duration $T_{\mathrm{load}}$ followed by $n$ survival cycles, the frequency is computed with denominator $T_{\mathrm{load}} + nT$ and tends to the expression above in the long-time limit; this differs from reloading every cycle. The common gating and calibration apply consistently to all families; one cannot change only the neutral reference clock while retaining the other families' old bare frequencies.

## S13.5 Survival records and background after exit

Repair and thermal processing reuse 2.3.6–2.3.7 and S3. The repair channel on the code is the identity, so any complete Kraus refinement must have the form $d_j I$: the identity channel's Choi matrix has rank one, and every vector in its decomposition lies on the same line, with $\sum_j |d_j|^2 = 1$. The fine thermal-survival columns are $\sqrt{r_\alpha} I$, acting as the identity on source, internal chains, nodes, and background, with $q = \sum_\alpha r_\alpha \le 1$. By Theorem S13.2, each cycle's inverse connection returns to the same preparation code.

Write $U_\nu = e^{-ic\theta D_\nu \otimes Y}$ and $U_B = e^{-ic\theta G}$. For any delivered matter–background joint state $\rho$, all fine histories of uninterrupted survival and their sum are

$$
\begin{aligned}
K_{\boldsymbol{j},\boldsymbol{\alpha}} J &= \left( \prod_{k=1}^{n} d_{j_k} \sqrt{r_{\alpha_k}} \right) J (U_\nu^n \otimes U_B^n), \\
\rho_{\mathrm{live}}^{(n)} &= q^n J (U_\nu^n \otimes U_B^n) \rho (U_\nu^{\dagger n} \otimes U_B^{\dagger n}) J^\dagger.
\end{aligned}
\tag{S159}
$$

This uses cycle-by-cycle multiplication and summation over complete outcomes; $q^n$ changes only the envelope. The initial state may contain matter–background entanglement; unready inputs still follow the formation orbit of S3. No claim is made that every initial state immediately has the same pure preparation.

**Proposition S13.4 (All exit histories and the background orbit).** In decoded logical coordinates, split matter into the survival subspace $L$ and exit subspace $E$. Thermal-failure columns $B_a : L \to E$ satisfy $\sum_a B_a^\dagger B_a = (1-q) I_L$, and old

exit states take the identity bypass. Matter action switches off on $E$, while the local background evolves by $U_B$ in both subspaces every cycle. The state first exiting in cycle $k$ and read in cycle $n$ is

$$\rho^{(n)}_{\mathrm{exit},k} = q^{k-1} \sum_a (B_a U_\nu^k \otimes U_B^n)\rho(U_\nu^{\dagger k} B_a^\dagger \otimes U_B^{\dagger n}), \qquad 1 \le k \le n. \tag{S160}$$

All outcomes form a trace-preserving instrument, and the unconditional background satisfies

$$q^n + (1-q)\sum_{k=1}^{n} q^{k-1} = 1, \qquad \boxed{\rho_B(n) = U_B^n \rho_B(0) U_B^{\dagger n}.} \tag{S161}$$

**Proof.** There are $k-1$ scalar successes before first failure; cycle $k$ completes the common activity before failing, so the matter accumulation is $B_a U_\nu^k$. The background completes $n$ active stages regardless of exit. The geometric series makes the traces sum to 1. Taking the partial trace over all matter outcomes, a complete trace-preserving local map leaves the other subsystem's reduced state unchanged, proving the background identity without requiring a separable input. Postselection on an individual failure can change conditional background statistics; the conclusion concerns all outcomes. ■

The local background extension has an actual effect. Take source $\phi = \sqrt{p}|0\rangle + \sqrt{1-p}|1\rangle$, ready work state $|0\rangle_W$, and bare background vertex $P_0^S \otimes \sigma_x^B$. The global reflection $2P_\phi P_{0,W} - I$ becomes $-I$ after exit into $|1\rangle_W$, returning the background to the bare vertex and giving exit probability $p\sin^2(gt)$ from $|0\rangle_B$. The local source reflection still gives $\sin^2(gpt)$, generally different. Intertwining within the code therefore does not replace testing the exit domain. The phase surviving through $nT$ is compared with the background over the same $nT$; an old carrier that already exited at $kT$ no longer has its mass defined using $nT$.

## S13.6 Spectral structure and scope

The main paper has already obtained $C^TC = \frac{9}{2}I + \frac{3}{2}\mathbf{1}\mathbf{1}^T$. Its phase invariance can also be written as an explicit equivalence. Put $P = \mathbf{1}\mathbf{1}^T/3$. Then

$$\begin{aligned} J_0 = \frac{1}{\sqrt{3}}\begin{pmatrix} 0 & 1 & -1 \\ -1 & 0 & 1 \\ 1 & -1 & 0 \end{pmatrix}, \qquad & R_\delta = P + \cos\delta(I-P) + \sin\delta J_0, \\ J_0^2 = -(I-P), \quad J_0 P = 0, \quad & R_\delta^T R_\delta = I, \\ C(\delta) = R_\delta C(0), \qquad & \mathcal{M}(\delta) = R_\delta \mathcal{M}(0) R_\delta^T. \end{aligned} \tag{S162}$$

The first two lines follow by matrix multiplication, and the last by direct substitution of the three cosine components. Thus the common formation angle only rotates eigenvectors and leaves the mass spectrum unchanged for the present chain weights. The three positive eigenvalues also satisfy

$$\varepsilon^3 - 6\Big(\sum_a h_a\Big)\varepsilon^2 + \frac{135}{4}\Big(\sum_{a<b} h_a h_b\Big)\varepsilon - \frac{729}{4} h_1 h_2 h_3 = 0. \tag{S163}$$

The coefficients come from the trace, sum of second-order principal minors, and determinant of $Q_\nu^{1/2} C^T C Q_\nu^{1/2}$, respectively. The mass spectrum here does not determine the mixing matrix of the weak readout; normal ordering remains the identification used in the main text's comparison with oscillation quantities.

For actual multiplier cutoff $R \in \mathbb{N}_{>0}$, define $z_b^{(R)} = \sum_{1 \le n \le R,\ n \equiv b\ (7)} n^{-3}$, $h_a^{(R)} = z_a^{(R)} z_{7-a}^{(R)}$, and $\mathcal{M}_R = C\,\mathrm{diag}(h_a^{(R)}) C^T$. By the positive-series tail bound of S5.3, $r_b = z_b - z_b^{(R)}$ satisfies $0 \le r_b \le 1/(2R^2)$. Hence

$$\begin{aligned} d_a &= h_a - h_a^{(R)} = z_a^{(R)} r_{7-a} + z_{7-a}^{(R)} r_a + r_a r_{7-a}, \\ 0 \le \mathcal{M} - \mathcal{M}_R &= C\,\mathrm{diag}(d_a) C^T \le 9 \max_a d_a\, I. \end{aligned} \tag{S164}$$

The final step uses $\|C\|^2 = 9$. Differences between corresponding ordered eigenvalues lie between 0 and $9 \max d_a$; mass errors acquire the common coefficient of S13.4. Cutoff $R$, address depth $K$, and matching dimension $N$ each have their own definition; finite cutoff can generally change the ratios of the neutrino mass eigenvalues.

**Limits of the action.** A complete return does not dispense with protection in the active window. If only a single marked phase gate is used, let $\Pi$ be a projection and let $E = J^\dagger U^\dagger \Pi U J$. Then

$$K = J^\dagger U^\dagger e^{-i\varphi \Pi} U J = I + (e^{-i\varphi} - 1)E, \qquad I - K^\dagger K = 4 \sin^2(\varphi/2) E(I - E). \tag{S165}$$

The identity follows from $e^{-i\varphi\Pi} = I + (e^{-i\varphi} - 1)\Pi$ and $0 \le E \le I$. Leakage out of the code generally remains, so this is not the $e^{-i\varphi E}$ of Theorem S13.2. If chain labels are copied into an uncontrollable environment, the original preparation reflection no longer applies directly, and all histories must be retained as in S4.

To compare explicitly with the original shell restoration, take only the entire unit shell as relay space here, so its uniform state is the shell state $e_0$ and $N$ is that shell's dimension; the embedded $P_4$ in an arbitrary deeper construction is not identified with the entire unit shell. If the original shell restoration $B_{ab} = \sqrt{g_a} |e_a\rangle\langle e_b|$, $B_0 = P_D$ is inserted between the two filters, starting from $e = e_0$ gives $P_e B_r P_x e = \delta_{r,(0,0)} \sqrt{g_0} e/N$, and the total double-success share becomes $g_0/N^2$. Retaining records does not automatically cancel this shell weight from the full channel. If $e^{-i\varphi Y}$ acts with probability $g_0$ and the identity otherwise, the coherence factor is $(1 - g_0) + g_0 e^{-2i\varphi}$, whose phase and modulus must both be calculated.

Padding to a fixed processing duration also affects the reading. If random completion histories are merged without resolving them, coherence is $z_\omega = \sum_h p_h e^{-i\omega\Theta_h}$, whose argument is generally not proportional to $\omega$ and cannot be replaced by mean waiting time. Neither of these two substitutions is the reference process of this section.

This section gives the complete conditional mass matrix of the delivered neutral carrier. With the same $\mathcal{M}$ and linear generator retained, the common activity ratio, port flags, and background energy scale change the overall scale; unequal chain weights, the transported object, multiplier coherence, noncommon deformation of the cyclic columns, and finite truncation can change the relative spectrum. Scalar survival probabilities change only the envelope. Entrance encoding, collective transport, complete return, and simultaneous protection retain their respective physical constructions; $\eta_\nu = 1$, derived from the common protocol, is no longer listed as

a separate neutral-family input. Loading of general incident states, actual decay products, and mixing and oscillations still require their corresponding processes.

## S14 Nucleon phase and electromagnetic share

For the complex multiplier $a_N \neq 0$ of a complete cycle, write $a_N/|a_N| = e^{i\Theta_N}$ on a continuously unwrapped phase branch. Its phase frequency is $-\Theta_N/\tau$, where $\tau$ includes all processing in that cycle. The normalized ratio $a_N/\operatorname{Re} a_N = 1 + i\tan\Theta_N$ is a different reading. The tangent prescription in the main text must therefore be separately implemented by actual phase writing or an effective generator; it does not follow automatically from taking a complex argument. In particular, if all other data are real and identical and the two complete self-adjoint matrices satisfy only $H_n = \overline{H_p} = H_p^T$, their characteristic polynomials coincide. In this special case, conjugate spectral phases cannot yield distinct self-adjoint eigenmasses, nor does the case exclude a separately defined phase readout.

The six-route electromagnetic share can be obtained from an explicit branched vertex. For nonzero $x \in \mathbb{Z}/p^K\mathbb{Z}$, write $x = p^j u$ and define address $r \in \mathbb{F}_p^\times$ by the unit part $u$ mod $p$. Every address in every shell contains $p^{K-j-1}$ points, so the uniform collective pattern $b$ gives

$$p_r = \langle b|P_r|b\rangle = \frac{1-p^{-K}}{p-1}, \qquad \varpi_b = \sum_r p_r^2 = \frac{(1-p^{-K})^2}{p-1}. \tag{S166}$$

Let the electromagnetic vertex be $V = g\sum_r P_r \otimes Q \otimes (c_r + c_r^\dagger)$. Its address ports are orthogonal at intermediate stages and return coherently at the endpoint, and the vertex, like the maintenance port, is protected by simultaneous reflection with $b$. Write $J_b\psi = b \otimes \psi$, $S_b = 2J_bJ_b^\dagger - I$; cancellation between the two arms then gives

$$\frac{V + S_bVS_b}{2} J_b = J_b \left[ gQ \otimes \sum_r p_r (c_r + c_r^\dagger) \right]. \tag{S167}$$

Proof: decompose $P_r b = p_r b + u_r$, where $u_r \perp b$; reflection cancels $u_r$ between the two arms. If all ports have the same spectrum and the unprotected intermediate states displaced from $b$ have equal energy, the protected and unprotected second-order responses are proportional to $\sum p_r^2$ and $\sum p_r$, respectively, so their ratio is $(1-p^{-K})/(p-1)$. The deep-layer limit of $p = 7$ is $1/6$. This follows under the stated branching and protection conditions; finite-depth corrections remain.

A share does not determine the sign or point-charge scale. Eliminating a positive-energy dynamical oscillator gives a negative second-order energy shift; the positive field-energy reading of a Coulomb-type constraint field uses a different elimination problem. The positive electromagnetic difference of the main text, the scale $\alpha_{\mathrm{EM}} M_B$, the depth, and whether address matching here is already included in $\alpha_{\mathrm{EM}}$ all retain their original correspondence conditions. This process explanation therefore cannot directly replace the exact $1/6$ in the main text by a new unconditional value.

# S15 Joint representation of quantum attributes

This section constructs a joint representation of charge, color, spin, and chirality using the matter identifications adopted in the main paper, and examines their relation to actual propagation, mass action, and weak contact. We set $\hbar = 1$.

## S15.1 Propagation matrices and rotations

Let the three position translations be mutually commuting unitary operators $L_i$, and define

$$\begin{gathered} D_i = \frac{L_i^\dagger - L_i}{2i}, \qquad T_i = P_b^+ \otimes L_i + P_b^- \otimes L_i^\dagger, \\ P_b^\pm = \frac{I \pm Z_b}{2}. \end{gathered} \tag{S168}$$

On the complete exchange port $\mathbb{C}^2_\tau$, the sum of the two oppositely directed exchanges is

$$\frac{g_i}{2}\begin{pmatrix} 0 & T_i - T_i^\dagger \\ T_i^\dagger - T_i & 0 \end{pmatrix} = g_i\, \tau_y \otimes Z_b D_i. \tag{S169}$$

The three edges use the respective port rotations

$$U_1 = I, \qquad U_2 = e^{i\pi\tau_z/4}, \qquad U_3 = e^{-i\pi\tau_x/4}, \tag{S170}$$

and couple simultaneously to the same protected complete port, giving

$$\mathscr{S} = \mathbb{C}^2_\tau \otimes \mathbb{C}^2_b, \qquad B = I_\tau \otimes Y_b, \tag{S171}$$

$$(A_1, A_2, A_3) = (\tau_y, \tau_x, \tau_z) \otimes Z_b. \tag{S172}$$

Consequently,

$$B^2 = A_i^2 = I, \qquad \{B, A_i\} = 0, \qquad \{A_i, A_j\} = 2\delta_{ij} I. \tag{S173}$$

These matrices follow from the simultaneous connections of the original exchanges specified above. Three finite-duration exchanges executed in sequence must instead be evaluated as their actual product.

Define

$$S_i = -\frac{i}{2} A_j A_k, \qquad \Gamma_\chi = -iA_1A_2A_3, \tag{S174}$$

where $(i, j, k)$ is cyclic. Direct multiplication gives

$$\begin{gathered} (S_1, S_2, S_3) = -\frac{1}{2}(\tau_y, \tau_x, \tau_z) \otimes I_b, \\ \Gamma_\chi = -I_\tau \otimes Z_b. \end{gathered} \tag{S175}$$

It follows that

$$\begin{gathered} [S_i, S_j] = i\epsilon_{ijk} S_k, \\ \sum_i S_i^2 = \frac{3}{4} I, \end{gathered} \tag{S176}$$

$$[S_i, A_j] = i\epsilon_{ijk} A_k, \qquad [S_i, B] = 0, \tag{S177}$$

and

$$\Gamma_\chi^2 = I, \qquad [\Gamma_\chi, A_i] = [\Gamma_\chi, S_i] = 0, \quad \{\Gamma_\chi, B\} = 0. \tag{S178}$$

Thus $S_i$ gives a spin-1/2 representation, while $\Gamma_\chi$ gives a chirality grading compatible with propagation. Once a spatial orientation has been fixed, write

$$P_L = \frac{I - \Gamma_\chi}{2}, \qquad P_R = \frac{I + \Gamma_\chi}{2}. \tag{S179}$$

These matrix relations hold exactly on the finite carrier. When the three propagation directions have the common long-wavelength calibration $g_i \ell_i = c_0$, the background is uniform, and boundary effects are negligible, the principal part of propagation is $c_0 \mathbf{A} \cdot \mathbf{k}$, and its spatial rotations are compatible with the rotations generated by $S_i$. The full dispersion on the finite lattice retains its actual spatial symmetries.

## S15.2 Joint representation of charge, color, and spinors

For fermionic matter branches $f$, take the complete carrier

$$\mathcal{H} = \mathcal{H}_{\text{pos}} \otimes \left[ \bigoplus_f (\mathcal{M}_f \otimes V_{c,f} \otimes \mathscr{S}) \right] \otimes \mathcal{H}_{\text{aux}}. \tag{S180}$$

Here $\mathcal{M}_f$ retains the formation and mass modes, $V_{c,f}$ carries the color representation used in the main paper, and $\mathcal{H}_{\text{aux}}$ retains auxiliaries and records. This decomposition does not require the actual state to be a product state.

Retaining the charge and color identifications of the main paper, define

$$Q = e \bigoplus_f q_f I, \qquad T^a = \bigoplus_f I_{\mathcal{M}_f} \otimes t_f^a \otimes I_{\mathscr{S}}. \tag{S181}$$

For lepton color singlets, take $t_f^a = 0$; for quarks use the triplet generators $t^a$, and for antiquarks use $-t^{a*}$. Their respective factors of action and the matrix relations of the preceding subsection give

$$\boxed{\begin{aligned} [Q, T^a] = [Q, S_i] = [Q, \Gamma_\chi] = 0, \\ [T^a, S_i] = [T^a, \Gamma_\chi] = [S_i, \Gamma_\chi] = 0. \end{aligned}} \tag{S182}$$

A joint spectral decomposition of $Q, T^3, T^8, S_3, \Gamma_\chi$ can therefore be chosen. The color generators among themselves, and the spin components among themselves, retain their noncommutative structure.

The need for internal degrees of freedom can also be deduced from propagation. The matrices $B, A_1, A_2, A_3$ generate all of $M_4(\mathbb{C})$, so

$$[X, B] = [X, A_i] = 0 \quad \Longrightarrow \quad X = x I_4. \tag{S183}$$

Nontrivial internal actions commuting with every free-propagation matrix must therefore be carried by a multiplicity space beyond the four-dimensional spinor.

This proves compatibility between the charge and color identifications of the main paper and the propagation spinor. The complete representation must still be connected to the formation state by an actual embedding and intertwining of the actions. The natural color representation $\mathbf{3}\oplus\mathbf{3}$ of the original six-dimensional species space does not itself contain a lepton singlet.

## S15.3 Mass action and changes in chirality

Denote the complete mass-phase operator of each family by $\mathsf{M}_f$, including the mass coefficient in the current background. The corresponding propagation is

$$h_f = \mathsf{M}_f \otimes B + \sum_i g_i A_i D_i. \tag{S184}$$

Since $\mathsf{M}_f$ acts on the internal mass space,

$$\boxed{[\Gamma_\chi, h_f] = 2\mathsf{M}_f \otimes \Gamma_\chi B.} \tag{S185}$$

The mass action thus connects the left- and right-chiral components. For nonzero mass, chirality is generally not conserved.

For a uniform mass eigenmode $m$, let

$$\begin{aligned} h &= mB + \mathbf{a}\cdot\mathbf{A}, \qquad E = \sqrt{m^2 + |\mathbf{a}|^2}, \\ \Pi_E^+ &= \frac{I + h/E}{2} \quad (E > 0). \end{aligned} \tag{S186}$$

Using $A_i = 2\Gamma_\chi S_i$, we have

$$\{\Gamma_\chi, h\} = 4\mathbf{S}\cdot\mathbf{a}. \tag{S187}$$

Multiplying on both sides by the positive-energy projector gives

$$\boxed{\Pi_E^+ \Gamma_\chi \Pi_E^+ = \frac{2\mathbf{S}\cdot\mathbf{a}}{E}\Pi_E^+.} \tag{S188}$$

For $|\mathbf{a}| > 0$, helicity is

$$h_{\mathbf{a}} = \frac{2\mathbf{S}\cdot\mathbf{a}}{|\mathbf{a}|}. \tag{S189}$$

The compression of chirality to the positive-energy subspace is therefore

$$\Pi_E^+ \Gamma_\chi \Pi_E^+ = \frac{|\mathbf{a}|}{E} h_{\mathbf{a}} \Pi_E^+. \tag{S190}$$

The two coincide in the massless case. For a massive particle, chirality cannot be treated as an additional constant binary label carried within the positive-energy subspace. The positive and negative propagation branches here are defined separately from the full matter conjugation between particles and antiparticles.

## S15.4 Weak chiral selection by a shared contact

Take an actual source $S$, target $T$, and reference $R$ on the unit shell. Let $\chi_3$ be the original real weak character, and let $M_3^{(X)}$ denote multiplication by this character on carrier $X$. The uniform source state $e_0$ and the weak direction $\psi_3$ satisfy

$$M_3^{(S)} e_0 = \psi_3, \qquad M_3^{(S)} \psi_3 = e_0. \tag{S191}$$

In the original coordinate decomposition of the target, $D_j^{(T)}$ projects onto group $j$, and

$$M_3^{(T)} = -Z_g \Gamma_\chi, \qquad Z_g = \sum_{j=0}^{2} (-1)^j D_j^{(T)}. \tag{S192}$$

Use the same original point contact,

$$\begin{aligned} H_c(t) &= v(t)(\Pi_{ST} + \Pi_{SR}), \\ \Pi_{SX} &= \sum_x P_x^{(S)} P_x^{(X)}. \end{aligned} \tag{S193}$$

The source visits the complete weak-kernel orbit $\ker \chi_3$. For each address pair, the total contact projector over the orbit is

$$\Pi_{\mathrm{w},SX} = \frac{I + M_3^{(S)} M_3^{(X)}}{2}. \tag{S194}$$

All address projectors commute, so the phases accumulated over the complete orbit add exactly.

The reference is coherently loaded by the group translation corresponding to target group $j$, so that its loaded multiplier is $Z_g Z_R$, where $Z_R = M_3^{(R)}$. Completing the two equal-area visits and unloading the reference gives

$$\begin{aligned} U_{\mathrm{pair}} &= e^{-i\alpha} \exp\left[ -\frac{i\alpha}{2} M_3^{(S)} Z_g (Z_R - \Gamma_\chi) \right], \\ \alpha &= \int v(t)\, dt. \end{aligned} \tag{S195}$$

Here $\alpha$ is the action area of each point visit. This identity retains every intermediate direction along the complete orbit and is used for the actual segmented contact protocol.

If the actual reference is supported in the $Z_R = +1$ sector, after omitting the common phase $e^{-i\alpha}$, the source input $e_0$ gives the two complete output operators

$$\boxed{\begin{aligned} K_{\mathrm{w}} &= -i Z_g \sin\alpha\, P_L, \\ K_0 &= P_R + \cos\alpha\, P_L. \end{aligned}} \tag{S196}$$

They satisfy

$$K_{\mathrm{w}}^\dagger K_{\mathrm{w}} + K_0^\dagger K_0 = I. \tag{S197}$$

The original weak direction therefore accepts only the left-chiral component, while the right-chiral component and the unaccepted part of the left-chiral component remain in the other output. The required phase and area relation between the two paths is fixed jointly by the symmetric double visit using the same contact.

A general reference gives

$$E_{\mathrm{w}} = \frac{\sin^2 \alpha}{2}(I - \Gamma_\chi Z_R). \tag{S198}$$

When the target and reference are initially independent, the acceptance effect on the target is

$$\boxed{\begin{gathered} E_T = \frac{\sin^2 \alpha}{2}(I - r\Gamma_\chi), \\ r = \mathrm{Tr}(Z_R \rho_R). \end{gathered}} \tag{S199}$$

The choice $r = 1$ selects the left side, $r = -1$ selects the right side, and $r = 0$ accepts both sides equally.

These results establish a conditional selection mechanism for the contact stage. Subsequent original spectral acceptance, label matching, and thermal exit retain their actual outputs. It remains to establish how formation and service processes themselves supply the required oriented reference and shared-contact configuration. This proposition does not identify the conditional selection with a derivation of the complete weak vertex, absolute weak strength, or natural selection of a handedness.

# S16 Formation structure and common family returns

The return of a formed carrier includes both formation records and current work auxiliaries. Formation records retain all paths and used work from the initial formation, while current work participates in the present loading, maintenance, and inverse. They enter the complete physical space of the family processes in Chapter 8 as distinct physical factors.

## S16.1 Complete return and persistent response

In the $Z_{\mathrm{mov}}$ basis of matter orientation, let $J_0, J_1$ be actual isometric loadings from one logical space into the same complete physical space, and let $X = X^\dagger$ be the exchange between the endpoints. Define

$$J = \mathrm{diag}(J_0, J_1), \quad K = J_0^\dagger X J_1, \quad X_{\mathrm{raw}} = \begin{pmatrix} 0 & -iX \\ iX & 0 \end{pmatrix}. \tag{S200}$$

**Theorem S16.1 (Persistent generator of a complete return).** For the entire actual preparation, put $P = JJ^\dagger$ and $S = 2P - I$. Two simultaneous arms of equal strength give

$$\begin{gathered} H = \frac{g}{2}(X_{\mathrm{raw}} + SX_{\mathrm{raw}}S), \qquad h = g\begin{pmatrix} 0 & -iK \\ iK^\dagger & 0 \end{pmatrix}, \\ HJ = Jh, \qquad e^{-itH}J = Je^{-ith}, \\ h^2 = g^2\,\mathrm{diag}(KK^\dagger, K^\dagger K). \end{gathered} \tag{S201}$$

**Proof.** Since $SJ = J$, $(X_{\mathrm{raw}} + SX_{\mathrm{raw}}S)J = 2JJ^\dagger X_{\mathrm{raw}}J$. This proves generator intertwining; self-adjointness gives intertwining for all times. Squaring gives the last identity. For the actual preparation $J = UJ_{\mathrm{in}}$, reversing its operations yields $U^\dagger e^{-itH}UJ_{\mathrm{in}} = J_{\mathrm{in}}e^{-ith}$. All failed outputs are included in $J_y$; the orthogonal complement of the code subspace retains $g(I-P)X_{\mathrm{raw}}(I-P)$. This proves the theorem.

The same maintenance reflection can be applied to several exchanges, whose logical compressions add. In both families considered here, $K$ is self-adjoint; separating the matter-orientation factor gives the internal mass matrix. The independent modification preparation has mean $F = 1 - a$, which enters the mass response through compression of $X$.

## S16.2 Link formation and complete work delivery

From this point take p=7. Let $D$ be the link direction and $E$ the complete 48-dimensional spectral work of Section 7.1.3. The single-address acceptance projector pulled back through the complete spectral action is $Q = U_s^\dagger \Pi_{\mathrm{addr}} U_s$, and the original ready state $a_E$ satisfies $w = \langle a_E|Q|a_E\rangle = \alpha_0$. Set $\ell = \sin\delta_2 \cos\delta_2$ and $e = \ell^2$. The success columns of one bare path and six corrected paths are

$$T_0 = \ell|1\rangle\langle 0| \otimes I_E, \qquad T_h = \frac{\ell}{\sqrt{2}}|1\rangle\langle 0| \otimes Q \quad (1 \le h \le 6), \qquad \mathsf{E} = eP_0^D \otimes (I_E + 3Q). \tag{S202}$$

For this spectral value $4e < 1$. The complete failure column and isometric output are

$$\begin{aligned} F_L &= P_1^D \otimes I_E + P_0^D \otimes [\sqrt{1-e}(I_E - Q) + \sqrt{1-4e}\, Q], \\ V\psi &= \sum_{h=0}^{6} |h\rangle T_h \psi + |f\rangle F_L \psi, \qquad V^\dagger V = I. \end{aligned} \tag{S203}$$

Write $U_L$ for the link unitary realized by the complete spectral unitary and port exchanges, with ready-input column V; the subsequent return applies its inverse. For the original $|0, a_E\rangle$, the total success share is $b_L = e(1+3w)$. A phase-corrected exchange of the entire path–spectral work with a new resource moves the used work into A and supplies the current ready state. Direction D and all spectator correlations are retained, giving

$$\begin{aligned} &\Gamma\psi = \sum_{h,d} |d\rangle_D |h\rangle_{A_h} [(\langle d| \otimes I_E) K_h \psi]_{A_E}, \quad K_h = T_h \ (h \neq f), \quad K_f = F_L, \\ &\Gamma^\dagger \Gamma = I, \qquad \Pi_Q^A P_0^D \Gamma = 0, \qquad \Pi_Q^A = \sum_{h=1}^{6} |h\rangle\langle h|_A \otimes I_{E_A}. \end{aligned} \tag{S204}$$

For the original D0 input, every failed output lies in D0; general inputs also retain an $f, D1$ component. Failure to supply new work enters an orthogonal bypass, while supply success determines the occupation probability of the loaded branch.

Both orientation endpoints use the same current link loading $L\psi = V(\psi \otimes a_E) \otimes |0\rangle_M$. A neutral working bit M coherently records current output D0 through $C_0 = P_1^{D_{\text{out}}} \otimes I_M + P_0^{D_{\text{out}}} \otimes X_M$. Summing all eight output columns gives

$$\begin{aligned} (V \otimes |0\rangle_M)^\dagger C_0 (V \otimes |0\rangle_M) &= P_1^D \otimes I_E + \mathsf{E}, \\ L^\dagger L = I_D, \qquad L^\dagger C_0 L &= P_1^D + b_L P_0^D =: F_D. \end{aligned} \tag{S205}$$

The D1 term comes from the identity component of the complete failure column. Initial formation $\Gamma$ and current loading L each access their own work auxiliaries; the inverse undoes only the current loading.

## S16.3 Down-type node visits, color retrieval, and complete return

Use the same odd source pattern, whole-group comparison, and even spectral round trip. Write P for the source reference-group projector and $Q_S = I - P$ for its complement, distinct from link spectral projector Q. The node success column $\mathsf{A}$ and complete failure columns $D_\beta$ satisfy $\mathsf{A}^\dagger \mathsf{A} + \sum_\beta D_\beta^\dagger D_\beta = I$. A first return makes one visit; only $Q_S$ with first-visit success makes a second visit, together with target-color exchange $\mathcal{S}_c$ and the current direction marker. With $L_c = C_0 L$, the complete nonzero visiting columns at the incoming endpoint are

$$\begin{aligned} T &= P \otimes \mathsf{A} \otimes I_c \otimes L + Q_S \otimes \mathsf{A}^2 \otimes \mathcal{S}_c \otimes L_c, \\ F_{1\beta} &= I \otimes D_\beta \otimes I_c \otimes L, \\ F_{2\beta} &= Q_S \otimes D_\beta \mathsf{A} \otimes \mathcal{S}_c \otimes L_c. \end{aligned} \tag{S206}$$

Isometry of L and Lc and completeness of one node visit give $T^\dagger T + \sum_\beta (F_{1\beta}^\dagger F_{1\beta} + F_{2\beta}^\dagger F_{2\beta}) = I$. Each column also retains all eight current link outputs. Both color-capacity failures carry ordinary L. For source–target success column $\mathcal{K}$, let $E_c = \mathcal{K}^\dagger \mathcal{K} = \sigma \sum_j p_j P_j^T$. Use $3E_c \le I$ and two orthogonal capacity-failure ports with columns $\sqrt{I - 3E_c}$. The two endpoints are then separately isometric, and capacity failures give no cross term.

Let $N_c$ collect the short return, first failures, and incoming capacity failure; let $C_c$ collect successes and failures of the second visit; and let $V_o$ be the complete outgoing color preparation. Then $N_c^\dagger C_c = 0$, and the actual endpoints are

$$\widetilde{V}_i = N_c \otimes L + C_c \otimes L_c, \qquad \widetilde{V}_o = V_o \otimes L, \qquad \widetilde{V}_y^\dagger \widetilde{V}_y = I. \tag{S207}$$

**Proposition S16.2 (Common down-type response).** For the common source and actual node preparation of Section 8.3,

$$\begin{aligned} \widetilde{V}_o^\dagger \widetilde{V}_i = \sigma \Big[ & \frac{3 r_3 p_0}{\sqrt{2}} P_0^T \otimes I_D \\ & + r_3^2 (p_1 P_1^T + p_2 P_2^T) \otimes F_D \Big] \otimes I_{C_T} = \mathcal{F}_d. \end{aligned} \tag{S208}$$

**Proof.** The short return has color contraction 3I and node response $r_3/\sqrt{2}$; identical link endpoints give $L^\dagger L = I$. Actual color exchange in the long return changes the contraction to I, two successful visits to the same node give $r_3^2$, and the link endpoints give $L^\dagger L_c = F_D$. Source-group comparison occurs once. Spectral failures are orthogonal to outgoing spectral preparation, and the two capacity failures are mutually orthogonal. They therefore contribute no cross kernel while remaining in the isometric output. Adding the contributions proves the result.

After including independent modification work in the complete preparation, Theorem S16.1 gives $H J_d = J_d g F \mathcal{F}_d Y$. The inverse removes the second-visit marker and color exchange, then the node and capacity preparation, and finally applies the original inverse of the current link. After the preceding operations are undone, the link column lies exactly in the image of V, where its actual inverse satisfies $V^\dagger V = I$. Current work returns ready, while the complete first-formation record remains.

## S16.4 Up-type record-controlled reentry and common terminal

The finite electromagnetic pool uses the alternating returns of Chapter 7. Write

$$\begin{gathered} r = \frac{(w_2/6)(2w_3/18 + w_2/6)}{3w_3/18 - 2w_2/6}, \qquad q_{+,L} = \sum_{j=0}^{L} (-r)^j = \frac{1 - (-r)^{L+1}}{1 + r}, \\ N_{E,L} = 6 + q_{+,L}(w_3 - 2w_2), \qquad \alpha_L = w_2 / N_{E,L}. \end{gathered} \tag{S209}$$

Stop at attempt $n = 0, \dots, L-1$ with weight $(1-r)r^n$, retaining weight $r^L$ at the final attempt; even and odd stops perform the original positive and negative outputs. The total weight of even stops is $q_{+,L}$. In the pool and matching notation of Section 7.3, the unnormalized retained state is $[P + q_{+,L}(w_3 P_b - w_2 R_d)]/(6 + w_3)$, with trace $N_{E,L}/(6 + w_3)$. Since $R_d v_m = 0$, same-address matching on the retained branch

followed by an independent $\chi_2$ visit gives $\alpha_L$. All preparation failures and stopping records remain. Set $L_{\mathrm{EM}} = 8$, giving $\alpha_8 = 0.00729733340150181257118548935\ldots$; r follows the original response prescription. This preparation cutoff is accounted for separately from the attempt count N below.

The up-type complete path combines its visits before a single coherent readout. Use the full six-dimensional species space of Section 7.1.3 as work space, write $\nu_k = \psi_{\chi_k}$, $\nu_0 = e_0$, and set $S_L = \zeta_* S$, $\zeta_* = \zeta(1/2)$. Then $S_L\nu_k = L_k\nu_k$ on nontrivial directions and $S_L\nu_0 = 0$; full-space transport $M_r\nu_k = \nu_{k+r}$ uses indices modulo six. The even closed path, real direction, and odd conjugate closed path give

$$\begin{gathered} A_\ell = S_L^\dagger S_L, \qquad A_{\mathrm{word}} = (M_{-1}^\dagger A_\ell M_{-1})(M_1^\dagger S_L M_1)A_\ell, \\ A_{\mathrm{word}}\nu_2 = |L_2|^2 L_3 |L_1|^2 \nu_2, \qquad C_{\mathrm{word}} = |\zeta_*|^2, \quad \Lambda = C_{\mathrm{word}}^2, \\ \kappa_c = \frac{|L_2|^2 L_3 |L_1|^2}{C_{\mathrm{word}}}, \qquad \Lambda|\kappa_c|^2 = S_u. \end{gathered} \tag{S210}$$

The amplitude capacity $C_{\mathrm{word}}$ is inherited from the complete preparation of the lepton double visit, with effect capacity $\Lambda$. To realize this complete path, write $S_L = \sum_a \beta_a D_a$, $\beta_a = \zeta_* b_a$, and convolve coefficients using unit-group multiplication. Since $D_a D_b = D_{ab}$, put $c = \beta^* * \beta$, $\beta_a^* = \overline{\beta_{a^{-1}}}$; the coefficients of $A_{\mathrm{word}} = \sum_a a_a D_a$ are $a = (c\chi_{-1}) * (\beta\chi_1) * c$. Amplitudes with the same net group action are added before any intermediate visit is read. Evaluation with the remainder bounds of Section 7.1.3 gives $B_{\mathrm{word}} := \sum_a |a_a| = 0.13575205849143\ldots < 1/5$, while $C_{\mathrm{word}} = 2.13263529140049\ldots > 21/10$. Add two cancelling identity paths with coefficients $\pm(C_{\mathrm{word}} - B_{\mathrm{word}})/2$, yielding eight terms $(d_i, D_i)$. The complete preparation and spectral action are

$$\begin{gathered} |u_c\rangle = \sum_i \sqrt{|d_i|/C_{\mathrm{word}}}\,|i\rangle, \qquad U_c = \sum_i \frac{d_i}{|d_i|} D_i \otimes |i\rangle\langle i|, \\ \langle u_c|U_c|u_c\rangle = A_{\mathrm{word}}/C_{\mathrm{word}}, \qquad \sum_{\mu \perp u_c} K_\mu^\dagger K_\mu = I - \frac{A_{\mathrm{word}}^\dagger A_{\mathrm{word}}}{C_{\mathrm{word}}^2}. \end{gathered} \tag{S211}$$

The preparation is normalized, every controlled block is unitary, and all seven failure outputs remain. Each group action preserves $\nu_2$, so the complete output is $\nu_2 \otimes f_c$, where $f_c$ is the normalized eight-path output state. The acceptance amplitude is $\kappa_c = \langle u_c|f_c\rangle$, and the actual acceptance intensity is $|\kappa_c|^2 = 0.001607677003956\ldots < \alpha_8$; the effect capacity connects it to $S_u$ in the main text.

Fix a group-zero target and let $J_w\chi = \chi \otimes |\nu_2, f_c\rangle$ retain every work output. With physical source S distinct from spectral work, use the three-port competition

$$\begin{gathered} K_c = \kappa_c P_0^S \otimes |\nu_2, u_c\rangle, \quad B_c = J_w\sqrt{\alpha_8 I - |\kappa_c|^2 P_0^S}, \quad R_c = \sqrt{q}\,J_w, \\ q = 1 - \alpha_8, \quad |\kappa_c|^2 < \alpha_8, \qquad K_c^\dagger K_c + B_c^\dagger B_c + R_c^\dagger R_c = I. \end{gathered} \tag{S212}$$

The source-domain competition retains the success column of the complete spectral path, while the square-root failure column gives the source state on the failed branch. Choose the entrance and success, failure, and continuation ports as orthogonal physical endpoints, with entrance loading $J_i = |\mathrm{in}\rangle \otimes J_w$. Put

$\mathcal{V} = |s\rangle K_c + |f\rangle B_c + |r\rangle R_c$, $L = \mathcal{V} J_i^\dagger$, $P_i = J_i J_i^\dagger$, and $P_f = \mathcal{V}\mathcal{V}^\dagger$. The joint exchange $U = I - P_i - P_f + L - L^\dagger$ satisfies $U J_i = \mathcal{V}$, is the identity on the orthogonal complement, and preserves the source-group projector.

The reentry edge is $\Pi_Q^A \otimes |\mathrm{in}, n+1\rangle\langle r, n| + \mathrm{h.c.}$; complementary records send continuation outputs to parking. Successes and competitive failures wait with their original histories. The actual loading columns are therefore

$$\begin{aligned} J_1\psi &= |s,0\rangle K_c\psi + |f,0\rangle B_c\psi + |p\rangle\sqrt{q}J_w\psi, \\ J_N\psi &= \sum_{n=0}^{N-1} q^{n/2}(|s,n\rangle K_c\psi + |f,n\rangle B_c\psi) + q^{N/2}|r,N\rangle J_w\psi, \\ J_{\mathrm{load}} &= (I - \Pi_Q^A)\otimes J_1 + \Pi_Q^A \otimes J_N. \end{aligned} \tag{S213}$$

Completeness gives $J_{\mathrm{load}}^\dagger J_{\mathrm{load}} = I$, while the projector $\Pi_s$ onto success at any attempt gives

$$J_{\mathrm{load}}^\dagger \Pi_s J_{\mathrm{load}} = R_N \otimes (|\kappa_c|^2 P_0^S), \qquad R_N = I + (G_N - 1)\Pi_Q^A, \quad G_N = \frac{1-q^N}{\alpha_8}. \tag{S214}$$

The current terminal visits the complete link and independent weak spectral work. Let $\Pi_{\mathrm{acc}} = \sum_{h=0}^{6} |h\rangle\langle h|$ project onto the seven successful paths of the current link, and let $\Pi_f$ project onto its failure, both acting as the identity on spectral work. Let $\Pi_W$ be the independent weak acceptance projector and $U_W$ its complete unitary. The same-position projector for matter position X and accompanying position R is $Q_{\mathrm{eq}}$; orientation $Z_{\mathrm{mov}}$ controls the actual propagation step. Weak occupation is supplied as in Section 7.4, and the independent visit has complete acceptance effect $a_W = w_3/N_W$. After common neighboring-position loading, one orientation endpoint meets the accompanying resource; the neutral marker flips when the current link fails with output D0, or when the link succeeds but the independent weak visit fails. Summing the complete link columns above gives overlap one for input D1, while the cross term for input D0 retains only the joint acceptance effect $b_L a_W$, giving

$$\begin{aligned} \mathcal{T}_X &= P_0^{Z_{\mathrm{mov}}} \otimes S_X + P_1^{Z_{\mathrm{mov}}} \otimes S_X^\dagger, \qquad Q_{\mathrm{eq}} = \sum_x P_x^X \otimes P_x^R, \\ R_{\mathrm{out}} &= \Pi_f P_1^D \otimes I_W + \Pi_{\mathrm{acc}} \otimes \Pi_W, \quad B_{\mathrm{out}} = I - R_{\mathrm{out}}, \\ C_u &= I + Q_{\mathrm{eq}} B_{\mathrm{out}}(X_M - I), \qquad \mathcal{A}_u = \mathcal{T}_X^\dagger C_u \mathcal{T}_X (U_L \otimes U_W). \end{aligned} \tag{S215}$$

$$\begin{aligned} (V^\dagger \otimes U_W^\dagger) R_{\mathrm{out}} (V \otimes U_W) &= P_1^D \otimes I_E \otimes I_W + e P_0^D \otimes (I_E + 3Q) \otimes Q_W, \\ Q_W = U_W^\dagger \Pi_W U_W, \qquad F_{cu} &= P_1^D + b_L a_W P_0^D, \qquad H_u J_u = J_u\, g h_c R_N F_{cu} Y. \end{aligned} \tag{S216}$$

The complete preparation $J_u$ includes the history loading, current $\mathcal{A}_u$, original node, modification work, and physical source. Success at any attempt enters the output exchange once:

$$\begin{aligned} X_{\mathrm{out}} &= \Pi_s \otimes B_N \otimes D_{\mathrm{mod}} \otimes I_{\mathrm{current}} \otimes Y, \\ \mathcal{S}_u = 2J_u J_u^\dagger - I, \qquad H_u &= \frac{g}{2}(X_{\mathrm{out}} + \mathcal{S}_u X_{\mathrm{out}} \mathcal{S}_u), \\ h_c &= |\kappa_c|^2 p_0 F/\sqrt{2}. \end{aligned} \tag{S217}$$

Compressing the output exchange to the complete preparation image and applying Theorem S16.1 gives the persistent generator in Eq. (S216); the baseline coefficient $h_c$ already contains the acceptance intensity of the complete spectral path.

**Proposition S16.3 (Three response rates on the formation image).** On the initial formation image, the common successor has the three response rates $h_c, G_N h_c, b_L a_W h_c$. For the original D0 input, the corresponding branch occupation probabilities are $e, 3ew, 1 - b_L$.

**Proof.** The output columns of S16.2 deliver D1 along all six corrected successful paths, so $\Pi_Q^A P_0^D \Gamma = 0$. Substituting the commuting record and direction projectors into Eq. (S216) gives the three response rates; taking the squared norms of the corresponding output columns of $\Gamma$ gives the stated occupation probabilities. The intersection of $\Pi_Q^A$ and D0 remains a sector of the complete action space, orthogonal to this formation image. This proves the proposition.

Loading and its inverse preserve coherent reversibility across the success, competitive-failure, parking, tail, and weak-work outputs. Path projectors of the formation record are preserved, coherence between paths evolves with the joint phases, and inverse preparation retrieves the current work. If new attempt outcomes are permanently read out, maintenance compatible with those records is used.

## S16.5 Background and motion of the same carrier

The source in down-type capacity failures is retained as well. An isometry E restores each abbreviated failure column $|f, c\rangle$ to $|f\rangle|\Phi\rangle_S|c\rangle$ and is the identity in the success sector. For the original ready columns r and base loading columns $v_y$, use the actual preparation reflections

$$d_{y,jc} = \frac{E(r_{jc} - v_{y,jc})}{\|r_{jc} - v_{y,jc}\|}, \qquad \mathcal{H}_y = I - 2\sum_{jc} |d_{y,jc}\rangle\langle d_{y,jc}|. \tag{S218}$$

These directional columns are orthogonal, and the overlap of the ready and base columns is the real capacity tail. The reflections therefore send ready columns to the required loading and act identically on their orthogonal complement. Write $R_y$ for the subsequent actual node, color, and marker sequence and $U_L$ for the common current link. Then $A_y = R_y \mathcal{H}_y U_L$. Since the link does not act on the source, its transported background projector is exactly

$$Q_{B,y} = A_y P_0^S A_y^\dagger = R_y \mathcal{H}_y P_0^S \mathcal{H}_y R_y^\dagger, \qquad J_y^\dagger Q_{B,y} J_y = p_0 I. \tag{S219}$$

Up-type loading instead preserves the source projector directly. Every column of the three-port competition intertwines $P_0^S$, so the complete historical loading $\mathcal{V}$ satisfies

$$P_{0,\mathrm{out}}^S V_h = V_h P_{0,\mathrm{in}}^S, \qquad \mathcal{V}^\dagger P_{0,\mathrm{out}}^S \mathcal{V} = P_{0,\mathrm{in}}^S. \tag{S220}$$

The second identity uses the isometric sum over all histories. Include the common capacity, tail, and gradient auxiliaries of Section 2.5.3 and apply the same complete preparation reflection to background and mass. Adding the two self-adjoint intertwining relations gives

$$(H_M + H_B)J = J\left[g\mathcal{F}Y + \frac{gp_0}{\Lambda}G_B\right], \quad \mathcal{F}_d^{\text{full}} = F\mathcal{F}_d, \quad \mathcal{F}_u^{\text{full}} = h_c R_N F_{cu}. \tag{S221}$$

The down-type loading endpoints are independent of logical position and block diagonal in the $Z_{\text{mov}}$ basis, hence compatible with the original translation and Z. Up-type encounter depends on the relative positions of matter X and accompanying resource R. Actual position loading stores the unknown earlier position in q:

$$\begin{gathered}\mathcal{L}|x,q,r,\nu\rangle = |x,r,q+x+d_\nu,\nu\rangle, \qquad \mathcal{L}S_X\mathcal{L}^\dagger = S_X S_R, \\ [\mathcal{P}, Z_{\text{mov}}] = [\mathcal{P}, S_X S_R] = 0.\end{gathered} \tag{S222}$$

Here $\mathcal{P}$ is the complete up-type loading. The common neighboring entrance $q = 0, d_\nu = \pm 1$ gives $F_{cu}$ above. At general positions the geometric factor is $I - (1 - b_L a_W)P_0^D P_{\text{neigh}}$. Place the same complete maintenance at both shared motion endpoints and activate the original oppositely directed exchange edges

$$\begin{gathered}T = P_0^{Z_{\text{mov}}} S + P_1^{Z_{\text{mov}}} S^\dagger, \\ \mathbb{G}_+ = \omega \begin{pmatrix} 0 & T \\ T^\dagger & 0 \end{pmatrix}, \qquad \mathbb{G}_- = -\omega \begin{pmatrix} 0 & T^\dagger \\ T & 0 \end{pmatrix}.\end{gathered} \tag{S223}$$

Take $S = S_X$ for down-type matter and $S = S_X S_R$ for up-type matter. Preparing the port as $(1, i)^T/\sqrt{2}$ combines the edges into the original difference propagation. Both endpoints carry all work and formation records and share a phase reference. Blockwise addition gives

$$\mathbb{H}\mathbb{J} = \mathbb{J}\left[g\mathcal{F}Y + \frac{gp_0}{\Lambda}G_B + 2\omega Z_{\text{mov}} D_x\right]. \tag{S224}$$

On a common background spectral branch, subtract the background term and use $g = 2\omega$. For an internal rate f compatible with translation,

$$E_\pm(k) = \pm\sqrt{g^2 f^2 + g^2 \sin^2(k\ell_x)}, \qquad \mu = |gf|, \quad c_0 = g\ell_x, \quad M_{\text{in}} c_0^2 = \mu. \tag{S225}$$

The last identity uses curvature at the origin or the long-wave-packet reading, with parameters measured per active window. In complete-cycle time, $c_{\text{obs}} = f_{\text{act}} c_0$ and $M_{\text{in,obs}} = M_{\text{in}}/f_{\text{act}}$, so $M_{\text{in,obs}} c_{\text{obs}}^2 = m^{\text{obs}} = f_{\text{act}}\mu$. The actual inverse first removes the motion port, then the loading of each family; the up-type process additionally applies $\mathcal{L}^\dagger$ to restore earlier positions.

## S16.6 Complete timing and physical readings

Initial formation, full-work exchange, and resource supply enter the startup duration. Loading, control, inverse operations, repair, and waiting enter the processing time of each cycle in which they actually occur. With a common active window for mass, background, and motion,

$$\tau = t_H + T_{\text{proc}}, \qquad f_{\text{act}} = t_H/\tau, \qquad m^{\text{obs}} = f_{\text{act}}|gf|, \qquad v_E = \frac{f_{\text{act}} g p_0}{\Lambda}. \tag{S226}$$

Resource phases are compensated by the actual program or included in the joint generator, with energy–charge compensation, controls, and switching work accounted for in the complete exchanges. The common period cancels the active fraction in mass ratios, while formation and supply probabilities determine branch occupations. Persistent readings here refer to the loaded code subspace; thermal repair and exit follow Section 2.3.6 on their respective domains.

Module inputs and physical attributes follow the correspondence of Chapter 5, while the complete actions in this section specify their paths and endpoints. Down-type and up-type matter each complete their return under a common successor; the charged-lepton and neutral processes are given in Section 8.2 and S13, respectively.

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